

Seeds of Market-Based Capitalism: Financial Innovation and China's Modernization *

First version: November 2024

This version: August 2026

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Job Market Paper

Abstract

This paper examines the historical and institutional roots of China's market-based capitalism through the lens of financial development. We study how Christian missionaries contributed to the emergence of modern financial and economic activity between 1857 and 1936. We then trace how these effects became dormant under the planned economy after 1949 and re-emerged following the market-oriented reforms of the 1980s. Using a staggered difference-in-differences design for the pre-1936 period, we find that missionary activity increased modern banks and enterprises. We use newly compiled data on mission schools and publications, Chinese bankers, and elites to test several mechanisms. We find evidence mostly consistent with two: practical education and the diffusion of knowledge through publications and translations. Missionaries, especially Protestants, introduced training in subjects such as mathematics, bookkeeping, economics, and English, helping form high-skilled local actors and connect them to foreign knowledge and institutions. They also translated Western texts into vernacular Chinese, helping diffuse useful knowledge relevant to modern economic and financial practices. Exploiting the coincidental timing of railway construction relative to national missionary migration, we document that these effects weakened during the planned economy but re-emerged after China's market reforms, shaping financial development and growth into recent decades. Banking infrastructure grew faster in less exposed areas between 1949 and 1978, suggesting that this re-emergence cannot be explained by physical capital. Instead, the return of high-skilled human capital and foreign investment help explain the re-emergence of these effects. Overall, the results suggest that missionaries transmitted not only human capital, but also institution-specific knowledge and transnational connections that helped plant the seeds of market-based capitalism in China.

*We thank Samuel Bazzi, Sara Lowes, Paul Niehaus, Gaurav Khanna, Craig McIntosh, Ruixue Jia, Ruiming Liu, Jared Rubin, Steven Pfaff, Nicola Limodio, Laura Muñoz Blanco, Martin Fernandez Sanchez, Daryl Ireland, and Steven Brownstone for helpful feedback. We also thank seminar participants at UC San Diego, the University of Bologna, and Boston University Center for the Study of Asia, as well as conference participants at SIOE, the HHDI-NIDE Workshop, IRES, ASREC, SOCAE, DevPEC, and the Migration and Development Conference. We gratefully acknowledge funding from the IHS Grant and the Cindy Vojtech Summer Research Prize in Economics.

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1 Introduction

What explains the origins and takeoff of market-based capitalism? This question has long been central to economics, but it is especially important for understanding China. Since market-oriented reforms began in 1978, China has rapidly integrated into domestic and international markets, private enterprise has expanded, and GDP growth has averaged over 9 percent annually, lifting nearly 800 million people out of poverty (World Bank, 2024). Yet this transformation followed three decades in which private markets and international economic connections were sharply restricted. Did the effectiveness of China's post-1978 reforms depend partly on economic conditions formed before the Communist period?

This paper investigates whether historical exposure to Christian missionaries may help explain why China's reforms were so effective in setting the economy on a steep and durable growth path. We examine how missionaries affected the adoption of modern banks and enterprises between 1857 and 1936¹, and how this historical exposure interacted with the market-oriented reforms of the 1980s to shape banking and economic activity through recent times. Our argument has three parts. Before 1949, missions helped build market-relevant skills and practical knowledge among Chinese actors, while also expanding international connections. From 1949 to 1978, under Mao's central planning, when production and finance were largely state-directed, the returns to much of this expertise fell sharply: people who remained in China had limited scope to use their pre-1949 skills and knowledge, some high-skilled people left the mainland, and access to foreign markets closed. After 1978, reforms that reduced state control over markets and reopened China to the world increased the value of surviving skills and knowledge. Reopening also allowed foreign-exposed talent to reconnect with mainland institutions and foreign capital to flow toward historically exposed places. Studying the emergence of modern finance in China highlights a broader complementarity between the local knowledge and human capital needed to adopt new ideas and the policy environment that allows them to be put to productive use.

A central and measurable institution of market-based capitalism is banking, which mobilizes savings and channels credit toward investment. We focus on financial innovation and the emergence of modern banks, which introduced important changes relative to traditional forms of finance in China. Following the Opium Wars of 1839–1842 and 1856–1860, China was forced to open to the West after a long period of relatively restricted foreign access. Trade expanded, foreign banks entered treaty ports,² and Chinese reformers began to build modern financial institutions. Traditional banks, including draft banks (*Piaohao*) and money houses (*Qianzhuang*), relied more heavily on personal networks, relational enforcement, and owners' capital. Together, they provided long-distance remittances and local credit, but their reliance on personal networks, unlimited liability, and limited capital bases constrained their ability to finance activity at larger scale. Modern banks, by contrast, included limited-liability joint-stock corporations and state banks, mobilized deposits and capital at larger scale, and relied more on formal governance, standardized accounting, contracts, and collateral (Cheng, 2003; Lin et al., 2021). Traditional Chinese finance was well suited to the relational networks and domestic financial system in which it developed. However, as emphasized by Greif et al. (2025), China and Europe developed dif-

¹The historical panel begins in 1857, when our modern-bank data begin, and ends in 1936, before the 1937 Japanese invasion disrupted economic activity and data coverage.

²Treaty ports were cities opened to foreign trade and residence under the treaties imposed on China following the Opium Wars.

ferent institutional arrangements for sustaining cooperation, and corporate forms facilitated cooperation beyond personal networks and at greater scale. China's opening increased the value of standardized organizations that could operate across distance and connect Chinese activity to international markets.

To study how market-relevant inputs formed historically, we combine newly available data from the China Historical Christian Database (CHCD) (Mayfield et al., 2024) with historical data on modern and traditional banks, bank balance sheets, enterprises, trade, railways, treaty ports, government investments, and biographies of Chinese bankers and elites. The CHCD allows us to construct detailed measures of exposure to recorded Christian churches, schools, hospitals, orphanages, publishing houses, and other institutions from 1550 to 1950. The panel structure is important because common missionary atlases tend to oversample prominent missions established early (Jedwab et al., 2022). Consistent with this, in our data, early missionary entry is concentrated in more central and developed places, but this selection becomes much weaker after the initial treaty-port period and when comparing locations reached by missionaries at similar times.

We study the historical roots of market development between 1857 and 1936 by exploiting the yearly expansion of missionaries across Chinese prefectures in a staggered difference-in-differences design. Because missionary presence eventually spread to most prefectures, the design mainly compares places reached by missionaries at a given point in time with places reached later. Locations reached at similar times are generally balanced on pre-existing characteristics, and we find no systematic pre-trends in modern banks, enterprises, traditional banks, railways, treaty ports, government investment, or other predictors of modernization.

Missionary presence led to a large and persistent increase in modern banking. After missionary arrival, prefectures reached by missionaries experienced a 31% increase in modern banks relative to comparison areas. The effect grows gradually after entry and appears across private, government, and foreign banks. Newly digitized bank balance-sheet data show that bank size also expanded: increases in loans and deposits are consistent with a larger role for modern banks in local finance. These results remain similar when we restrict the sample to the post-1900 period, when missionary activity expanded much more broadly inland. Importantly, given the economic importance of treaty ports (Jia, 2014), the results are also similar when we exclude them from the sample. Because missionaries eventually expanded far beyond these initial hubs, the results suggest that missionary activity helped reshape the geography of modern finance in China's interior.

We also document how this process interacted with preexisting financial institutions, suggesting both complementarity and substitution depending on the specific functions of traditional banks. After missionary arrival, *Piaohao* initially expanded but later declined as modern banks rose. Their specialization in long-distance remittances put them in more direct competition with modern banks, which eventually substituted for some of their functions. *Qianzhuang*, by contrast, specialized more in local lending and could supply local information and networks of trust to modern banks, which sometimes used them as intermediaries for part of their lending. They therefore remained more complementary to modern banks and survived much longer. These patterns suggest that missionary exposure did not simply replace Chinese institutions. It helped Chinese actors combine new organizational forms with existing financial demand and locally embedded networks.

Finally, we examine broader market effects. Missionary activity also increased enterprises by 27%,

with the effect concentrated in private firms, and is associated with stronger trade-oriented activity. These broader effects could reflect both a direct role of missionaries in modernization and a downstream response to the expansion of modern banking. These findings are consistent with existing evidence on missionaries and economic development in China (Bai and Kung, 2015) and on missionary connections to international trade (Chen et al., 2025b). Both effects appear with a lag relative to the banking response, consistent with at least part of the broader market effect operating downstream of finance.

Adopting a new organizational form required access to new knowledge and the skills to operate and adapt it locally. Mission organizations could serve as intermediaries in this process. Through schools, publications, translations, and international links, they combined access to foreign knowledge with Chinese-language ability, long local residence, and extensive Chinese participation.³ One important channel was practical education. Missionary schools, especially Protestant ones, taught subjects such as mathematics, science, bookkeeping, economics, and English. These skills were useful for working in, managing, and expanding modern financial institutions. English also helped Chinese actors work in foreign firms and banks, often as intermediaries, where they could learn foreign financial practices and later carry them into Chinese firms and banks.

Using newly digitized biographies of Chinese bankers, we identify bankers with Christian-school backgrounds or other recorded Christian connections. These bankers account for roughly 27% of matched historical banker observations, and their local share rises after missionary arrival. We also use 14,763 biographies of prominent Chinese elites, covering individuals in business, government, science, and other professions. Missionary arrival is followed by sustained increases in elites with Christian-school backgrounds and in elites who study abroad. Together, the banker and elite results suggest that mission education helped form high-skilled Chinese intermediaries with access to foreign institutions and experience. The banking response is also stronger in places with missionary schools and is driven by Protestant rather than Catholic missions, consistent with Protestant missions' emphasis on schooling and with the broader link between Protestantism and human capital (Becker and Woessmann, 2009; Bai and Kung, 2015; Bai, 2019).

The second channel is the diffusion of useful knowledge through publications and translations. Missionaries produced and translated educational, scientific, and economic materials into Chinese, often in vernacular or bilingual formats. These texts made modern economic and institutional practices accessible beyond the classroom. The effect of missionary presence on modern banks is substantially larger in locations with missionary publication or printing-press activity and largest where schools and publications were both present.⁴ Publications made foreign knowledge more accessible, while schools helped people use it. We also find little support for the banking effect being driven by missionaries' own capital needs or by the weakening of Confucian clans, which have been shown to substitute for financial markets in China (Chen et al., 2022b).

To trace these historical differences through the Mao period and subsequent reforms, we connect the historical panel design to a long-run cross-sectional design using the total number of mission institutions

³Appendix Section E.1 documents missionaries' language training, inland residence, and reliance on Chinese personnel, which helped combine foreign access with the local knowledge needed to translate and adapt it.

⁴These comparisons exploit heterogeneity in mission activities, which often overlapped and was not randomly assigned. We therefore interpret them as informative about the mechanism rather than as separately identifying the contribution of each activity.

established by 1936 as our measure of historical exposure. Because missionary presence accumulated endogenously across places, we adapt the strategy developed by [Sequeira et al. \(2020\)](#) to instrument for missionary exposure. Their study develops a strategy for isolating exogenous migration flows based on the coincidental timing of push and pull factors, which we adapt to missionary migration in China. In our setting, the idea is simple: railway expansion made prefectures more accessible to missionaries, but its effect was larger when unusually many missionaries happened to be entering China. We therefore compare prefectures with similar overall railway exposure whose railway expansion happened to occur just before periods of high rather than low national missionary inflows. The identifying variation comes from this timing rather than from differences in railway access itself, which we control for directly in the regression.

This approach allows us to trace the historical arc of missionary influence on banking across political regimes. In the final years of the historical period, the estimates reproduce the positive effect on modern banks and are close in magnitude to the difference-in-differences estimate. After 1949, however, the missionary-exposure gradient attenuates sharply and remains muted during the Mao period. This attenuation did not occur because historically exposed prefectures lost bank offices. Instead, the state expanded banking across China proportionally faster in the least mission-exposed prefectures, compressing the inherited geography of banking. This pattern is consistent with the broader goal of reducing regional differences under the planned economy ([Jia, 2014](#)). Such equalization can help explain why the historical banking advantage dissipated under Mao. But the fact that the same geographic gradient re-emerges after market reform suggests that this redistribution of physical banking infrastructure was not enough to erase the underlying differences across places.

The historical gradient re-emerges as China dismantles central planning and reopens domestic and international markets, and remains positive through 2022. The reforms unfolded in stages. By 1984, rural decollectivization had shifted production decisions from communes to households and the banking system had been reorganized into a two-tier structure separating central-bank functions from lending by specialized state banks; China's accession to the WTO in 2001 further deepened its integration into international markets ([Lin, 1992](#); [Erten and Leight, 2021](#)). In our IV estimates, the missionary-exposure gradient in banking begins to rise in the 1980s, with the clearest changes around these two episodes. By 2022, the estimated effect corresponds to roughly 15% of the mean number of bank offices. The same IV also shows positive effects on broader market development: historical missionary exposure increases GDP throughout 1992–2019 and the number of industrial enterprises after reform, both by 16%, with stronger effects in industries that depend more heavily on external finance. Together, these results show that historically exposed places had an advantage when markets and international connections were allowed to operate, lost that advantage under closure and state allocation, and regained it when China reopened.

We next examine why the historical advantage of mission-exposed places weakened under Mao and re-emerged after reform. Before 1949, these places had produced more internationally oriented, high-skilled people and developed more trade-oriented activity. Biographical data show a larger technical diaspora from these origins: mainland-born engineers and scientists from mission-exposed prefectures are disproportionately observed in Taiwan during the Maoist period. Because these places were also more connected to trade, they had more to lose when China closed to international exchange. The scope

to use pre-1949 expertise narrowed, part of that talent moved outside the mainland, access to foreign markets closed, and bank offices were redistributed toward previously underbanked areas. Together, these changes help explain why the earlier advantage of mission-exposed places weakened during the Mao period.

After reopening, surviving human capital, practical knowledge, and earlier international connections could become economically valuable again. Some people carrying pre-1949 financial expertise remained economically active after reform. Among elites who survived into the reform period, those with foreign exposure from mission-exposed origins are also more likely to have recorded ties to mainland political, government, and scientific institutions, while historically mission-exposed prefectures receive more foreign direct investment. These patterns, together with the re-emergence of banking and broader market activity, provide suggestive evidence that the historical “seeds” were not simply physical bank infrastructure or persistent schooling. They included Chinese human capital, institution-specific knowledge, and connections to external markets whose value depended on the policy environment. Consistent with this interpretation, we find less support for explanations based on the direct survival of mission schools or publications, persistent religiosity, or greater Mao-era state investment.

This paper makes three contributions to our understanding of the origins and evolution of market-based capitalism. Our first contribution is to the literature on the deep roots of development. This literature studies how specific historical shocks and interventions reshape institutions, human capital, beliefs, and social organization in ways that influence economic outcomes after the original exposure has ended ([Acemoglu et al., 2001](#); [Nunn, 2008, 2009](#); [Dell and Olken, 2020](#); [Lowe and Montero, 2021](#)). We extend this literature in two respects. We study the historical origins of a specific organizational innovation, modern banking, and shed light on how the effects of historical exposure evolve over time. Where historical persistence is often documented by relating a past exposure to a later outcome, our setting allows us to observe formation, suppression, and reactivation across political regimes. We thus contribute to a small number of studies that explicitly argue that persistence can vary over time and emphasize the role of political leaders ([Charnysh, 2015](#); [Ochsner and Roesel, 2024](#)), the political context ([Rozenas and Zhukov, 2019](#); [Cantoni et al., 2019](#)), and collective memory ([Fouka and Voth, 2023](#)). We add a human-capital and network-formation channel, showing that talent may be necessary for capitalist development and innovation, but that its effects still depend on the political context. This complements work on China’s post-1978 reforms ([Lin, 1992](#); [Song et al., 2011](#); [Xu, 2021](#); [Erten and Leight, 2021](#)): the rapid and spatially uneven effects of these policies have roots in the historical and institutional context. This perspective also relates to work showing that the effects of development policies depend on the environment in which they are implemented ([Bau et al., 2025](#)). Traditional and modern institutions can also coexist, as shown by the complementary relationship between traditional and modern banks, suggesting that such interactions may be important for development and innovation.

Second, we contribute to work on the diffusion and adoption of technologies, ideas, and organizational forms, which emphasizes human capital, absorptive capacity, locally usable know-how, and complementary managerial and organizational knowledge ([Nelson and Phelps, 1966](#); [Comin and Hobijn, 2010](#); [Giorcelli, 2019](#); [Verhoogen, 2023](#); [Juhász et al., 2024](#); [Giorcelli and Li, 2026](#)). We show that both specialized human capital and access to new knowledge were required for the adoption of a new financial technology in China. [Juhász et al. \(2026\)](#) show that access to knowledge codified in the

vernacular was central to technology absorption in Meiji Japan, where the state coordinated the large-scale codification of foreign knowledge. In our setting, missionary organizations and Chinese actors instead helped translate and adapt foreign knowledge without a centralized state program, amid contested foreign influence and the later suppression of market institutions. In China, [Ma \(2021\)](#) shows that Jesuit exposure increased scientific production among Chinese literati, with the effect weakening after the Jesuits' expulsion. We study a later and broader diffusion process involving bankers, entrepreneurs, and other elites, and connect this exposure to the adoption and later re-emergence of a new organizational form. By emphasizing economically relevant specialists, our findings also relate to evidence on the importance of upper-tail knowledge for industrialization ([Squicciarini and Voigtländer, 2015](#)) and to the debate over institutions and human capital ([Acemoglu et al., 2002](#); [Glaeser et al., 2004](#)). In our setting, these forces are complementary: modern banks provided a new organizational technology, while education, translation, and practical experience supplied the skills needed to operate it.

Third, we contribute to the literature on religion and development. Religious beliefs and organizations can affect development through human capital, technology, finance, public-goods provision, and political institutions ([Becker et al., 2024](#); [Lowes et al., 2026](#)). The role of religion in capitalist development nevertheless remains debated. Classic work links Protestantism to Europe's economic development ([Weber, 2001](#); [Becker and Woessmann, 2009](#)), while other studies show how religious law, political authority, and community organization can hinder credit and financial institutions ([Rubin, 2011](#); [Kuran, 2012](#); [Pascali, 2016](#)). Most empirical work on Christian missions emphasizes their long-run effects on human capital and income, political participation and institutions, and cultural and social outcomes ([Woodberry, 2012](#); [Nunn et al., 2014](#); [Wantchekon et al., 2015](#); [Valencia Caicedo, 2019](#); [Bergeron, 2019](#); [Deng et al., 2025](#)). Missionary printing provides an important example of how missions could also transmit information beyond formal schooling ([Cagé and Rueda, 2016](#)). We shift attention toward market-based development and the adoption of modern banking. Education remains important in our setting, but it operated alongside translations, publications, foreign-language skills, and business networks that helped Chinese actors access and adapt knowledge useful for operating new organizations. Within the literature on missions in China, existing studies document effects on urbanization, women's education, and long-run income ([Chen et al., 2014](#); [Bai and Kung, 2015](#); [Ma et al., 2022b](#)), while more recent work links anti-Christian conflict to lower foreign investment and missionary connections to international trade and information flows ([Huang et al., 2024](#); [Chen et al., 2025b](#)). We complement these findings by showing how missionary exposure contributed to the local adoption of modern banking by Chinese actors and by tracing this differential through Republican-era expansion, Maoist suppression, and post-reform re-emergence.

Our focus on modern banking also connects to work on the historical development of finance in China, which emphasizes legal institutions, culture, communications technology, transport, patriotism, and upper-tail human capital.⁵ More broadly, this connects to seminal work on finance and growth ([King and Levine, 1993](#); [Rajan and Zingales, 1998](#)) and to evidence that bank access and banking reform affect lending quality, structural transformation, and development ([Jayaratne and Strahan, 1996](#); [Burgess and Pande, 2005](#); [Heblich and Trew, 2019](#)). This finance-based channel also offers one possible reconcil-

⁵See [Cheng \(2003\)](#) for historical background and, among recent work, [Levine et al. \(2023\)](#) on legal institutions, [Sun et al. \(2024\)](#) on patriotism, [Chen et al. \(2022b\)](#) on Confucianism, [Lin et al. \(2021\)](#) on telegraphs, [Zhao et al. \(2023\)](#) on railways, and [Lin et al. \(2024\)](#) and [Chen et al. \(2025a\)](#) on upper-tail human capital.

iation between missionaries' limited direct reach relative to China's size and the sizable downstream effects reported in the China literature. The relevant knowledge did not need to reach a large share of the population: a relatively small group of bankers, managers, and entrepreneurs with specialized skills and international connections could help a pivotal institution take root.⁶ Once established, banks could mobilize savings and allocate capital well beyond the individuals directly exposed to missionary organizations.

The remainder of the paper proceeds as follows. Section 2 develops the conceptual framework and provides historical background on China's financial transition and missionary expansion. Section 3 describes the data. Section 4 presents the historical staggered difference-in-differences design and results. Section 5 presents the long-run instrumental-variables design and results. Section 6 examines how the historical effects could dissipate and re-activate. Section 7 concludes.

2 Conceptual and Historical Background

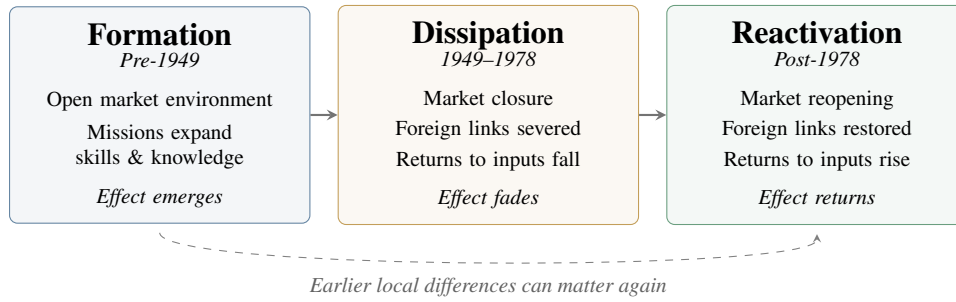
This section provides the conceptual and historical foundations for our empirical analysis. We begin by providing a general framework for understanding how spatial differences in development can evolve nonlinearly yet persist over time. We then describe the history of financial institutions in post-Opium War China and missionary activity, which helps explain why modern banks were a key institution during this period and why missionaries may have affected financial innovation.

2.1 Conceptual Framework

This paper revolves around a central conceptual claim: a complementarity between local market-relevant inputs and the policy environment. Missionary exposure could increase local human capital, practical knowledge, organizations, and networks relevant for market activity. A favorable policy environment raises the return to these inputs; when markets and international exchange are suppressed, differences in these inputs may generate little observed market activity even if some of them survive. Figure 1 helps organize the historical pattern we study: formation before 1949, dissipation under Mao, and reactivation after reform. Importantly, reactivation does not require all original inputs to survive unchanged. Earlier differences need only persist sufficiently, or shape the initial response to reform, for a geographic gradient to re-emerge and subsequently be sustained. The following section provides historical background for this framework.

⁶Related evidence from the United States shows that bank branching improved the geographic allocation of capital and generated persistent productivity gains (Quincy and Xu, 2025).

Figure 1: Conceptual Framework: Formation, Dissipation, and Reactivation



Notes: Local market-relevant inputs include human capital, knowledge, organizations, and networks. The policy environment indicates whether conditions are more or less favorable to markets and international exchange.

2.2 China's Financial Institutions

The institutional transition at the center of this paper is the movement from traditional Chinese finance to modern banking. Before the First Opium War, China's financial market was dominated by two traditional institutions: *Piaohao* (ticket houses) and *Qianzhuang* (money shops). These institutions provided remittances, loans, and currency exchange services. *Piaohao* managed long-distance remittances for the Qing government and large merchants through extensive branch networks (Cheng, 2003). *Qianzhuang* were smaller, locally operated money shops that accepted deposits, exchanged currencies, extended short-term merchant credit, and connected local merchants to the broader financial system.

Modern banks entered China after the First Opium War. The first were privately owned foreign banks established in treaty ports, where they were active in import–export finance, political loans, and other commercial ventures (Moazzin, 2022).⁷ Chinese entrepreneurs and government officials soon recognized the value of these institutions. As foreign trade expanded and modern enterprises became more common, Chinese-run modern banks began to appear. The first Chinese-owned modern bank, the Imperial Bank of China, was founded in 1897 under official sponsorship. It was organized as a joint-stock corporation with limited liability and formal rules governing deposits, loans, and banknote issuance (Cheng, 2003).⁸

The key difference between traditional and modern banks was their organizational structure and ability to operate at scale. Traditional banks relied more heavily on kinship and native-place networks, local information, and relational enforcement. These features supported trust, including across the long-distance branch networks of *Piaohao*. But dependence on owners' capital and inside networks constrained their scalability. Modern banks included joint-stock corporations and state banks. They mobilized deposits and capital at larger scale, used more formal governance, and adopted standardized reporting, contracts, collateral practices, and telegraphic transfers (Cheng, 2003; Lin et al., 2021). Telegraphic transfers accelerated settlement, while rapid reporting improved the monitoring of distant branches. In joint-stock banks, limited shareholder liability facilitated risk sharing and larger institutions. At the same time, weaker personal exposure to losses and delegated management introduced

⁷Several elements of the European banking model later introduced into China developed in late medieval and Renaissance Europe, including deposit banking, book-entry transfers, bills of exchange, and merchant credit (de Roover, 1948; Spufford, 1988). Joint-stock ownership and limited-liability corporate forms developed later.

⁸The Imperial Bank was officially sponsored but not fully state-owned. The first state bank, the Bank of the Board of Revenue, followed in 1904. Modern Chinese banks subsequently included private joint-stock, state, and mixed public–private institutions.

risk-taking and agency trade-offs. These arrangements expanded banks' capacity not only to mobilize funding, but also to transform it into trade and industrial credit. Together, these features made modern banks better suited for an economy increasingly connected to trade, industry, and larger capital markets. Table 1 summarizes these differences.

Table 1: Traditional and Modern Banks in China

Feature	Traditional banks	Modern banks
Organization and governance		
Institutional form	Proprietorships; kin-based partnerships	Joint-stock corporations; state banks
Owners' liability	Unlimited	Generally limited
Agency and control	Owner- and network-based monitoring	Boards, managers, and formal reporting
Financial intermediation		
Funding base	Owners' capital; short-term funds	Share capital; larger deposit base
Core intermediation	Remittances, exchange, merchant credit	Payments, trade finance, industrial credit
Screening and security	Reputation, guarantors, local information	Accounts, contracts, and collateral
Transfers and reporting	Mail transfers; branch correspondence	Telegraphic transfers; rapid reporting
Economic trade-offs		
Main advantage	Relational enforcement; local knowledge	Scale, diversification, standardization
Main constraint	Capital and network dependence	Agency costs; risk-taking incentives

Notes: The traditional-bank category pools *Piaohao* and *Qianzhuang*; “kin-based” refers broadly to family, lineage, and native-place ties rather than exclusively to family ownership. Entries describe broad tendencies rather than mutually exclusive practices. *Piaohao* operated long-distance remittance networks, whereas *Qianzhuang* were more locally embedded and sometimes intermediated between modern banks and local merchants. Traditional institutions also used sophisticated accounting, guarantees, and, in some cases, collateral; modern banks also continued to rely on relationships. The central distinction is the greater standardization and scalability of modern ownership, reporting, and lending practices. Appendix Table A.6 separates the two traditional institutions.

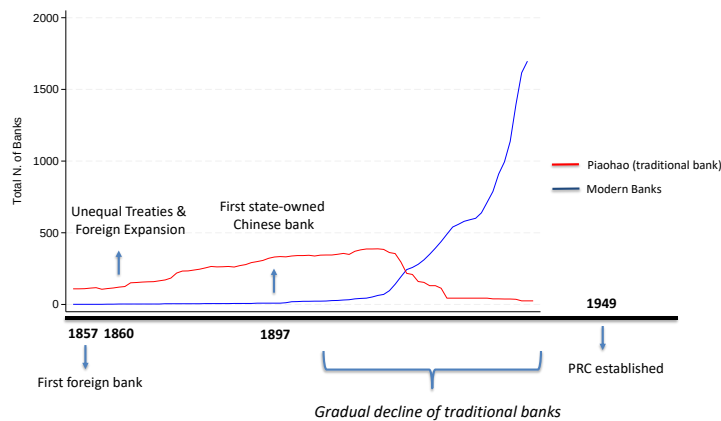
Between 1911 and 1936, modern banks expanded rapidly in response to rising financial demand. The collapse of the Qing government weakened *Piaohao*, whose business depended heavily on official remittances and ties to the state. Modern banks replaced some functions of *Piaohao* by offering lower-cost remittance services. At the same time, modern banks often lent to and through *Qianzhuang*, which survived longer because their local information and networks of trust complemented the more standardized operations of modern banks. Figure 2 shows the aggregate transition: modern banks rose quickly, while *Piaohao* declined.

2.3 Missionary Expansion in China

Christian missions expanded in China during the same period as this transition to modern institutions.⁹ Before the treaty-port era, missionary activity had little formal protection and depended on Qing and local approval (Liu et al., 1980). The treaty-port system created after the First Opium War gave missionaries greater access to coastal cities. The treaties concluded around the Second Opium War marked a further turning point: after 1860, missionaries obtained broader rights to enter inland regions and to build churches, residences, hospitals, and schools (Fairbank and Liu, 1980; Chen et al., 2014; Ma et al.,

⁹Christianity had a much earlier Jesuit presence in China. Matteo Ricci entered China in the late sixteenth century and settled in Beijing in 1601. This presence later receded: Christianity was proscribed in 1724, missionaries outside Beijing were expelled or forced underground, and the papacy suppressed the Society of Jesus in 1773. Chinese Catholic communities nevertheless survived, although foreign missionary infrastructure remained sparse until renewed expansion in the 1840s (Bays, 2011; Mungello, 2012).

Figure 2: Traditional and Modern Banks During China's Opening



Notes: The figure plots the aggregate number of modern banks and *Piaohao* over time. The x-axis is calendar year and the y-axis is the total number of banks. Modern bank data come from the Bank of China Economic Research Office and the *National Banking Yearbook*. *Piaohao* data come from Huang (2002) and Liu (2018).

2022a).¹⁰

Missionary expansion was not smooth. Christianity remained culturally and politically contentious, and anti-Christian conflict culminated in the Boxer Rebellion of 1900, in which over 240 foreign missionaries and 20,000 Chinese Christians were killed (Ren, 1903). This hostility formed part of the broader conflict surrounding Western influence and the unequal treaties. At the same time, China's military defeats and growing interest in Western science, technology, and institutions made access to foreign knowledge increasingly valuable, placing missionaries in the unusual position of being both contested foreign actors and important intermediaries with the West.¹¹ The Boxer Rebellion was a severe but relatively temporary interruption to missionary expansion. Missionary activity resumed rapidly after its suppression, and the following two decades saw substantial further expansion. Figure ?? illustrates the geographic spread of missionary activity over this period. Christian believers increased from 95,943 in 1900 to 366,527 in 1920, while missionary presence reached 94% of China's counties (Stauffer, 1922; Liang, 1988).

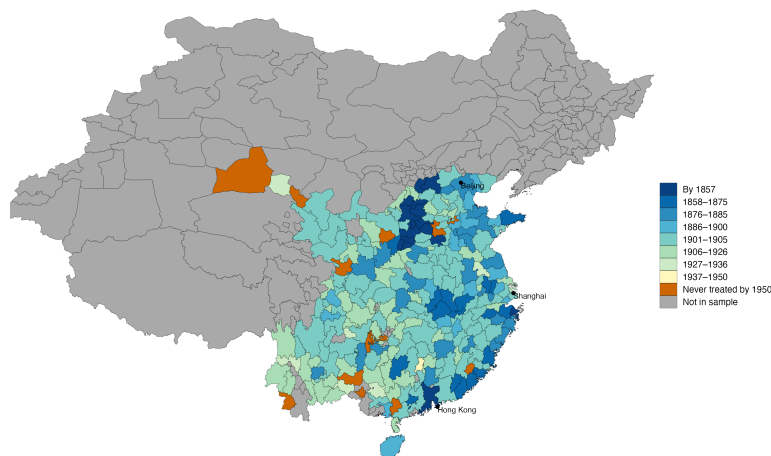
Missionary traditions also differed in ways that matter for our interpretation. Catholic missions often required converts to renounce traditional practices such as ancestor worship, while Protestant missions placed more emphasis on schools, vernacular publications, and practical training, and were more closely associated with ideas of hard work and material success (Fairbank and Liu, 1980; Liang, 1988; Bays, 2011). These differences motivate our Protestant–Catholic comparisons and also connect to the broader Weberian hypothesis linking Protestantism to economic development (Weber, 2001; Becker and

¹⁰The *Convention of Peking* of 1860 confirmed and expanded these rights. By the late nineteenth century, around 1,500 missionaries and more than 80,000 Protestant converts were recorded in China. Conversion remained the primary objective, but schools, hospitals, and material support also served educational and medical purposes while attracting students and potential converts; some schools provided food, housing, medical care, clothing, or stipends alongside religious and academic instruction (Spence, 1990; Nunn, 2010).

¹¹China's defeats in the Opium Wars exposed Western military advantages, while later reliance on Western-officered forces such as the Ever-Victorious Army made the value of foreign expertise particularly visible. Missionary influence also reached prominent individuals and reform efforts: Protestant tracts shaped Hong Xiuquan's religious vision and the Taiping movement; Sun Yat-sen, founder of the Republic of China and the Kuomintang (KMT), encountered Christianity, democracy, and republican government in mission schools; and mission-trained figures and missionary translators contributed to Self-Strengthening projects through machinery procurement, foreign-language education, and technical translation (Spence, 1990).

Woessmann, 2009).

Figure 3: Timing of Missionary Expansion



Notes: The map shows the first recorded arrival of any missionary institution in each prefecture through 1950. Colors identify arrival cohorts; orange prefectures have no recorded missionary institution by 1950, and gray prefectures are not in the analysis sample. The historical difference-in-differences analysis uses arrivals through 1936; the 1937–1950 cohort is shown descriptively to complete the historical expansion. Appendix Figure A.3 presents the same diffusion at selected dates. Mission data come from the CHCD, and prefecture boundaries are based on CHGIS 1911.

2.4 Missionary Education and the Diffusion of Useful Knowledge

Missionaries could affect modern banking and other new organizations through several channels. One important channel was the transmission of practical skills and knowledge useful for operating and adapting these institutions. China already had an advanced education system, but much formal schooling was oriented toward the civil service examination, a system of bureaucratic recruitment that had operated for more than 1,300 years (Bai and Jia, 2016; Bai, 2019). This system emphasized classical texts, literary training, and preparation for official careers. It produced a highly educated elite, but was less directly connected to the practical skills used in modern banks and enterprises, such as accounting, foreign languages, commercial law, and applied mathematics.

Missionary education introduced a different type of human capital. Protestant missionaries in particular established institutions ranging from village primary schools to high schools, professional schools, practical-training programs, colleges, and universities. For our mechanism, the most relevant margin was not basic literacy alone, but the upper-secondary, higher-education, and applied-training institutions that prepared students for work in new organizations. By 1935, Protestant missionaries had founded 16 of the 80 colleges operating in China, 20% of the total, and Protestant colleges offered especially broad training in the natural sciences and applied fields (Ma et al., 2022a). Mission high schools, colleges, and training programs taught subjects such as mathematics, science, bookkeeping, economics, and English (Bays, 2011; Bai and Kung, 2015; Bai, 2019). These skills were useful for working in, managing, and expanding modern banks and enterprises. English training was particularly valuable because it provided access to foreign texts and prepared students for employment in foreign firms and banks, including as intermediaries known as compradors.¹² Missionary schools were small relative to China’s entire edu-

¹²Compradors were Chinese agents who connected foreign firms and banks to Chinese clients and markets through local

cation market, but they were disproportionately important in modern higher education and specialized practical training. They could therefore have high economic value at a time when new financial and commercial institutions were being adopted.¹³

Missionaries also diffused useful knowledge through publications and translations. Missionary presses produced religious texts, school materials, newspapers, scientific works, and bilingual or vernacular publications. A concrete example is the American Presbyterian Mission Press, also known as the Huamei Press. Founded in 1844 and moved to Shanghai in 1860, it trained Chinese workers in modern printing and integrated typesetting, typesetting, printing, binding, sales, and distribution. Skills and personnel from this ecosystem later contributed to Chinese-owned commercial publishing. The founding group of the Commercial Press, established in Shanghai in 1897, came from the missionary-school and mission-press network and drew on the Huamei Press model.¹⁴ These materials made foreign knowledge more accessible to Chinese readers and helped transmit ideas about science, education, commerce, and modern institutions (Dennis et al., eds, 1911; Stauffer, 1922; Liang, 1988).¹⁵

This publication channel complements the education channel. Schools trained students directly, while translations and printed materials allowed knowledge to circulate beyond the classroom. Together, these channels explain why missionary exposure could affect the local adoption of modern banks. Missionary organizations did not simply transplant foreign institutions. They helped Chinese students, workers, and entrepreneurs access and adapt external knowledge, creating skills and networks to support banks and other modern organizations.

3 Data and Descriptive Statistics

This section describes the datasets used in our analysis, focusing on missionary diffusion and modern banks, and presents additional data sources. All data are aligned with 1911 prefectural boundaries from CHGIS (2016), a standard reference in this field (Bai and Jia, 2016; Lin et al., 2021; Dong and Zhang, 2024). Table A.3 reports summary statistics for outcomes from the historical difference-in-differences samples, Table A.4 reports missionary treatment, mechanism measures, and the principal predetermined controls used in the balance tests, and Table A.5 reports the variables in the main long-run instrumental-variables samples.

knowledge and networks. Some later used this experience in Chinese enterprises and financial institutions. See Spence (1990) and Appendix Table E.4.

¹³As qualitative evidence, we reviewed 44 interviews with Christian missionary workers held by the Claremont Colleges Special Collections. The interviews describe mission schools and curricula, including English instruction, girls' education, adult and practical training, and graduates entering Chinese banks. The interviews are used as qualitative illustrations rather than as a representative sample. Appendix Section E.2 presents this evidence alongside historical photographs of classroom instruction, a college library, adult worker training, and practical field instruction; see Appendix Figures E.1 and E.2. It also reports additional historical examples in Appendix Section E.3.

¹⁴Appendix Figure E.3 reproduces an 1895 photograph of the American Presbyterian Mission Press. The Commercial Press's founding group included Xia Ruifang, Bao Xian'en, Bao Xianchang, and Gao Fengchi. The Bao brothers studied at the Presbyterian Qingxin school and worked at the Huamei Press. Gao came from the same school and printing network and remained at the mission press during the Commercial Press's early years. Xia learned English and printing at Qingxin before working as a typesetter in foreign-run presses. The Commercial Press's official history describes the firm as deliberately drawing on the Huamei Press model. We interpret this link as the diffusion of personnel, skills, and organizational practices, not as missionary ownership of the Commercial Press (McIntosh, 1895; Lu, 2017).

¹⁵This information channel is related to recent work on Jesuit missions and international trade, which emphasizes missionary letters and publications as information flows from China to Europe. Our mechanism concerns the internal diffusion of skills and knowledge useful for financial modernization (Chen et al., 2025b).

3.1 Missionary Institutions

We use the *China Historical Christian Database* (CHCD) for detailed geographic data on Christian institutions in China from 1550 to 1950. This dataset, a collaborative effort by theology researchers at the Center for Global Christianity and Mission at Boston University, merges data from various sources for a comprehensive study of Christianity in China. For more details, see [Mayfield et al. \(2024\)](#).¹⁶

The current baseline CHCD v3.0.0 extract contains 20,213 institution-location records. After matching to 1911 prefectures and counting each institution once, it covers 5,885 unique institutions. For supplementary descriptive and school-heterogeneity measures, we also add 971 new prefecture-matched school records digitized with CHCD collaborators from historical school and mission directories.¹⁷

The CHCD records the locations of missionary institutions and, when available, the start and end years of each institution-location relationship. We use those locations and dates to build a prefecture-year panel. Our main treatment is the first year in which any missionary institution is recorded in a prefecture.¹⁸ Appendix Table A.1 details the distribution of institution types, and Appendix Figure A.2 shows the cumulative expansion of missionary institutions over time. In the main analysis, we treat all missionary institutions equally, while using school data to explore heterogeneity.

Figure 3 shows the expansion of missionary presence at the prefecture level. When a missionary institution is established, the prefecture is marked as “treated.” Prefectures shown in gray are excluded due to limited data for remote areas.¹⁹ Before 1900, missions were concentrated along the coast and the Yangtze River. After 1900, they spread rapidly inland, and by 1936 only a few prefectures lacked missionary presence. We call these “never treated” prefectures. Our results remain robust even when these are excluded.

The CHCD also includes person, corporate-entity, and event nodes that can be linked to institutions and prefectures. We use these data mainly to describe missionary flows and to recover subdenomination detail. For the broad Protestant–Catholic split, we instead use the institution tradition field, because the CHCD team treats institution affiliation as more stable than individual affiliation.²⁰

In the linked person data, 54% of named missionary personnel are women. Appendix Section E.1 compares this with historical evidence on the broader Chinese mission workforce.

¹⁶Here is the complete list of CHCD sources. Appendix Figure A.1 describes the CHCD data structure. Appendix Table A.2 reports a short external check against CMS/Stauffer Protestant missionary counts in 1920.

¹⁷The directory sources documented in the CHCD handoff material include Educational Directory of China snapshots for 1914, 1916, 1917, 1918, and 1920, as well as 1936 handbooks. The added records mainly affect education descriptives and school heterogeneity; the any-mission arrival treatment is essentially unchanged because most affected prefectures already had earlier CHCD missionary presence.

¹⁸We use institutions because they capture a stable local organizational presence. In the language used by CHCD collaborators, a mission station is a place where a mission organization sent missionaries for evangelization. It is less formal than a hospital or university, but still represents a local base where missionaries lived or worked, not only individual movement. In the 5,885 prefecture-matched institution records, 1,112 lack a raw relationship-level start year. After supplementing the raw field with institution-node dates and the linked-relation date recovery used in the cleaned data, 966 records (16.4%) still lack a usable start-year.

¹⁹See Appendix Figure A.3 for more detailed maps of missionary diffusion.

²⁰The CHCD v3.0.0 node file contains 37,394 person nodes, 1,213 corporate-entity nodes, and 163 event nodes. Among the 5,560 prefecture-matched institutions classified as either Protestant or Catholic, 5,001 are Protestant (89.9%) and 559 are Catholic (10.1%). Recent work also emphasizes that measuring Catholics in China is difficult and may understate Catholic presence ([Zhao and Ma, 2025](#)). See Appendix Table B.1 for the institution-based broad split, Appendix Table B.2 for people-based subdenominations, and Appendix Figure B.3.

3.2 Modern Banks in China

For our historical analysis (1857–1936), we obtain bank data primarily from the *National Banking Yearbook* by [Bank of China Economic Research Office \(1937\)](#). Between 1934 and 1937, the Republic of China government conducted a nationwide bank survey, using the banking system’s extensive branch network to collect information on the ground and provide a comprehensive account of the historical development of banking in China. The survey records all banks and bank offices still operating at the time, together with older establishments that had already ceased operations. The establishment-year field reaches back to 1857.²¹ The dataset records names, establishment dates, prefectures, addresses, ownership (Chinese government-owned, private, government-private joint, and foreign banks), and details on head offices and branches.

We supplement our analysis with foreign bank data from [Huang \(1995\)](#). Since the census mainly covers Chinese-owned banks, Huang’s collection of enterprise directories (1861–1948) fills in gaps for foreign banks by providing details on bank names, operational dates, countries of origin, main branch locations, assets, and founders.²² We used the Google API to obtain geographic coordinates for each modern bank’s location, manually verified the data, and matched these coordinates to 1911 prefectures defined in [CHGIS \(2016\)](#).

The data show a rapid but uneven spatial expansion of modern banking. As shown in Figure 2, modern banks grew quickly as *Piaohao* declined. In the full historical panel, 9 of 247 prefectures had a modern bank by 1897. This increased to 57 by 1915, 82 by 1926, and 168 by 1936. Appendix Figure A.4 maps this expansion using 1911 prefecture boundaries. Banks were initially concentrated in treaty-port and lower-Yangtze prefectures, then spread more widely inland during the Republican period.

For the long-term analysis, we use annual county- and district-level counts of licensed financial-institution offices from 1949 to 2022, compiled from the China Financial Supervision and Administration Bureau website.²³ We aggregate these counts across institution types and assign each modern county or district to a 1911 prefecture by its centroid.²⁴

3.3 Other Data

Additional details on these data sources are provided in Appendix Section B.

Historical Biographies (1857–1949). We hand-collected and coded biographies of directors, founders, and managers of Chinese-funded banks from *Jindai Zhongguo Yinhangye Jigou Renming Dacidian* ([Jiang, 2014](#)). The resulting person-level data link banking positions to education, Christian connec-

²¹The data are therefore not a pure survivor sample. About 5% of bank rows are explicitly flagged as ceased operations, in liquidation, or otherwise closed, and some report closure or reorganization years. Because closure dates are not recorded systematically, smaller establishments that disappeared before the survey, especially in earlier periods, may still be underrecorded. We therefore interpret the outcomes as total recorded modern-bank establishments rather than a complete opening–closure panel.

²²[Huang \(1995\)](#) collected dozens of enterprise lists published in major Chinese cities between 1861 and 1948, such as the North China Desk Hong List, The Shanghai Directory, and others.

²³See <https://www.cbirc.gov.cn>.

²⁴According to the Ministry of Justice of the People’s Republic of China, financial institutions include policy banks, commercial banks, financial asset management companies, credit unions, postal savings institutions, trust and investment companies, financial leasing companies, and foreign financial institutions. The recent registry’s institutional definition may therefore differ from those of the historical modern-bank records. Using both sources nevertheless provides a broad sense of financial activity across the historical and reform periods, without implying that their institution counts are perfectly comparable.

tions, comprador experience, and careers across Chinese and foreign financial institutions.²⁵ We also use the Republican China Elite Database assembled by Wang et al. (2026) from Xu's *Minguo Renwu Dacidian* (Xu, 2007). The source volume was converted by optical character recognition, manually cleaned, parsed into person-level fields, and linked by native place to 1911 prefectures. Its 16,445 biographies record education, overseas study, and careers, but represent prominent individuals, not the general population.

Historical Financial Records (1823–1940). We digitized two complementary sources of modern-bank balance sheets. Republican Ministry of Agriculture and Commerce tables for the 1910s list head offices and branches separately; we geocode each location to 1911 prefectures and aggregate deposits, capital, banknote issuance, and reserves (Ministry of Agriculture and Commerce, 1912–1924). The *National Banking Yearbook* provides richer bank-level accounts for 1932–1936, including loans, assets, income, cash, securities, and paid-in capital (Bank of China Economic Research Office, 1937). We also reconstruct branch histories of *Piaohao* from Huang (2002) and Liu (2018), and geocode *Qianzhuang* records from the *National Banking Yearbook*. The dated opening and closing records for *Piaohao* allow us to fully track entry and exit rather than only surviving institutions.

Other Historical Outcomes (1840–1936). We geocode domestic-enterprise head offices from Du (1991, 2019) and complement them with foreign-enterprise directories from Huang (1995). The domestic source records registered, relatively capitalized firms and therefore underrepresents smaller businesses.²⁶ Prefecture-level domestic and international trade outcomes come from an unusually detailed 1934–1936 census of product types, production, prices, and sales areas. The Republican government commissioned local postal offices to collect information at the county and township levels (Ministry of Communications, 1937).

Historical Controls. We geocode railway stations and opening years from Ma et al. (1983) and reconstruct prefecture-level railway access over time. Geographic and economic controls (including historical population, coastal and river access, rice suitability, provincial-capital status, and treaty-port status) come from Bai and Jia (2016); selected specifications add genealogy-book counts from Dong and Zhang (2024) as a measure of clan intensity.

Long-Run Economic Outcomes. Beyond the post-1949 financial-institution office counts described above, we use 1-km gridded GDP for 1992–2019 from Chen et al. (2022a), summing grid cells directly within 1911 prefecture polygons rather than passing through modern administrative units. Industrial-enterprise outcomes for 2007–2013 come from the National Bureau of Statistics' Annual Survey of Industrial Production. We assign geocoded firms directly to 1911 prefectures, using county-centroid and area-weighted city matches only when a direct spatial match is unavailable. Foreign direct investment in 2017 comes from the *China City Statistical Yearbook*; population adjustments use historical population and the National Bureau of Statistics' 2000 Population Census and 2015 one-percent sample survey.

Shaanxi Provincial Records (1949–1998). We transcribed the Shaanxi Provincial Bureau of Statistics' county compilation for 1949–1990, which provides education, fiscal, investment, infrastructure, enterprise, and market outcomes unavailable in a comparable national panel (Shaanxi Provincial Bureau of Statistics, 1991). We harmonize changing county, district, and county-level-city names to 1911 counties

²⁵Appendix Figure B.1 illustrates how narrative biographies were converted into coded fields.

²⁶The source includes shipping, industrial, and mining enterprises capitalized above 10,000 silver dollars and financial enterprises capitalized above 50,000 silver dollars.

using reviewed Chinese-name aliases. We separately digitize dated references to Christianity, closures, and repression from Shaanxi county and city gazetteers through 1998. These data permit a finer county-level replication within one province, although they do not substitute for the national analysis.

Other Long-Run Measures. The Chinese Engineers Relational Database provides structured biographies, education histories, jobs, and locations for engineers in Republican China and postwar Taiwan (Pelzer et al., 2025); the Republican-era elite biographies described above measure foreign exposure and later institutional roles. Contemporary education comes from population Census and intercensal microdata from 1990 to 2015; religious affiliation, values, and financial behavior come from the China Family Panel Studies (Xie and Hu, 2014) and the 2007 *Spiritual Life Study of Chinese Residents*. We link their reported locations to the corresponding historical geography; Appendix Section F.4 describes the samples and specifications.

4 Historical Analysis

Estimating Equations

Our baseline historical panel includes 247 prefectures between 1857 and 1936, of which 232 receive missionaries during the sample period.²⁷ We exploit the gradual expansion of missionaries described in Section III and define treatment based on the timing of missionary arrival. We follow Sant’Anna and Zhao (2020) to construct heterogeneous-robust estimates in an event-study regression. For simplicity, consider first the following static standard TWFE model, valid under heterogeneous treatment effects:

$$Y_{it} = \alpha_i + \theta_t + D_{it}\beta_{post} + \varepsilon_{it} \quad (1)$$

Where i and t stand for prefecture and year, respectively. D_{it} is a dummy variable that takes the value 1 for all years after the first missionary institution entered that prefecture, and 0 otherwise. D_{it} is always 0 for prefectures where missionary institutions never entered (i.e., “never treated”). Y_{it} represents the total number of modern banks or enterprises. The parameter of interest, β_{post} , measures the change in the outcome variables of the treated prefectures compared to the yet-to-be-treated and never-treated prefectures, conditional on the set of prefecture and year fixed effects. By design, we exclude prefectures that were already treated before 1857 (i.e., “always treated”), and we cluster standard errors by prefecture.

In practice, our estimation is more closely represented by the following TWFE event-study model:

$$Y_{it} = \alpha_i + \sum_{r \neq -1} 1[R_{i,t} = r]\beta_r + \varepsilon_{it} \quad (2)$$

Where we define $G_i = \min\{t : D_{i,t} = 1\}$ to be the earliest period at which unit i receives treatment, $R_{i,t} = t - G_i$ is the time relative to treatment (e.g., $R_{i,t} = 0$ in the first treated period for unit i), and the summation runs over all possible values of $R_{i,t}$ except for -1 , the last pre-treatment period. However, even this specification can encounter “forbidden comparison” issues under heterogeneity across cohorts

²⁷This is not the total number of Chinese prefectures, which ranged between 250 and 300 at the time. We included prefectures in the sample for which we have data on modern banks, enterprises, missionaries, and covariates. This yields coverage similar to that of other historical studies of China, due to data availability.

(Sun and Abraham, 2021). Given that in our setting we are estimating ATTs over an 80-year period and for 42 cohorts, we believe that the doubly-robust Sant’Anna and Zhao (2020) estimator is suitable for our baseline model, as it will allow us to correctly estimate β_r even if the treatment effect for a cohort treated in 1861 is very different from that of a cohort treated in 1934.²⁸ The distribution G_i is shown in Appendix Figure C.1.

Understanding the First Stage and Balance

A key assumption in our model is parallel trends between control and treated units in the staggered setting.²⁹ This imposes that, in the counterfactual scenario where treatment had not occurred, the average outcomes for all adoption groups would have evolved in parallel in the post-period. In our case, this means that for each cohort of treated prefectures, we would expect the total number of modern banks (or enterprises) to have evolved similarly to not-yet-treated cohorts if they had not received treatment. While we do not observe the counterfactual outcome, we can examine how prefectures that receive missionaries in a given time frame compare to the next-treated cohort of prefectures. If we observe differences in economic and social indicators across these prefectures, it could suggest that they were already following distinct trends.

A key concern from the literature on missionaries is that missions might have been established in places that were positively selected (Jedwab et al., 2022). However, it is important to note that most studies on missionaries lack access to panel data, so we know relatively little about the path of expansion and the relative selection year-to-year, especially when considering economic dynamics rather than static levels. Mission access changed repeatedly, from treaty-port restrictions to later inland expansion and the Boxer Rebellion. We therefore test whether these changes created selection bias.

A first simple exercise is to examine what predicts the establishment year of missions in the cross-section (the variable that defines treatment cohorts G_i). We regress this variable on a series of predetermined economic and social indicators at the prefecture level, including province fixed effects. The time-varying variables in this specification are measured before the relevant missionary entry window. They include modern banks, enterprises, government investment during the Self-Strengthening Movement, railways, and treaty ports, together with geographic and economic controls from Bai and Jia (2016).³⁰

The results from these regressions are shown in Table 2. Appendix Figures C.2 and C.3 report the corresponding one-by-one coefficient plots. Columns 1 and 2 reveal significant selection when considering the entire time period and all prefectures, except for those always treated. Prefectures with

²⁸Note that since our estimation does not include covariates, we are assuming unconditional parallel trends and not taking advantage of the outcome model and propensity score matching. This makes this estimator very general and similar to other estimators proposed in the literature. In short, our estimator is calculating many group-time average treatment effects on the treated, $ATT(g, t) = \mathbb{E}[Y_{i,t} - Y_{i,g-1} \mid G_i = g] - \mathbb{E}[Y_{i,t} - Y_{i,g-1} \mid G_i = g']$, for any $g' > t$, which gives the average treatment effect at time t for the cohort first treated in time g . These ATT’s are then aggregated to obtain event-study estimates as an average of the treatment effect l periods after adoption across different adoption cohorts (Roth et al., 2023): $ATT_l^\omega = \sum_g \omega_g ATT(g, g + l)$. Intuitively, our estimator can be viewed as an aggregation of the multi-period analog of equation 2, with comparison groups being the not-yet treated in subsequent periods and the never-treated.

²⁹More precisely: $\mathbb{E}[Y_{i,t}(0) - Y_{i,t'}(0) \mid G_i = g] - \mathbb{E}[Y_{i,t}(0) - Y_{i,t'}(0) \mid G_i = g']$, for all $t, t' \geq g_{min}$, where g_{min} is the first period in which a unit is treated

³⁰Modern banks, enterprises, and government investments are observed in 1880, treaty ports in 1860, and railways in 1890. For the post-1900 specification, time-varying controls are measured in 1900. Population and provincial-capital status are measured using the earliest available pre-period values.

Table 2: Predictors of Mission Establishment Timing and Status: Prefecture-Level Estimates

Variables	(1)	(2)	(3)	(4)	(5)
	Establish. Year All	Post 1900	Cohort Comparisons		
	1860-1900	1900-1904	1904-1920		
Total Modern Banks	9.156 (8.755)	-1.899 (5.715)	-0.196 (0.244)	-0.069 (0.364)	0.106 (0.251)
Total Enterprises	0.851 (2.399)	0.096 (0.522)	-0.051 (0.038)	0.024 (0.034)	-0.020 (0.023)
Ln (investment gov. factory)	0.016 (0.413)	0.332 (0.696)	0.026 (0.018)	-0.086* (0.045)	0.063 (0.056)
Total railways	-5.237* (3.004)	-2.163 (3.654)	0.837*** (0.230)	-0.056 (0.174)	0.025 (0.099)
Total Treaty Ports	-25.857*** (9.315)	-0.629 (2.666)	0.695*** (0.158)	0.152 (0.171)	0.319 (0.275)
Whether there's main river	-1.008 (1.814)	0.195 (1.371)	0.058 (0.059)	0.073 (0.088)	-0.072 (0.107)
Provincial capital	-9.202*** (2.391)	-4.765** (2.241)	0.190** (0.078)	0.195 (0.146)	0.257 (0.217)
Crop suitability: Rice	-0.660 (1.225)	0.861 (1.002)	-0.014 (0.040)	-0.051 (0.064)	0.068 (0.081)
Ln population	-2.464** (1.128)	-1.428 (0.883)	0.039 (0.037)	0.116** (0.057)	0.034 (0.068)
Coastal	-0.240 (3.100)	0.260 (3.421)	-0.054 (0.101)	0.072 (0.213)	0.062 (0.475)
Province FE	Yes	Yes	Yes	Yes	Yes
Obs.	214	156	215	157	77
Adj. R sq.	0.330	0.105	0.342	0.0752	0.208
Mean Dep. Var.	1901	1907	0.250	0.448	0.635
S.D. Dep. Var.	13.75	8.173	0.433	0.498	0.482
P-value F-stat.	1.56e-05	0.452	3.59e-08	0.180	0.494

Notes: The historical panel includes 247 prefectures, of which 232 receive missionaries during the sample period. Regressions are at the prefecture level. Column 1 regresses G_i , the variable defining treatment (i.e., the establishment year of missions), on a series of covariates. The dependent variable for column 2 is the same but only for prefectures where missions were established post-1900. For columns 3-5, the dependent variable is whether a mission was established in the given range of years, and the sample includes only prefectures receiving missionaries in that time frame and in the subsequent period. Controls are from [Bai and Jia \(2016\)](#) and are measured in the pre-period. Banks, enterprises, and government investments are observed in 1880, treaty ports in 1860, and railways in 1890 (all measured in 1900 for column 2). Obs. refers to the number of observations; Adj. R^2 is the adjusted R^2 . Fixed effects may vary; refer to the table. Standard errors are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. Appendix Figures C.2 and C.3 report the corresponding one-by-one coefficient plots.

more railways, treaty ports, population, or provincial capital status are more likely to have received missionaries earlier. This finding aligns with historical accounts and the missionary literature, which indicate that missionaries were initially confined to a few major cities and treaty ports, with their freedom of movement only being established after the Second Opium War in 1860. Following this, missionaries aimed to reach as many prefectures as possible to convert all ethnic groups, eventually extending their reach to more remote and less populated areas ([Bays, 2011](#)). Column 2 focuses on prefectures treated after 1900, when selection is much weaker. Due to this observation, for robustness, we will later restrict the sample to post-1900 periods only.

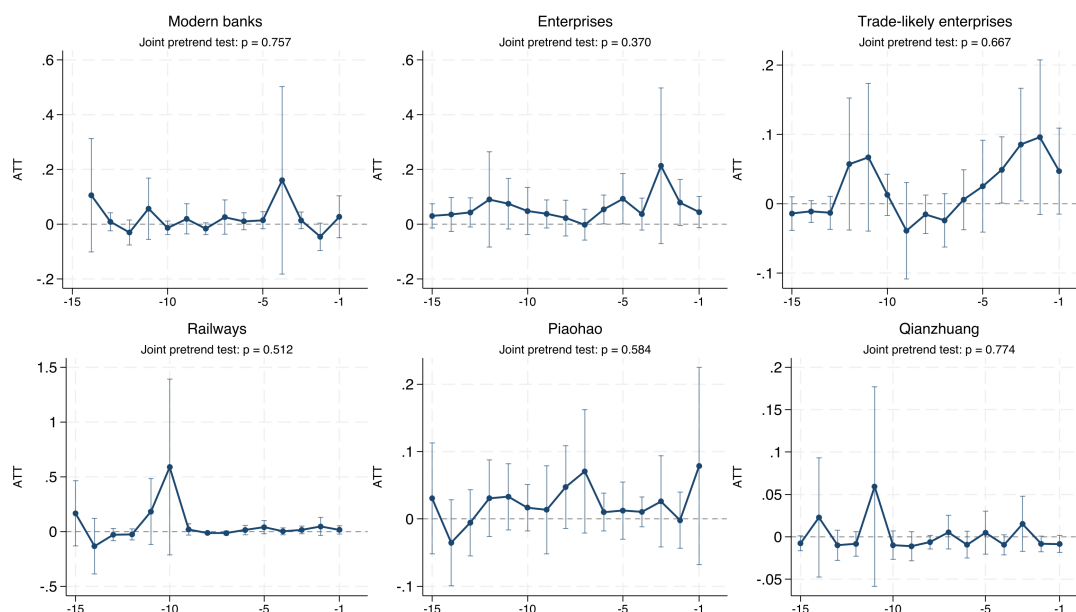
Next, following [Colonnelli and Prem \(2022\)](#), we compare cohorts of prefectures that receive treatment within a given range of years to those that will receive it in the future and those never treated. We regress the indicator of whether a missionary institution entered a prefecture for different treatment co-

horts on the same X 's, now measured at the beginning of the time period of the treated cohorts. Columns 3–5 of Table 2 split the sample into four time periods and confirm the patterns observed in columns 1 and 2. While the estimation provides a weighted average of group treatment effects across different time periods, which is difficult to exactly replicate in a balance table, we can also generate narrower cohorts to compare a given treated cohort with the next cohort. Appendix Table C.1 uses a finer partition for these comparisons.³¹

Appendix Table C.1 shows that treatment and control groups are generally balanced, with only a few variables statistically different from zero. We could be facing power issues due to the limited sample size, but the results suggest that the selection we previously observed was largely driven by comparisons over long periods, when prefectures treated early were very different from those treated decades later. When comparing cohorts closer in time, treated and not-yet-treated prefectures appear more similar.

Consistent with the balance evidence, we also find no systematic pre-trends in the outcomes and predetermined covariates for which the historical panel contains sufficient pre-treatment variation. Figure 4 reports the annual pre-treatment coefficients across historical outcomes. Appendix Figure D.14 reports the corresponding evidence for government investment and treaty ports, for which annual data are not consistently available.

Figure 4: Annual Pre-Treatment Coefficients Across Historical Outcomes



Notes: The panels report annual Callaway–Sant’Anna event-study coefficients for event years $T - 15$ through $T - 1$, where estimable, with 95% confidence intervals. The reported p -values are covariance-based joint Wald tests that the $T - 15$ through $T - 2$ coefficients equal zero. Outcomes remain in their native units. Appendix Section D.3 explains the construction of trade-likely enterprises. *Piaohao* and *Qianzhuang* are the two types of traditional banks examined in the paper.

One interpretation is that mission expansion depended on the interaction between local suitability and external missionary supply. Local factors, such as a new railway connection or demand for schools,

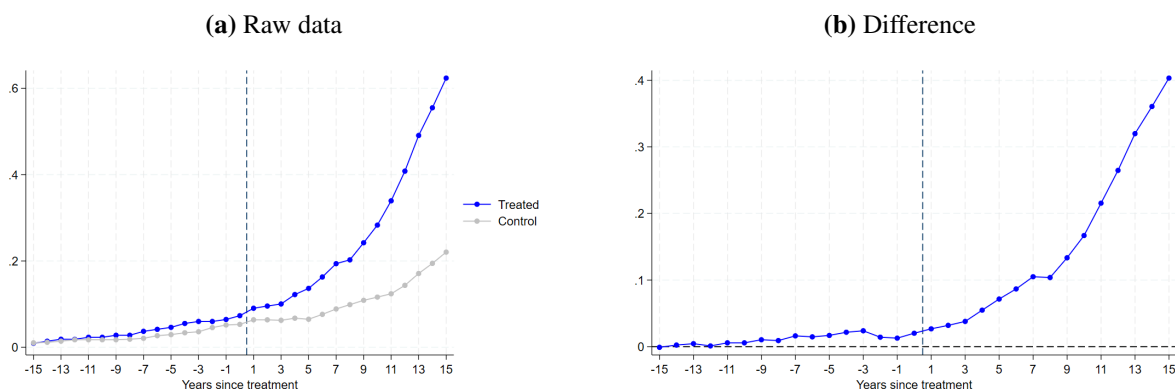
³¹Appendix Table C.1 splits the sample into eight narrower cohort comparisons: 1860–1880, 1880–1890, 1890–1900, 1900–1903, 1903–1904, 1904–1905, 1905–1910, and 1910–1920. The cohort boundaries are chosen to divide prefectures as evenly as possible across the comparison windows. Appendix Figure C.4 reports the corresponding one-by-one coefficient plot.

could make several prefectures plausible mission targets around the same time. Which of these places received missionaries first could then depend on external factors, such as missionary inflows, denominational budgets, conferences, revivals, wars, or funding cycles, that were not driven by the local economic trends of a specific prefecture.³² This interpretation is related to the source of variation used in the long-run IV analysis below, although the DiD estimates themselves do not rely on that instrument.

Results

We begin with modern banks, our main historical outcome. Panel A of Figure 5 displays the raw data over a window of $[-15, 15]$ years surrounding the arrival of missionaries, using the 247-prefecture historical panel between 1857 and 1936. The control group consists of the next cohort of treated prefectures and the never-treated group for each year. While not perfect, this comparison closely mimics the variation underlying our estimator. Treated and control prefectures follow similar trends before missionary arrival and then begin to diverge after treatment. Panel B plots this difference directly.

Figure 5: Raw Data: Total Modern Bank Dynamics



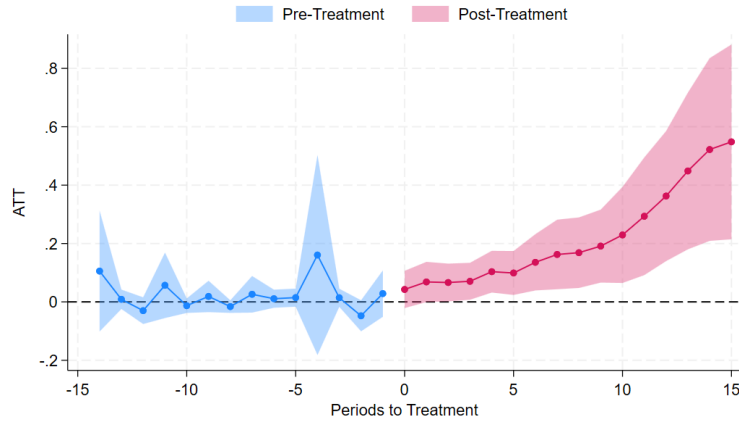
Notes: Panel A displays the raw data for total modern banks over a window $[-15, 15]$ years surrounding missionary arrival, using the 247-prefecture historical panel between 1857 and 1936. The control group consists of the next cohort of treated prefectures and the never-treated group for each year. Panel B plots the treated-control difference in the raw data.

To identify these dynamics more precisely, Figure 6 plots the event-study estimates from equation 2. The pre-treatment coefficients are close to zero, while the effect becomes positive and grows gradually after missionary arrival. The post-period average ATT is 0.22, which corresponds to a 31% increase over the mean number of modern banks in the sample.

One concern is that modern banks are rare in the early part of the sample, so flat pre-trends may partly reflect limited pre-period variation. Appendix Section D.1 addresses this concern in two ways. First, a design-based minimum detectable effect exercise uses the estimated pre-period coefficients and covariance matrix to ask how large hidden pre-trends would need to be detected. The exercise shows that the design is informative against several smooth pre-trend shapes, although very late pre-trend activation is harder to detect. Second, the result is robust to alternative outcome transformations, including $\log(y + 1)$, growth rates, an extensive-margin indicator, new-bank flows, and province-conditioned outcomes.

³²Francis Price Jones, for example, had studied Arabic expecting to serve in North Africa. When World War I stopped his mission board from sending missionaries there, it offered him immediate placement in China instead. Source: The Claremont Colleges Library.

Figure 6: Missionary Presence and Total Modern Banks: 15 Periods



Notes: The figure reports event-study estimates from Equation 2, using 15 periods before and after missionary arrival. The dependent variable is the total number of modern banks. The average post-treatment ATT is 0.22. The dependent-variable mean is 0.7 overall, 1.7 in 1920, and 5.8 in 1936. Appendix Table D.1 reports the pre- and post-treatment averages and alternative bank outcomes.

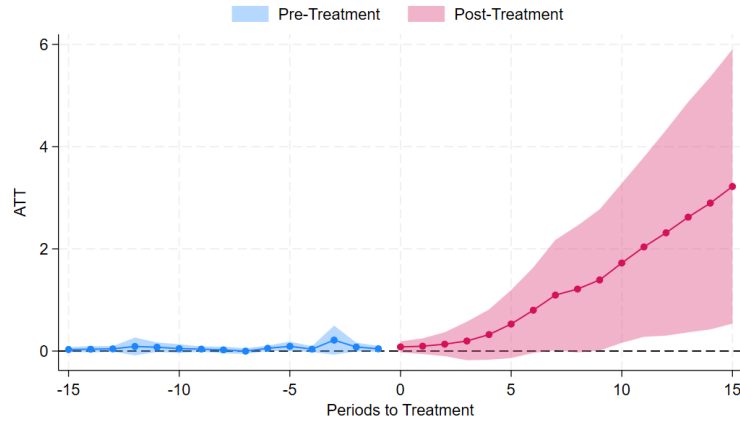
The effect is also not limited to the aggregate number of banks. Appendix Figure D.6 shows similar patterns for private, government, and foreign banks. Supporting balance-sheet evidence in Appendix Tables D.3 and D.5 also points in the same direction: missionary exposure is associated with larger deposits, loans, assets, and related measures of bank size. These results suggest that missionary presence increased modern banking activity, not just recorded establishments.

We next ask whether missionary exposure also affected other forms of market activity. The number of enterprises, particularly private ones, is a natural outcome. One possibility is that modern banks relaxed financial constraints and supported enterprise formation. A second is that missionary exposure directly affected enterprises and other market-related outcomes through the practical training provided by mission schools and the commercially relevant ideas spread through publications and translations, as described in Section 2. Figure 7 presents the results for total enterprises over the same 15-period window. As with modern banks, there are no visible pre-trends before missionary arrival. After treatment, missionary presence has a delayed positive effect on enterprises. This delayed timing is consistent with a downstream response operating through modern banks, but it does not rule out a more direct education and knowledge-diffusion channel. The event-study aggregation reported in Appendix Table D.1 gives a post-treatment ATT of 1.29, about 27% of the mean.

Appendix Figure D.8 and Appendix Table D.8 report additional market outcomes. Missionary exposure is associated with stronger growth in trade-oriented activity and exports in 1936, consistent with modern banks supporting broader market exchange. The trade effects also appear with a lag, which reinforces the interpretation of trade as a downstream market response. Appendix Figures D.10 and D.11 then study the transition from traditional to modern finance. The clearest substitution pattern is for *Piaohao*: missionary arrival is followed by the rise of modern banks and the eventual decline of these older remittance institutions. The *Qianzhuang* results are reported separately because their local networks appear more complementary to modern banks.

Finally, Appendix Section D.5 reports the remaining identification checks. The results are robust to restricting the sample to the post-1900 and post-1905 periods, excluding treaty ports, using longer

Figure 7: Missionary Presence and Total Enterprises: 15 Periods



Notes: The figure reports event-study estimates from Equation 2, using 15 periods before and after missionary arrival. The dependent variable is the total number of enterprises. The average post-treatment ATT is 1.29. The dependent-variable mean is 4.8 overall, 7 in 1920, and 17 in 1936. Appendix Figure A.5 reports the evolution of total enterprises over time, and Appendix Table D.1 reports the corresponding pre- and post-treatment averages.

event windows, balancing the number of pre- and post-treatment periods, adding railway controls, and examining group-specific ATTs. Appendix Figure 4 reports annual pre-treatment coefficients for modern banks, enterprises, trade-likely enterprises, railways, *Piaohao*, and *Qianzhuang*; Appendix Figure D.14 reports the corresponding evidence for government investment and treaty ports.

Mechanisms

Why would exposure to missionary institutions contribute to the development of modern banks? We focus on two related mechanisms, supported by available data: practical education and the diffusion of useful knowledge through publications and translations. Both channels speak to the same idea. Modern banks were a new institutional technology in China, and missionary institutions helped form the skills and knowledge that made this technology usable locally. The exercises below should be interpreted as suggestive mechanism evidence, since missionary schools, publications, and other mission activities often arrived together. We also examine and find less support for several alternative explanations.

Education One of the primary channels through which Christianity can foster market-oriented development is through the dissemination of ideas and education (Becker and Woessmann, 2009; Bryan et al., 2020). Jedwab et al. (2022) survey fifty studies on missionaries published over the past decade and find that nearly all of them associate colonial Christian missions with improved human capital outcomes. Education has also been identified as a key mechanism in other studies that find positive effects of missions on Chinese development (Bai and Kung, 2015; Ma et al., 2022b). In our setting, the relevant channel is a specific form of practical education. Missionary schools, especially Protestant ones, taught subjects such as mathematics, science, bookkeeping, economics, and English. These were skills needed to work in, manage, and expand modern financial institutions. We examine education in four ways.

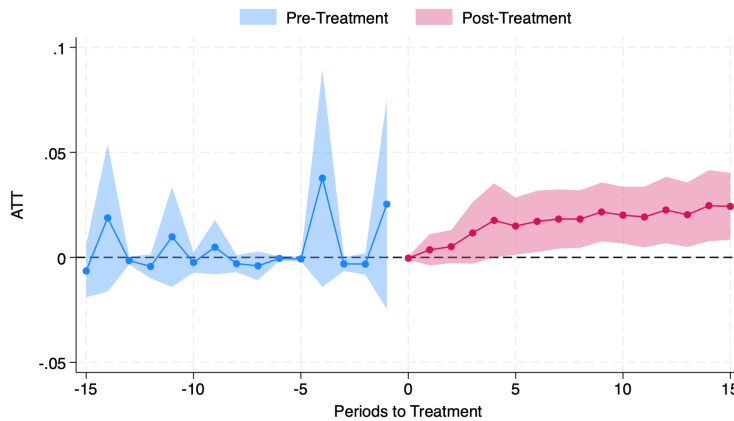
First, we focus on skilled education by using our banker dataset to assess how missionary institutions influenced the rise of Christian-affiliated bankers. If missionary education had a significant effect on the financial sector, we would expect a higher share of bankers who were either affiliated with Christianity

or educated in Christian schools in areas with missionary presence.³³

To test this, we digitized a book that compiled the resumes of notable bankers and created a variable to represent their relationship with Christianity (Jiang, 2014). We coded bankers as “Christian-affiliated” if they were born into a Christian family, converted to Christianity, attended a church school, or worked in a Christian institution. We then matched the names of the banks where these bankers were employed with our bank data for the period 1857–1936.³⁴ The presence of Christian bankers is significant, even though the data include only Chinese individuals and exclude foreigners: across the years, there are 693 Christian bankers out of 2,576 total bankers, yielding a ratio of Christian bankers to total bankers of 26.90%. The annual cumulative ratio was substantially higher in the early period: it exceeded 50% from 1865 through 1893 and peaked at 70.91% in 1885–1888. For intuition on the magnitude of this ratio, we compare it with the proportion of Christian converts. According to data provided by Mattingly and Chen (2022), the average number of Christians in China’s prefectures in 1920 was 2,730, with an average population of 1,371,070, implying that Christians made up 0.20% of the population. Appendix Figure E.8 shows the ratio of Christian bankers to the total number of bankers over time.

Figure 8 then estimates the effect of missionary presence on the share of Christianity-related bankers. The share increases after missionary arrival, consistent with the idea that missions helped train or connect individuals who later worked in modern banking.

Figure 8: Missionary Presence and the Share of Christianity-Related Bankers



Notes: The figure reports event-study estimates from Equation 1, using the share of Christianity-related bankers as the outcome. The average post-treatment ATT is 0.016. The dependent-variable mean is 0.025 overall, 0.052 in 1920, and 0.113 in 1936.

Second, we turn to Republican-era elites to ask whether missionary presence contributed to high-skill Chinese human capital and connections to foreign education and networks. Among the 14,763 biographies matched to our historical prefectures, 1,282 (8.7%) have a coded Christian-school background, our proxy for Christian educational affiliation. Appendix Figure B.2 reports its level and share across school-age cohorts. Because the biographies do not consistently report when and where schooling

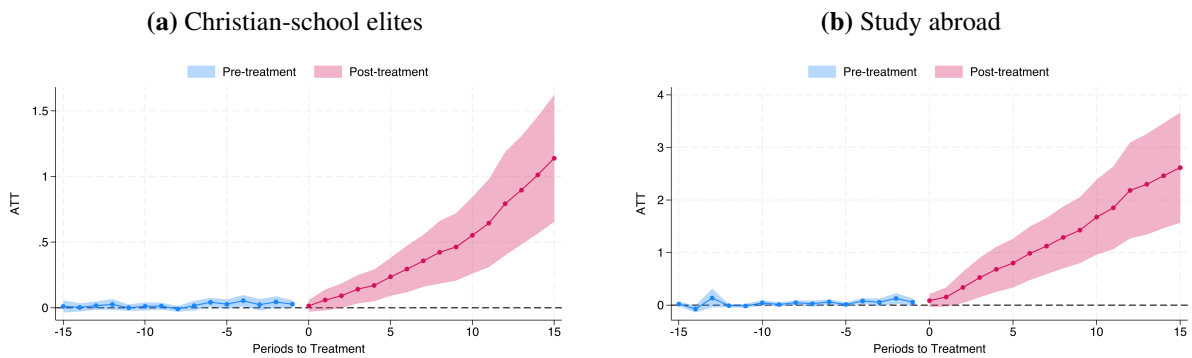
³³For example, Lin Zifeng (1892–1971) was a prominent banker and Baptist church leader in the Republic of China. His father was a Baptist pastor, and Lin became a Christian at the age of 14, attending the Baptist Jieyang Truth High School. In 1922, Lin founded the K. W. Bank in Guangzhou, which was reorganized as the K. W. Savings Bank in June of the same year. Between 1924 and 1931, he opened a branch in Shanghai with a share capital of one million Hong Kong dollars. See Appendix Section E.3 for additional historical examples.

³⁴Currently, the year is therefore imputed from the bank data.

occurred, we construct two prefecture-year proxies.³⁵

Panel A of Figure 9 assigns elites with a Christian-school background to their native prefecture at age 15, allowing us to ask whether missionary exposure during school age predicts later Christian educational affiliation. Panel B uses biographies that report whether and when elites studied abroad and asks whether missionary arrival increased the likelihood that they left China for foreign study. The event studies show limited differential accumulation before missionary arrival and sustained increases afterward in both outcomes. Subject to the age and location assumptions in Panel A, the results are consistent with missions helping form highly skilled Chinese intermediaries and strengthening access to foreign education and networks.

Figure 9: Missionary Presence and Chinese Elite Formation



Notes: The treatment is the first year of missionary presence in the elite's native prefecture, harmonized to 1911 boundaries. Both panels report cumulative prefecture-year stocks and are estimated separately using Equation 1. Panel A cumulates biographies with a coded Christian-school background, assigning each to the native prefecture at birth year plus 15. This is a proxy for school-age mission exposure, not a direct observation of local school attendance. Panel B cumulates biographies coded as having studied abroad, assigning each to the first four-digit departure year parsed from the study-abroad year field. Post-window increases are 158% in Panel A and 134% in Panel B.

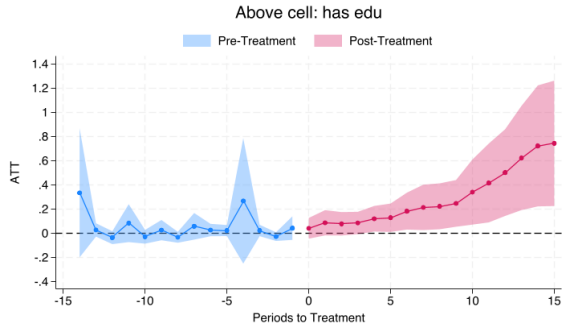
Third, we consider education more broadly by distinguishing missionary exposure that included schools from exposure that did not. With the expanded school data, the effect on modern banks is positive both in prefectures with missionary schools and in prefectures without them. This suggests that education is important, but it is not the only channel. The timing is also informative: the school difference becomes larger later in the event window, which is consistent with the time needed for education to affect the local supply of bankers and managers. For the enterprise outcome, the school channel carries most of the bite, suggesting that practical education mattered especially for the broader diffusion of market activity. Appendix Table E.3 summarizes these magnitudes, and Appendix Figure E.9 reports coefficient-by-coefficient difference tests for the main school and publication splits.

Fourth, using the CHCD institution tradition field, we categorize mission presence as Protestant or Catholic. If the effect operates through practical education, we expect it to be driven mainly by Protestants, who placed stronger emphasis on schooling and conversion through education (Bai and Kung, 2015; Bai, 2019). Figure 11 supports this interpretation. Protestant missionary arrival is associated with a positive effect on modern banks, while Catholic missionary arrival is not. This result is also potentially consistent with Weber's broader Protestant-ethic hypothesis (Weber, 2001; Becker and Woessmann, 2009).

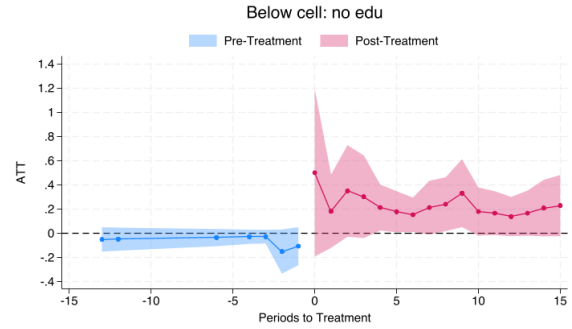
³⁵Appendix Section B describes the construction of the Christian-school indicator and foreign-study departure-year proxy.

Figure 10: Missionary Presence and Modern Banks: Places With and Without Missionary Schools

(a) Modern banks on missions with schools



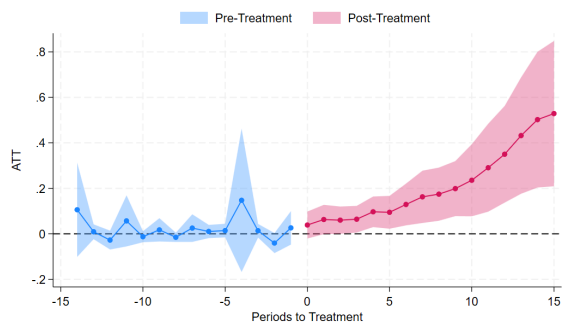
(b) Modern banks on missions without schools



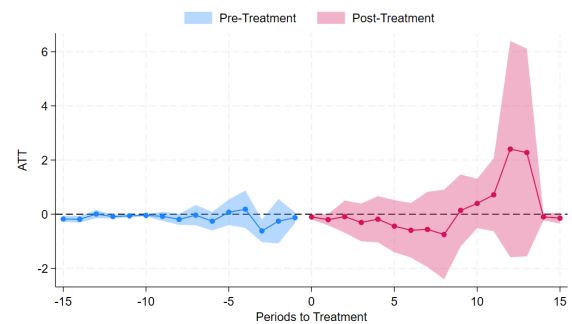
Notes: The figure uses the expanded school definition that includes the additional digitized missionary schools. Panel A includes treated prefectures with at least one missionary school; Panel B includes treated prefectures without a missionary school. Never-treated prefectures are used as controls in both panels. Split-group sizes: 135 treated prefectures with a missionary school and 97 treated prefectures without a missionary school.

Figure 11: Missionary Presence and Modern Banks: Protestant and Catholic Missions

(a) Protestant arrival



(b) Catholic arrival



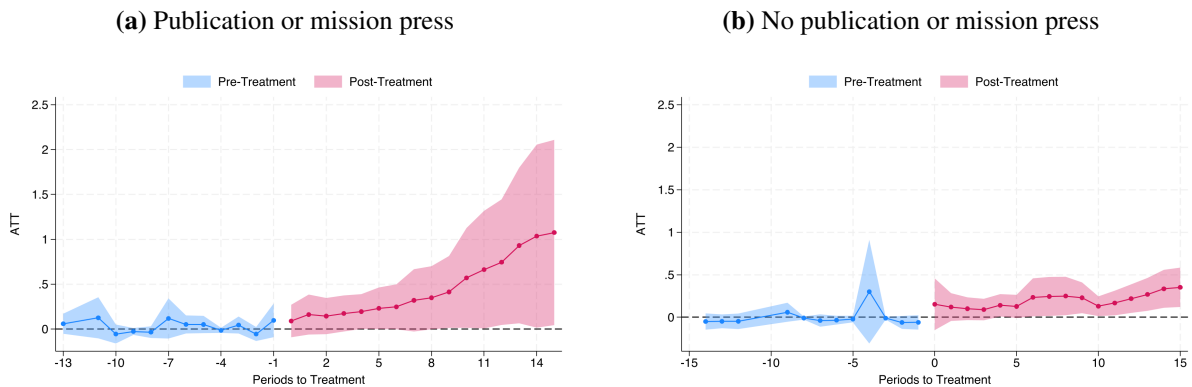
Notes: The treatment in Panel A is defined as the year of arrival of a Protestant missionary institution. The treatment in Panel B is defined as the year of arrival of a Catholic missionary institution. The outcome is the total number of modern banks.

Publications and translations The second mechanism is the diffusion of useful knowledge through publications and translations. Missionaries produced and translated Western texts into Chinese, including religious texts, school materials, newspapers, scientific works, and economic or educational materials. Many of these publications were written in vernacular or romanized Chinese, making them accessible beyond the classical elite.³⁶ This channel complements the education channel: schools trained students directly, while translations and printed materials helped knowledge circulate outside the classroom.

We exploit 677 located records of missionary publications spanning 74 prefectures, together with evidence on missionary printing presses from McIntosh (1895).³⁷ Among the underlying publications, 67.0% were religious or missionary in focus, 42.3% were in vernacular or romanized Chinese, 28.4% included a Western language, 15.1% included English, and 16.5% were educational or student-oriented. A smaller share, 5.8%, were scientific or technical. Appendix Table E.2 reports the full descriptive breakdown, and Appendix Figure E.4 plots the growth of missionary publications over time.

We then test whether locations with missionary publication or printing-press activity experienced stronger effects on modern banks. Figure 12 splits treated prefectures by whether they have any located CHCD publication record, regardless of whether its start year is known, or a McIntosh mission press. The effect is much stronger in places with publication or press activity and flatter elsewhere. This is consistent with publications and translations helping diffuse the knowledge needed to adopt and operate modern financial institutions.

Figure 12: Missionary Publication or Printing-Press Activity and Modern Banks



Notes: The outcome is the total number of modern banks. The left panel reports treated prefectures with any located CHCD publication record, regardless of whether its start year is known, or a McIntosh mission press; the right panel reports the other treated prefectures. Split-group sizes are 62 and 170 treated prefectures, respectively. Adding mission-press locations changes the publication-only split by one treated prefecture.

The education and publication channels also appear complementary. In the split by schools and publications, the largest effect is in prefectures that had both a missionary school and missionary publication activity. Prefectures with neither still show a positive effect, but the effect is smaller. This reinforces the

³⁶This emphasis was especially pronounced among Protestants. Their doctrine placed particular weight on lay believers reading Scripture directly, rather than receiving it primarily through clergy as in Catholicism, which encouraged Bible translation, vernacular printing, and literacy. (Gallego and Woodberry, 2010; Woodberry, 2012).

³⁷The CHCD records cover religious periodicals (including newspapers, magazines, journals, bulletins, newsletters, and school or society papers) while McIntosh locates mission presses. They are not a universal bibliography and may omit smaller or early titles. We therefore interpret the split as the geography of recorded publication or press activity, not a complete title count; see Appendix Figures E.5 and E.6.

interpretation that education and publications amplified missionary exposure, while also indicating that they do not fully exhaust the channels through which missionaries affected modern banks. Appendix Figure E.7 presents the complementarity split, and Appendix Table E.3 gives average post-treatment differences.

Additional channels and alternative explanations The mechanism evidence should not be read as only a human-capital story. The broader institutions literature distinguishes explanations based on skills from explanations based on rules, organizations, and institutional practices (Acemoglu et al., 2001, 2002; Glaeser et al., 2004). In this setting, missionary schools and publications helped move institution-specific knowledge, but other channels could matter.

We examine three additional channels. The first is a possible network channel through compradors. Compradors were Chinese intermediaries who worked with foreign firms and banks and could transmit foreign financial practices into Chinese institutions. Working as a comprador also required knowledge of English, which at the time was taught predominantly in missionary schools. Our banker biography data provide descriptive evidence consistent with this channel. Appendix Table E.4 shows that comprador-background bankers are a small group, but disproportionately connected to Christianity and to Chinese finance: 28% are Christianity-related, 63% later founded at least one Chinese bank, *Qianzhuang*, *Yinhao*, or *Piaohao*, and 72% worked at both foreign and Chinese banks.³⁸ We interpret this as suggestive evidence that missionary exposure may also have operated through networks of intermediaries connecting foreign and Chinese finance, allowing them to learn through direct exposure and helping them establish firms, banks, and other market-oriented institutions.

The second is a competing explanation based on clans. Recent work argues that Confucian clan institutions and financial markets can act as substitutes (Chen et al., 2022b), and missionary activity may weaken kin-based institutions (Schulz et al., 2019; Henrich, 2020). This raises the possibility that our results reflect missionaries weakening clans, rather than directly facilitating modern banks. Appendix Figure E.10 shows that the effect on modern banks is similar across high- and low-clan prefectures. Appendix Table E.5 then reports a horse-race exercise between missionary exposure and clan intensity, using the IV strategy introduced in Section 5. The missionary effect remains positive, while the clan effect becomes small and statistically insignificant.

Third, missions may themselves have demanded banking services to finance schools and hospitals. A scale comparison suggests a bounded role. The estimated effect of 0.22 banks, or 31% of the outcome mean, corresponds to roughly 490,000–877,000 yuan in loans or assets using the average 1932–1936 branch balance sheet. By comparison, the foreign-funded half of the Hwa Mei Hospital construction cost was about 110,000 yuan (13–22% of that scale), annual foreign support for Fenyang Hospital was about 12,000 yuan (1–2.5%), and the annual operating gap of a missionary middle school was 2,500–5,100 gold dollars (less than 1.1%).³⁹ Mission capital needs could therefore have generated some early bank use, but the available benchmarks are too small to explain the full banking effect.

³⁸In our banker biography data, 1,041 bankers are recorded, of whom 43 are flagged as comprador-related.

³⁹These are order-of-magnitude comparisons because the historical sources use imperfectly comparable currencies and mix construction costs, annual operating flows, and bank balance-sheet stocks. Bank scale comes from the 1932–1936 *National Banking Yearbook* balance sheets (Bank of China Economic Research Office, 1937); the project benchmarks come from the Ethel Lacey Hylbert interview on Hwa Mei Hospital, a historical account of Fenyang Hospital, and Ralph E. Diffendorfer’s *Missionary Education in China*.

Summary. Overall, the mechanism evidence suggests that missionaries mattered through more than one channel. Education created practical human capital relevant for banking and broader enterprise formation. Publications and translations diffused useful knowledge about modern economic and financial practices. The clan exercises suggest that the results are not simply driven by missionaries weakening kin-based institutions, while the comprador evidence points to a possible intermediary network connecting foreign and Chinese finance. The capital-needs comparison suggests that missionaries' own demand for financial services could explain some early bank use, but not the full banking effect. Taken together, the evidence is consistent with missionaries transmitting institution-specific knowledge: not just schooling, but also practices, networks, and organizational models that helped modern banks take root.

5 Long-Run Analysis

Did the skills, knowledge, and connections formed under missionary exposure before 1936 continue to matter when China returned to market-oriented development? To test this hypothesis, we examine the relationship between historical missionary exposure and bank development over this later period. Specifically, we aim to estimate the following model:

$$Y_{i,p} = \xi_p + \psi \text{TotMiss Inst}_{i,p} + \nu_{i,p}$$

Where i denotes a prefecture inside a province p . Our outcome of interest is the total number of modern banks in a given recent year, and our treatment of interest is a measure of exposure to missionaries over the whole time period, $\text{TotMiss Inst}_{i,p}$, which is the total number of missions ever built or established between 1850 and 1936.⁴⁰ This cross-sectional regression presents obvious endogeneity issues that we cannot overcome directly by exploiting panel techniques, as we did using the difference-in-difference analysis. Specifically, over the whole time period, the total number of missions established in a given prefecture is endogenous and likely to be correlated with both the error term and the outcome. Our solution is to construct an instrumental variable aimed at predicting the total number of missionary institutions. Following [Sequeira et al. \(2020\)](#), we exploit the interaction between national missionary migration flows and railway connection timing to instrument for missionary exposure. In this section, we show the estimating equations, explain the logic behind the instrument both graphically and descriptively, provide examples, and show the results.

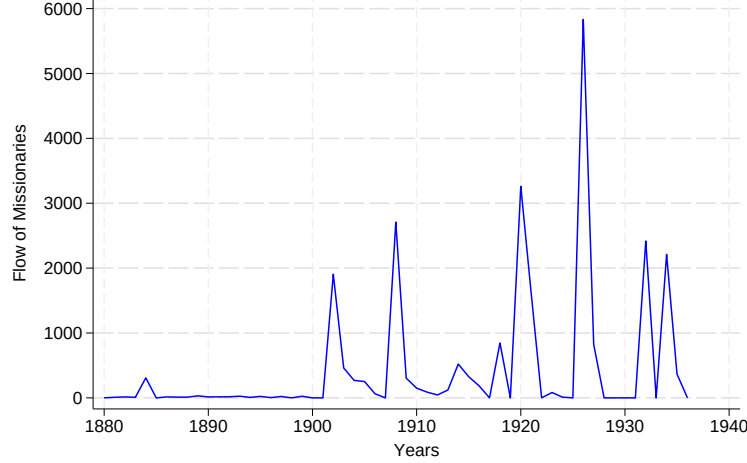
Estimating Equations

This IV strategy is based on two key observations regarding missionary immigration patterns during this period. Upon arriving in China, missionaries' preferred modes of transportation, whenever available, were rivers and the railway system to travel inland to their intended destinations. Consequently, a prefecture's connection to the railway network affected the number of missionaries settling there and the

⁴⁰This approach ensures consistency with the period used in the DiD analysis and allows us to examine outcomes in the 1930s. The results remain unchanged when using the broader timeframe of 1850-1950 instead.

subsequent accumulation of missions.⁴¹ The second observation relates to the significant year-to-year fluctuations in the total number of missionary immigrants to China within our study period, as shown in Figure 13. As a result, prefectures connected to the railway network during periods of high missionary immigration typically experienced an increase in missionary arrivals and, consequently, in the total number of missions established over the entire time period.

Figure 13: Annual Flow of Missionaries in China, 1880–1936



Notes: Flow of Missionaries at time t is defined as the total number of missionaries present in China in $t + 1$ minus the total present in t . Negative changes are set to zero, and the series is winsorized at the 1st and 99th percentiles to match the missionary-flow shifter used in the zero-stage regression. Source: China Historical Christian Database (CHCD).

To illustrate the source of variation in the instrument, we begin with the following auxiliary “zero-stage” equation:

$$\begin{aligned} \text{MissInst Stock}_{i,t} = & \alpha_t + \alpha_i + \delta \text{RRstock}_{i,t-1} \\ & + \beta \text{Missionary Flow}_{t-1} \times \text{RRstock}_{i,t-1} + \varepsilon_{i,t} \end{aligned} \quad (3)$$

Where i indexes prefectures and t years (1850-1936). $\text{MissInst Stock}_{i,t}$ is the total missionary institutions in prefecture i and year t . $\text{Missionary Flow}_{t-1}$ is the flow of missionaries arriving in China between $t - 1$ and t and $\text{RRstock}_{i,t-1}$ is total number of railways in year $t - 1$. α_i is prefecture fixed effects and α_t is year fixed effects.

Central to our identification strategy is the interaction between the aggregate flow of missionaries into China and whether a prefecture was connected to the railway network:

$$\beta \text{Missionary Flow}_{t-1} \times \text{RRstock}_{i,t-1}.$$

This interaction captures the differential effect that railway connection had on immigrant settlement during periods of high aggregate missionary immigration compared to periods of low aggregate immi-

⁴¹This does not conflict with the identification used in the staggered DiD. The two designs exploit different margins of missionary expansion. The DiD uses the timing of first missionary entry and requires that prefectures treated around similar periods were not already on different trajectories; the balance and pre-trend tests support this assumption. The IV instead exploits variation in the intensity of subsequent missionary exposure. Railway access need not strongly determine the exact year of first entry for its interaction with national missionary inflows to generate meaningful differences in the number of missionaries and missions accumulated over time.

gration. Consequently, we expect β to be positive.

The two variables that comprise the interaction term are also included in equation (3), capturing the effect of being connected to a positive number of railway lines when no missionaries are entering China, and missionary flow (absorbed by year fixed effects).

Then, to aggregate the zero stage at the prefecture level, we conduct the following exercise. After estimating equation (3), we calculate the missionary stock in each prefecture and year as predicted by the interaction between the aggregate inflow of missions and connection to the railway network:

$$\widehat{\text{MissInst Stock}}_{i,t} = \hat{\beta} \text{Missionary Flow}_{t-1} \times \text{RRstock}_{i,t-1}$$

where $\hat{\beta}$ is the estimate of β from equation (3). We then sum this prefecture year-specific predicted missionary stock over the whole time period (1850-1936):⁴²

$$\widehat{\text{TotMiss Inst}}_i = \sum_{t=1}^T \widehat{\text{MissInst Stock}}_{i,t}$$

Where T is the total number of periods. Finally, our 2SLS equations are equation (4), the first stage, and equation (5), the second stage:

$$\begin{aligned} \widehat{\text{TotMiss Inst}}_{i,p} = & \zeta_p + \mu \widehat{\text{TotMiss Inst}}_{i,p} \\ & + \omega \text{RR Duration}_{i,p} + \text{RR Density}_{i,p} + \varepsilon_{i,p} \end{aligned} \quad (4)$$

$$\begin{aligned} Y_{i,p} = & \xi_p + \psi \widehat{\text{TotMiss Inst}}_{i,p} \\ & + \pi \text{RR Duration}_{i,p} + \text{RR Density}_{i,p} + \nu_{i,p} \end{aligned} \quad (5)$$

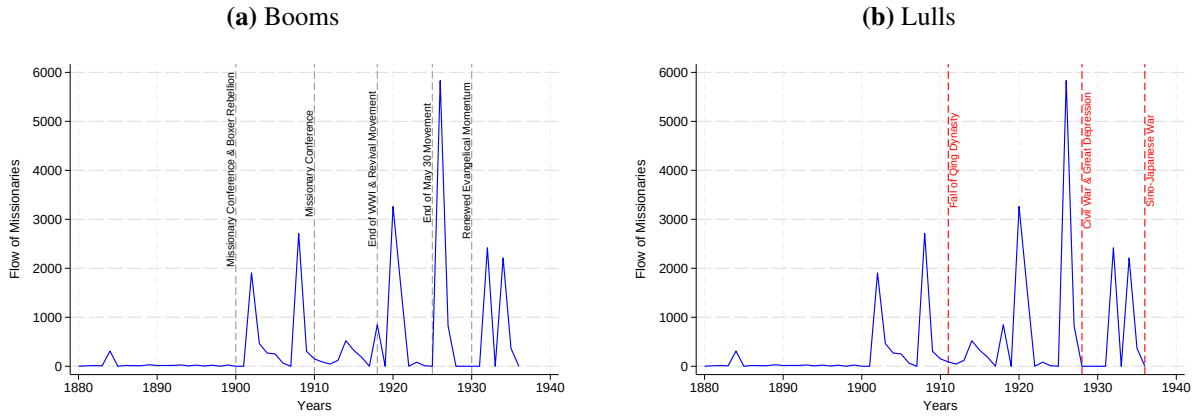
Where i indexes prefectures and p indexes provinces, ζ_p and ξ_p are province fixed effects, and $Y_{i,p}$ is our outcome of interest, measured in a given recent year. We use the total number of banks for each year in separate regressions from 1949 to 2022, and GDP estimated through night lights from 1992 to 2019. $\text{RR Duration}_{i,p}$ denotes the number of years a prefecture has been connected to the railway network as of 2000, while $\text{RR Density}_{i,p}$ is defined as: $\frac{\sum_{t=1}^T \text{RRstock}_{i,t}}{\text{Area in km}^2}$, representing the total number of railway lines per km^2 . This metric serves as the prefecture-level equivalent of the same term included in the zero-stage equation.

We include these controls, already introduced in the zero-stage, because the railway network influenced missionary numbers beyond its interaction with missionary flows. For instance, prefectures connected to the railway early could have a higher number of missions established by the end of the study period. Similarly, a greater number of railway lines could be associated with an increased presence of missions over time.

For this reason, our IV strategy cannot rely on railway stock and length of connection as sources of variation, but instead needs to directly control for these factors. In fact, conditional on the length of connection to the railways and the total number of railway lines per km^2 , we are identified by the fact that prefectures connected prior to high migration periods received more missions than those connected

⁴²Because $\hat{\beta}$ is common to all prefectures, it only rescales the instrument. Omitting it would leave the Kleibergen-Paap first-stage F -statistic, fitted values, and 2SLS estimates unchanged. Estimating this common scalar therefore does not require a generated-regressor correction (Sequeira et al., 2020).

Figure 14: Missionary Booms and Lulls: Annual Missionary Flow, 1880–1936



Notes: The conference markers are the 1900 Ecumenical Conference on Foreign Missions in New York and the 1910 World Missionary Conference in Edinburgh. The May Thirtieth Movement was a 1925 anti-imperialist protest wave in China; renewed evangelical momentum refers to early-1930s revival campaigns. The plotted missionary-flow shifter sets negative annual changes to zero and is winsorized at the 1st and 99th percentiles. Source: China Historical Christian Database (CHCD).

prior to a lull. What matters is therefore the coincidental timing of railway construction in interaction with missionary flows. To be more precise, our source of exogenous variation comes from the zero-stage interaction: number of railways $i,t-1 \times$ national missionary flows *from* $t-1$ *to* t .

The exclusion restriction is satisfied if this interaction affects our outcome of interest, such as the total number of banks in 2000, solely through its impact on the total number of missions. An example of a violation of this assumption would occur if a prefecture decides to build a railway line in period $t-1$ specifically to attract more missionaries in period t . Note that internal movements of people and resources due to the railway do not pose an issue because we are comparing places with the same exposure to the railway over the entire time period, with the only variable being the exact timing of railway connections. For this reason, this IV can only be used when it is possible to aggregate a long historical time period to study more recent outcomes. Potential issues could arise from economic or social conditions within China that affect both the exact timing of railway construction in a given prefecture and the flow of missionaries in a given year. For example, a war might delay railway expansion and missionary flows. However, what matters is that the war or event specifically shifts railway construction from $t-1$ to t , or from t to $t-1$, rather than causing general delays. In the next sections, using a combination of balance and descriptive statistics, we argue that this is unlikely.

Sources of Variation for Identification

The first source of variation used for identification is the yearly missionary migration flows. While multiple factors influence these flows, we can identify some key push and pull factors that may contribute to the spikes and lulls observed in Figure 13. Figure 14 illustrates events that may correlate with missionary migration. We note that some of these spikes coincide with factors related to the Christian church's priorities, such as the conclusion of missionary conferences aimed at boosting missionary presence overseas or major revival movements.

In fact, during this period, missionary migration was primarily driven by religious revivalist movements and economic conditions in the missionaries' home countries, rather than by the economic situa-

tion in China. In the nineteenth century, Western countries, particularly in Europe and the United States, experienced several religious revivalist movements, such as the Second Great Awakening, which fueled a strong desire for overseas missionary work (Shelley, 2004). Around 1830, there was a prevailing belief in British society that, as a Christian nation, it had a responsibility to spread Christianity to overseas territories (Shelley, 2004). Similarly, between 1880 and 1920, American society viewed overseas missions as a means to demonstrate American power, leading to an unprecedented wave of student volunteers for missionary work abroad (Wang, 1997). The end of World War I saw a significant missionary revival movement, especially in countries with sufficient resources to send missionaries abroad. A similar pattern followed the end of the Great Repression.

Additionally, missionary migration was constrained by the economic power of their home countries. A stable and continuous source of funding was essential for missionaries to sustain their work abroad. For instance, prior to World War I, British missions were the predominant force in China. However, during the war and the subsequent decline of British influence, the strength of British missions diminished. By 1930, the American Missionary Society's assets in China had grown to \$41.9 million, four times those of the British, reflecting the shift in missionary power due to war efforts (Dan, 1996).

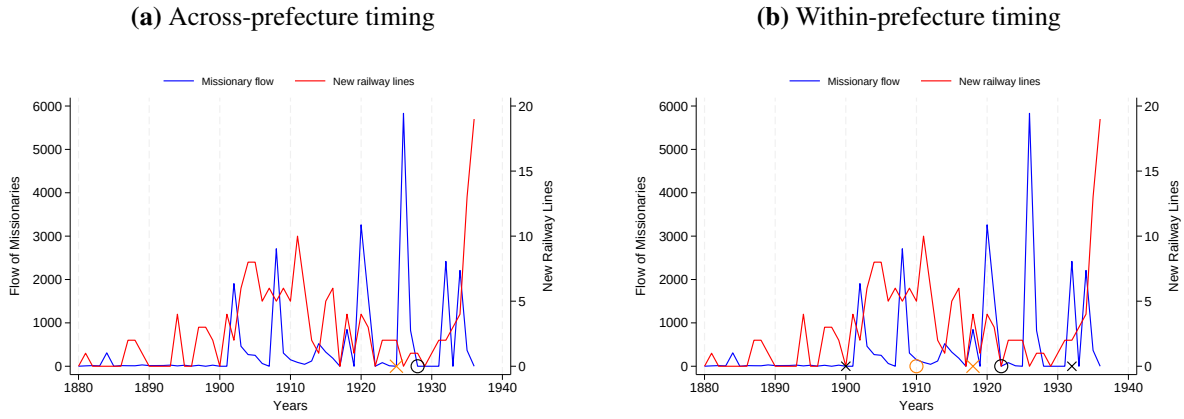
Other foreign immigrants in China followed a different trajectory than the missionaries. During this period, foreigners in China also included merchants, foreign government officials, and explorers. Due to security concerns, treaty restrictions, and economic incentives, merchants and government officials primarily operated within treaty port areas, while the only foreigners who ventured into the inland were a small number of explorers.

Nevertheless, it is important to note that other factors may have influenced these flows. For example, another pattern that emerges is the presence of spikes following the end of conflicts against Christians, such as the Boxer Rebellion or the May 30th Movement. Additionally, panel b shows that some of the lowest missionary migration periods coincide with the onset of political and social instability in China. These factors may threaten identification if correlated with the exact timing (year by year) of railway line completion.

The second source of variation is the number of new railway lines built annually in China. Figure 15 overlays this data with missionary flows to graphically illustrate the interaction used in the IV. Panel A illustrates the type of comparisons our IV makes, where an *X* represents a prefecture that gets connected to a new railway line the year before a missionary migration boom, while a circle represents a prefecture connected the year after. Implicitly, our IV relies on these comparisons, within provinces. As an example, consider the 1934 boom, which involved 2,210 new missionaries. In our data, Jiangning Prefecture in Jiangsu Province, connected just before the boom, received 57 new missionaries (2.58% of the total) and established 5 new missions (a 12% increase from a base of 42 missions). In contrast, Haizhou Prefecture in the same province, connected just after the boom, received 1 new missionary (0.05% of the total) and established 0 new missions (a 0% increase from a base of 2 missions).

Since only 30% of Chinese prefectures were connected to the railway by 1950, many comparisons also resemble those depicted in Panel B. In Panel B, the same prefecture, represented by the color black, is first connected prior to a boom, then receives a new railway line prior to a lull, and another one before a subsequent boom. The same logic applies to the “orange” prefecture. Essentially, these comparisons are made for each year and each prefecture that gets a new railway line, and then they are aggregated at

Figure 15: New Railway Lines and Missionary Flow, 1880–1936



Notes: The legend identifies the blue national missionary-flow series and the red number of new railway lines. A new railway line is either an added stop on an existing line or a newly constructed line. Black markers represent one prefecture and orange markers another prefecture in the same province; X's denote boom-year connections and circles denote lull-year connections. The missionary-flow series is the regression-consistent shifter used in the zero stage. Sources: China Historical Christian Database (CHCD) and [Ma et al. \(1983\)](#).

the prefecture level.

Threats to Identification

A first threat to our identification strategy is the correlation between missionary flows and both early railway connections and railway density. To account for this, in the construction of the instrument, we control for years of connection (as of 2000) and the number of railway lines per square kilometer. For example, Jiangning Prefecture has been connected to the railway network for 92 years with 7 railway lines, whereas Haizhou Prefecture has been connected for 95 years with 6 railway lines. The difference between these two prefectures, 3 years of connection and 1 railway line, will be accounted for in the first stage regression allowing the comparison to rely solely on the exact timing of railway expansion and missionary flows.

Secondly, the two interaction terms (connection date and railway density) could be correlated with each other. For example, in Jiangning Prefecture, the coincidental timing of railway expansion was driven by a deliberate desire to attract more missionaries. Moreover, political unrest in China might affect both the timing of railway construction and missionary flows. To address this issue, we can implement several strategies. First, we examine the balance between “boom Prefectures” and “lull Prefectures”. We define boom Prefectures as those connected to a railway line one or two years prior to when missionary flows are above the median within a given decade, and lull Prefectures as those connected in a below-median flow year or one year afterward. Table 3 presents the results of a pairwise t-test difference test using OLS, with clustering at the prefecture level (since the same prefecture may appear multiple times). All variables are measured at the beginning of the period, in 1880. We note that there are no significant differences between boom and lull places.

Another important consideration is our reliance on the exact year of completion for each railway line. Even if there are factors that could bias our estimates, these factors would need to simultaneously affect both railway construction and missionary flows with precise timing. Given the likely unforeseen delays in railway construction - our data indicate an average completion time of three years with a

Table 3: Balance Between Boom and Lull Prefectures

Variable	(1) Lull Prefecture		(2) Boom Prefecture		(1)-(2) Pairwise t-test	
	N/Clusters	Mean/(SE)	N/Clusters	Mean/(SE)	N/Clusters	Mean difference
Total Modern Banks	129	0.132	109	0.183	238	-0.052
	54	(0.072)	47	(0.104)	60	
Total Enterprises	129	0.419	109	0.606	238	-0.187
	54	(0.199)	47	(0.330)	60	
Ln (investment gov. factory)	129	2.292	109	2.377	238	-0.084
	54	(0.802)	47	(0.928)	60	
Total railways	129	0.000	109	0.000	238	0.000
	54	(0.000)	47	(0.000)	60	
Total Treaty Ports	129	0.140	109	0.183	238	-0.044*
	54	(0.064)	47	(0.076)	60	
Whether there's main river	129	0.682	109	0.651	238	0.031
	54	(0.074)	47	(0.081)	60	
Provincial capital	129	0.442	108	0.435	237	0.007
	54	(0.089)	46	(0.091)	59	
Crop suitability: Rice	129	2.097	109	2.088	238	0.009
	54	(0.188)	47	(0.185)	60	
Ln population	129	14.517	109	14.518	238	-0.001
	54	(0.133)	47	(0.134)	60	
Coastal	129	0.310	109	0.349	238	-0.039
	54	(0.088)	47	(0.092)	60	

Notes: Boom prefectures are prefectures connected to a railway line one or two years before national missionary flows are above the median within a given decade. Lull prefectures are prefectures connected in a below-median flow year within a given decade or one year afterward. Variables are measured at the beginning of the period, in 1880, except time-invariant geographic variables. The table reports means for each group and the pairwise difference from an OLS regression with standard errors clustered at the prefecture level, since a prefecture can appear more than once. N/Clusters reports the number of prefecture-year observations and distinct prefecture clusters. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

standard deviation of six years - such precise alignment is improbable. For example, we would have to assume that the Chinese Civil War of 1927, which likely correlates with a lull in migration between 1928 and 1931, specifically delayed the completion of a railway in a prefecture from 1926 to 1927 (or vice versa).

Finally, the timing of railway connections may be correlated with a prefecture's geographic location. To address this, we include prefecture fixed effects in our zero-stage regression and control for latitude and longitude as a robustness check in the first stage.

Results

We begin by presenting the zero-stage estimating equation (3) and the first-stage equation (4) in Table 4.⁴³

The interaction between national missionary flows and lagged railway stock is positive and statistically significant in the zero stage. Thus, a given national missionary inflow led to more local missionary

⁴³All results are presented using winsorized variables at the 99th percentile to address extreme outliers. The findings remain consistent when using the 99.5th percentile, not winsorizing, or excluding extreme outliers altogether. For detailed tables and figures, refer to the robustness section.

Table 4: Auxiliary Zero-Stage and First-Stage Estimates

Variables	Zero-stage	First stage
	(1) MissInst Stock	(2) Tot Miss Inst
RRstock (t-1)	1.5672*** (0.3465)	
Miss Flow x RRstock / 1,000	0.03738** (0.01456)	
Predicted Tot Miss Inst		3.3997*** (0.8816)
RR Duration		0.0690** (0.0279)
RR Density		-685.2861*** (128.9182)
Prefecture FE	Yes	No
Time FE	Yes	No
Province FE	No	Yes
Obs.	21,242	247
Adj. R sq.	0.722	0.240
Mean Dep. Var.	3.10	10.28
S.D. Dep. Var.	6.06	9.06
K-P first-stage F	—	14.87

Notes: MissInst Stock $_{i,t}$ is the total number of missionary institutions in prefecture i and year t . Missionary Flow $_{t-1}$ is the flow of missionaries arriving in China between $t - 1$ and t , and RRstock $_{i,t-1}$ is the total railway stock in year $t - 1$. Their interaction is divided by 1,000 for readability. Predicted Tot Miss Inst is the prefecture-level sum over time of predicted missionary institutions from the zero-stage regression in Equation 3. RR Duration $_{i,p}$ is the number of years connected to the railway network as of 2000, and Density $_{i,p}$ is $\left(\sum_{t=1}^T \text{RRstock}_{i,t}\right) / (\text{area in km}^2)$. The missionary stock, flow shifter, and railway variables use the 1st–99th percentile convention. Fixed effects are reported in the table. Robust standard errors are in parentheses. K–P F is the Kleibergen–Paap first-stage F-statistic; the zero-stage interaction statistic is not a weak-instrument diagnostic. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

institutions in prefectures with greater railway access. Column (2) reports the first stage: predicted cumulative missionary exposure predicts actual missionary institutions, with a Kleibergen–Paap F -statistic of 14.87.

We next estimate the second-stage regressions using the final six years of the historical modern-bank dataset and annual data on total banks from 1949 to 2022. Because the level of banking activity changes substantially across this period (see Appendix Figure F.1), we standardize the bank outcome within each year to facilitate comparisons of the coefficient path.⁴⁴ Figure 16 reports the resulting year-specific estimates, and Appendix Table F.2 reports selected years in regression-table form.

The historical estimates from 1930 to 1936 are positive and statistically significant at the 10% level. In raw-bank units, they imply increases of approximately 51% in 1930 and 34% in 1936. The 1936 magnitude is close to the 31% estimate from the historical difference-in-differences design. The IV therefore reproduces the main historical relationship despite relying on a different source of variation.

The missionary-exposure gradient re-emerges after the market-oriented reforms. The coefficient becomes positive during the 1980s and remains positive through the end of the sample.⁴⁵ The result is

⁴⁴Results using the total number of banks, rather than the standardized outcome, are reported in Appendix Figures F.3 and F.4.

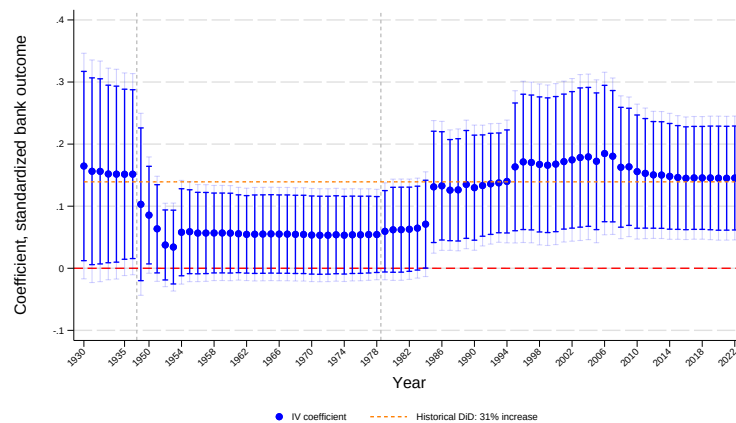
⁴⁵The 1990 estimate is significant at the 5% level and the 2022 estimate at the 1% level. In raw units, they imply increases in the total number of banks of approximately 14.2% and 14.7%, respectively.

also robust to adding alternative population measures as controls and to expressing bank outcomes per capita (Appendix Table F.3 and Appendix Figure F.5).⁴⁶

By contrast, the missionary-exposure gradient is muted between 1949 and 1978. This does not mean that banking offices disappeared. Private and foreign banks were closed, merged, or brought under state control, while the state expanded a nationwide network of local banking offices. These offices, however, operated as part of the planned economy rather than as market-oriented financial intermediaries.⁴⁷ An increase in recorded branches during this period therefore does not represent the same type of financial development as the growth of modern banks before 1949 or after reform.

The disappearance of the mission effect under Mao does not appear to arise simply because the state built fewer banks in historically mission-exposed areas. Rather, bank expansion was proportionally faster in the least mission-exposed prefectures, suggesting that state-directed expansion into previously underbanked areas may have compressed the inherited spatial distribution of banking. In the same 247-prefecture sample used for the baseline IV, raw recorded banks rose from 1,610 in 1936 to 4,164 in 1960. They increased by 213.2% in prefectures at or below the empirical 25th percentile of missionary

Figure 16: 2SLS Estimates: Standardized Modern and Total Banks, 1930–1936 and 1949–2022



Notes: The figure reports coefficients from Equation 5 estimated separately by year, using the historical modern-bank data used in the DiD design and recent county- and district-level bank counts assigned to 1911 prefectures by each modern unit's centroid. The annual outcome and IV-side variables use the 1st–99th percentile convention. Blue points are IV estimates; dark and light intervals show 90% and 95% confidence intervals. The red dashed horizontal line marks zero, the orange dashed horizontal line maps the historical DiD estimate of a 31% proportional increase into 1936 standardized-outcome units (0.139 standard deviations), and the vertical dashed lines mark the establishment of the People's Republic in 1949 and the beginning of market reforms in 1978. Historical modern-bank means are 2.4 in 1930 and 5.8 in 1936, so the raw-unit IV estimates imply increases of approximately 51% and 34%, respectively. Recent total-bank means are 145 in 1990 and 707 in 2022, so the corresponding increases are approximately 14.2% and 14.7%. Appendix Table F.2 reports the 1936 and 2020 estimates in regression-table form. Source: China Financial Supervision and Administration Bureau.

⁴⁶We do not use these population-adjusted specifications as our baseline because missionary exposure may itself affect population growth, migration, and urban concentration, making later population a potential outcome of treatment. Moreover, no single population measure is an obviously preferable denominator for the per-capita outcomes, and reliable prefecture-level population estimates are particularly scarce for the Mao period.

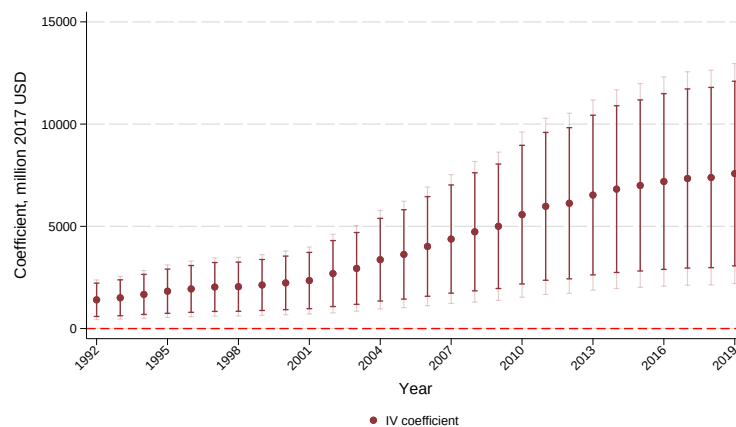
⁴⁷After 1949, most banking functions were consolidated under the People's Bank of China, which combined central-bank, commercial-bank, treasury, and government-accounting functions. Its local offices, together with state-guided rural credit cooperatives, handled deposits, cash distribution, payments, state-enterprise and commune accounts, and credit allocated according to government plans.

exposure, compared with 153.0% in the remaining prefectures. The network also became much less concentrated in the pre-Mao banking centers: prefectures in the highest quartile of the 1930s bank distribution accounted for 86.2% of recorded banks in 1936 but only 30.3% in 1960. The IV estimates likewise show no evidence that mission-exposed prefectures received fewer banks in absolute terms, but suggest slower proportional growth in those places. Appendix Figure F.2 and Appendix Table F.1 report these results.

This distinction also suggests that physical banking infrastructure was not sufficient to generate market-oriented financial development. Mao-era offices provided payments and administrative capacity, and their expansion may have had value for the planned economy. But the earlier financial geography did not reappear merely because banking offices existed. It returned only after banks again operated in an environment that permitted market exchange, commercial lending, private enterprise, and international connection. Appendix Section F.14 examines whether the gradient changes around major reforms. The clearest and most stable breaks occur around the 1984 completion of decollectivization and reconstruction of the two-tier banking system and around China's WTO accession in 2001; estimates around other dates are smaller or more sensitive to the chosen window.

The long-run relationship also appears in aggregate economic activity. Figure 17 reports estimates using gridded revised real GDP from 1992 to 2019. The annual coefficients are positive and statistically significant throughout the sample period. Although the estimates rise in levels as average output expands, their magnitude relative to mean prefecture GDP remains stable: the average estimate is approximately 16.1% of the annual mean, and the 2019 estimate is 16.5%.⁴⁸

Figure 17: 2SLS Estimates: Real GDP, 1992–2019 (Millions of 2017 USD)



Notes: The figure reports coefficients from Equation 5 estimated separately by year. GDP is constructed by summing 1-km grid cells directly within accepted 1911 prefecture polygons. The mission stock, instrument, railway controls, and annual GDP outcomes use the 1st–99th percentile convention. Maroon points and dark/light intervals show the IV estimates and 90%/95% confidence intervals. All 28 annual estimates are positive and statistically significant at the 1% level. The average estimate is approximately 16.1% of annual mean prefecture GDP, and the 2019 estimate is 16.5% of that year's mean. Appendix Table F.2 reports the 2019 estimate in regression-table form. Source: Chen et al. gridded revised real GDP and electricity consumption, 1992–2019, based on night lights.

⁴⁸Appendix Table F.4 reports corresponding 2019 IV estimates with two alternative population controls and with two standardized per-capita GDP outcomes. Appendix Figure F.6 reports the annual per-capita path using the fixed 2015 Census denominator.

The enterprise evidence points in the same direction. Historical missionary exposure predicts more industrial enterprises after reform, with similar 2007–2013 standardized estimates for private and all enterprises. The increase in private enterprises is larger for those with high external-finance dependence. Appendix Figures F.7 and F.8 report these results, and Appendix Section F.3 explains the enterprise mapping and external-finance-dependence classification.

Finally, we conduct a finer within-province validation in Shaanxi. This exercise is possible because we separately transcribed the provincial statistical compilation and harmonized changing county, district, and county-level-city names to 1911 county boundaries. The resulting data allow us to reproduce the long-run bank exercise at the county level, absorb historical-prefecture fixed effects, and reconstruct the railway-based instrument within a single province.

The exercise is necessarily suggestive because it covers only one province. It is nevertheless useful because it relies on substantially finer geographic variation than the national prefecture analysis. Early post-1949 financial-institution coverage is too sparse to estimate the Mao-period IV after including historical-prefecture fixed effects and railway controls. The OLS coefficient path, however, shows the same attenuation during the Mao period and recovery after reform. After 1978, the association between historical missionary exposure and standardized financial institutions is positive in both OLS and IV, and the county-level first stage remains strong. The GNP results are qualitatively consistent: the missionary-exposure gradient in log GNP is visually flat in the Mao-era benchmark years and rises sharply after opening begins in 1978.⁴⁹ Appendix Figures F.9 and F.10 report the coefficient paths, and Appendix Table F.5 summarizes the pooled estimates.

Robustness Checks

We perform several robustness checks on our main result. First, we use a recentering exercise following [Borusyak and Hull \(2023\)](#) to check that the IV is not simply capturing average exposure to missionary inflows through the railway network. The IV gives more predicted missions to prefectures whose railways were present in years with larger national missionary inflows. A related concern is that prefectures with longer or denser railway histories would have received more exposure to any average flow process, independent of the realized timing of missionary inflows. We therefore simulate counterfactual histories of national missionary flows, holding each prefecture’s railway path fixed, and subtract the exposure each prefecture would receive under these simulated flow histories.⁵⁰ This removes the average-flow-times-railway component and keeps the part coming from the realized alignment between missionary-flow spikes and railway connections. Appendix Section F.13 shows that the long-run estimates are very similar: for standardized banks in 2020, the coefficient falls from 0.145 to 0.127 under the full-flow permutation and remains significant at the 1% level. For GDP in 2019, recentering reduces the coefficient from 7,581 to 6,548 under the full-year permutation and 6,616 under the 10-year block permutation; both estimates remain significant at the 1% level. The first stage is also not weakened: the K–P F-statistic rises from 14.87 to about 16.7 for banks and from 15.01 to about 16.8–16.9 for GDP.

Second, since the exact functional form of the railway numbers in the first stage - which is cross-sectional rather than panel - is unknown, we modify how we control for railways in the first stage.

⁴⁹We use county GNP reported in the Shaanxi Provincial Bureau of Statistics’ 1949–1990

⁵⁰The national flow distribution is fixed; only the timing of booms and lulls changes.

We believe that our main specification, which includes railway duration and railway density, already precisely captures the railway duration introduced in the zero-stage. However, the results remain robust when adding non-linear railway controls (such as railway bins - Appendix Figures F.25 and F.26), mileage of railway lines per km^2 (Appendix Figure F.27), cubic polynomials of latitude and longitude (Appendix Figure F.28), the number of railways in 1936 (instead of the sum over time - Appendix Figure F.29), and replacing density with bins (Appendix Figure F.30).

Third, we alter the variation introduced in the zero-stage or exclude certain observations from the sample. For example, we exclude all treaty ports (Appendix Figure F.31), drop the top three railway outliers (instead of winsorizing - Appendix Figure F.32), use different or no winsorization (Appendix Figures F.34 and F.35), and use all the years in the zero-stage instead of the ones used in the DiD (1850-1950 instead of 1850-1936 - Appendix Figure F.33). We also use fixed and early historical prefecture characteristics as placebo outcomes in the IV second stage, confirming that the instrument is not predicting them systematically (Appendix Figure F.16).

Fifth, we identify which missionary-flow years contribute most to the variation in the exact instrument and then re-estimate the headline specifications after removing each positive-flow year in turn; although the instrument's variation is concentrated in the 1920s and early 1930s, no single year materially changes the estimates or first-stage strength (Appendix Figure F.17 and Appendix Table F.14).

6 Long-Run Mechanisms: Seeds and Reactivation

The previous sections showed how missionary exposure, particularly through schooling and translations, contributed to financial innovation and market development before 1949, and how these effects disappeared under Mao and reemerged after reform. We argued that missionaries planted seeds that survived the Mao era, and that the policy environment determined whether these market-related inputs could be put to use. But what exactly were these seeds, and what mechanisms could explain their persistence and reactivation? This is inherently difficult to establish. In this section, we provide suggestive evidence on two potential channels and briefly assess several alternative explanations.

High-skilled people and networks Mission exposure helped form practical human capital before 1949. Chinese people carried this human capital within and outside mainland China, and some of it was internationally oriented. Under Mao, the returns to much of this expertise fell sharply. The disruption of higher education during the Cultural Revolution further limited the supply of newly trained human capital: regular university admissions were suspended, higher education operated only in restricted form for much of the period, and the national entrance examination was restored only in 1977. When reform increased demand for financial, managerial, technical, and internationally oriented skills, older surviving experts and Chinese who had trained or worked abroad therefore represented potentially valuable sources of knowledge and experience. We examine three ways in which this human capital could have survived the Mao period: through a high-skilled diaspora outside mainland China, through foreign-exposed elites who later reconnected with mainland institutions, and through individuals who remained in China.

The first channel is a high-skilled diaspora. Using data on engineers and scientists (see Section

3), we identify scientists and engineers observed in Taiwan in 1974 and trace their places of origin in mainland China. Mission-exposed places are disproportionately represented in this diaspora: the linked sources identify 4,640 mainland-origin individuals observed in Taiwan in or after 1974. Appendix Table F.6 shows that origin in a prefecture with a mission by 1880 predicts a 2.45 percentage-point increase in near-reform Taiwan presence, relative to a mean of 3.0 percent; origin in a prefecture with a mission school by 1900 predicts a 5.36 percentage-point increase. Mission-exposed places thus produced a disproportionately large high-skilled diaspora.⁵¹

Second, some of this internationally oriented human capital could reconnect with mainland China after reopening. Using the same data on Republican-era elites showing that mission exposure predicted a greater likelihood of studying abroad, we examine whether these internationally exposed elites subsequently reappeared in mainland institutions. While foreign-exposed elites were generally less likely to hold mainland institutional roles, conditional on foreign exposure, those from mission-exposed places were more likely to do so: the interaction rises from 14.77 percentage points during the Mao period to 54.64 points after reform (Appendix Table F.7).⁵² This pattern is consistent with international connections fostered in mission-exposed places becoming more relevant once China reopened.⁵³

Shiing-Shen Chern provides one example of this path. He left mainland China in 1948 and built his career in the United States, but returned after China reopened, helped establish Nankai University's mathematics institute, and eventually returned permanently.⁵⁴

But survival did not necessarily require leaving China. Some of the people carrying pre-1949 expertise remained in the country through the Mao period and became economically important again after reform. Rong Yiren is a prominent example. Educated at St. John's University, a missionary-founded institution, Rong managed his family's textile enterprises before 1949. He remained in China through the Mao period, when the scope for his pre-1949 commercial and financial expertise was sharply restricted, and in 1979 became the founding chairman of the China International Trust and Investment Corporation (CITIC), a state investment company. He became known as the "Red Capitalist" and served as a key intermediary between the reform government and foreign capital.⁵⁵

Our banker biographies provide more systematic evidence of this type of survival. We find that 13.8% of historical bankers were still active during the Mao period (29% of whom were Christian-affiliated) and 6.1% were still active after 1978.⁵⁶ These are conservative measures because they capture only the original bankers, not the knowledge or networks transmitted to later generations. Still, they show directly that some people carrying pre-1949 financial expertise survived into the periods in which the historical effects disappeared and reemerged.

⁵¹This pattern may partly reflect the general targeting of elites for repression and links between Christianity, missionary education, and the Nationalist government. The Nationalist elite included prominent Christians: Chiang Kai-shek converted to Christianity, and his wife, Soong Mei-ling, came from a prominent Christian family. See the [Office of the Historian, U.S. Department of State](#).

⁵²The biographies follow the same Republican-era individuals, not their descendants, and each period is restricted to elites known to be alive. Because reform-period survivors were necessarily old, the estimate captures late-life institutional roles rather than transmission to later generations.

⁵³Related evidence from [Chen et al. \(2026\)](#) shows that, after local opening, diaspora firms disproportionately entered prefectures tied to their representatives' likely ancestral origins.

⁵⁴On Chern's work at Nankai and permanent return, see [Nankai University](#).

⁵⁵See [McNeill's \(2005\) obituary in *The Independent*](#).

⁵⁶These percentages are proxies based on keywords in the biographies, with an LLM used to identify activity in each period and random batches manually validated.

Foreign-facing commercial connections The evidence on Chinese human capital is only part of the story. Mission-exposed prefectures also developed more trade-oriented activity before 1949. Closing the economy meant that many of these links between China and the rest of the world could no longer be put to use. Reopening could make them valuable again.⁵⁷ More broadly, this can be interpreted as a form of market memory. Places that had developed trade routes, commercial connections, and financial systems before 1949 may have retained some advantage when the policy environment again allowed markets to operate in similar ways. One implication is that foreign capital should flow disproportionately toward historically exposed locations once international exchange resumes. This interpretation is related to work by [Huang et al. \(2024\)](#), who show that anti-missionary conflict reduced investment in modern firms and banks and left a persistent penalty in contemporary FDI. Their estimates identify the negative effect of anti-missionary conflict, rather than a positive effect of missionary presence on FDI.

The FDI evidence is consistent with this prediction. Historical missionary exposure predicts higher foreign direct investment in 2017 in both OLS and IV specifications (Appendix Table F.8). The IV estimates imply approximately 14 percent more total FDI and 6 percent more FDI per capita for each additional historical mission. Thus, the reactivation visible in banking is accompanied by renewed foreign capital flows toward historically mission-exposed places.

Alternative explanations We find less support for three alternative explanations. First, the results do not appear to reflect the direct persistence of missionary organizations, which could no longer operate after 1949 in their historical form. Their human-capital output, however, may have persisted or been rebuilt. In the 2015 Census, OLS estimates are positive and precise across the education ladder, while the standardized IV estimate is near zero for middle school and positive but imprecise from senior high through postgraduate education (Appendix Figure F.13). Within Shaanxi, the mission-exposure gradient rises after reform for secondary but not overall enrollment (Appendix Figures F.14 and F.15). These patterns provide little support for broad schooling persistence, but are consistent with persistence or rebuilding of upper-tail human capital.

Second, we find limited evidence for broad religious or cultural persistence, although these channels are harder to study. Historical missionary exposure predicts somewhat higher Protestant and Christian affiliation, generalized trust, and greater ownership of savings and financial assets, but measures of religious participation, prayer, donations, and broader values do not move consistently (Appendix Tables F.9–F.12). The generalized-trust result is consistent with related work on missionary exposure and the scope of morality ([Bergeron, 2019](#)), and with evidence of higher trust and trustworthiness in places with greater historical Protestant activity in China ([Deng et al., 2025](#)). These patterns do not establish a persistent capitalist culture or broad preference changes, but leave room for narrower changes in beliefs or trust.

Third, the results do not point to greater Mao-era state capacity or public investment in historically exposed places. In the Shaanxi county data, IV estimates for fiscal expenditure, fiscal revenue, fixed-asset investment, electricity, machinery, and health infrastructure are generally null or negative, while

⁵⁷ A 1989 account of Shanghai's aroma industry offers one illustration. It describes pre-1949 links with Western fragrance houses, the cultivation of aromatic inputs including geranium, vetiver, patchouli, and pine-needle oils, and renewed sales and technical ties with international firms after 1978. The author characterized Shanghai as inheriting an earlier foreign-trade tradition in the industry ([Wu, 1989](#)).

positive estimates are concentrated in industrial enterprises, retail sales, and township enterprise activity (Appendix Table F.13). This pattern is more consistent with a market-facing advantage than with greater state infrastructure.

7 Conclusion

This paper studies the historical and institutional roots of China's market-based capitalism through the lens of financial development. Using newly compiled data on Christian missionary institutions, modern banks, enterprises, traditional financial institutions, and long-run banking outcomes, we show that missionary exposure helped China adopt modern banks before 1936 and shaped the response of local economies to the 1980s market reforms.

Our historical analysis exploits the staggered expansion of missionaries across Chinese prefectures between 1857 and 1936. We find that missionary presence increased modern banks by 31% and enterprises by 27%. These effects are not driven by pre-existing trends in banks, enterprises, railways, treaty ports, government investment, or traditional banks. The results are also robust to alternative transformations of the bank outcome, different bank types, balance-sheet measures, and a range of sample restrictions.

The evidence points to two main mechanisms. First, missionaries, especially Protestants, supplied practical education in subjects such as mathematics, bookkeeping, economics, science, and English. These skills were useful for operating modern financial institutions. Consistent with this channel, we find a high share of Christian-affiliated bankers, stronger effects in places with missionary schools, and larger effects for Protestant missions. Second, missionaries helped diffuse useful knowledge through publications and translations. The effects on modern banks are stronger in locations where missions engaged in publication activity, suggesting that printed materials and translations helped make modern economic and institutional practices more accessible.

We also document how missionary exposure interacted with China's transition from traditional to modern finance. Modern banks expanded especially in places with pre-existing traditional financial activity, suggesting that they emerged where financial demand was already present. Over time, they substituted for *Piaohao*, whose remittance functions became less central after the fall of the Qing state. By contrast, *Qianzhuang* survived longer and appear more complementary to modern banks, likely because they retained local information and networks of trust that modern banks initially lacked.

The long-run analysis shows that these historical effects did not disappear permanently. Using an instrumental variable based on the interaction between railway expansion and national missionary inflows, we find that missionary exposure predicts modern banking development before 1949, weakens during the Maoist period, and re-emerges after the market reforms of the 1980s. We also find effects on per capita GDP after 1992, with a stronger increase after China joined the WTO. Suggestive mechanism evidence links the Mao-era dissipation to high-skilled diaspora from exposed origins, trade closure, and the redistribution of bank offices; it links post-reform reactivation to surviving or rebuilt Chinese human capital, the renewed mainland presence of foreign-exposed elites, and foreign investment flowing toward exposed places. We find less support for direct persistence through mission schools or publications, broad religiosity, or state infrastructure. National reopening was necessary for these mobile inputs to

earn local returns, while bank infrastructure alone was insufficient.

Overall, the results suggest that missionaries transmitted more than human capital. They also helped diffuse institution-specific knowledge: the skills, practices, networks, and organizational forms that made modern banks usable in the Chinese context. In this sense, missionaries helped plant the seeds of market-based capitalism, not by replacing Chinese institutions, but by interacting with local conditions and helping new financial institutions take root. This helps explain why some places were better positioned to respond to China's reform-era opening and why the roots of China's modern market economy partly predate the reforms.

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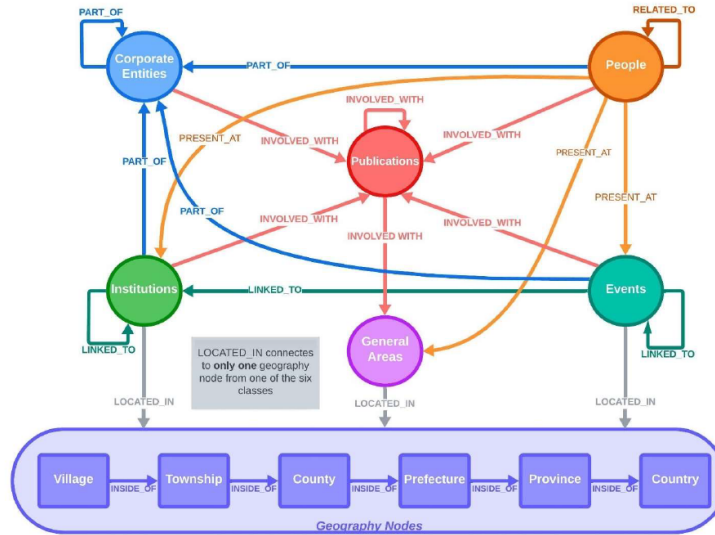
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A Diffusion of Missionary Institutions, Banks, and Enterprises

A.1 Missionary Data Structure and Institution Types

Figure A.1: Data Structure of CHCD



Notes: The CHCD includes nodes for people, corporate entities, institutions, publications, general areas, and events, along with six types of geographic nodes. This schema illustrates possible relationships among these nodes. Person nodes represent human beings. Corporate-entity nodes represent organizations that do not have a direct geographic footprint. Institution nodes include churches, hospitals, and schools. Publication nodes represent textual documents.

Table A.1: Category of Religious Institutions

Institution Category	Freq.	Percent
Ecclesial	4,880	71.18%
Education	1,353	19.73%
Medical	517	7.54%
Other	52	0.76%
Rescue	23	0.34%
Press	16	0.23%
Business	12	0.18%
Scientific	3	0.04%
Total	6,856	100.00%

Notes: The table counts each baseline CHCD v3.0.0 institution once after matching to 1911 prefectures and adds 971 new directory-based CHCD records used in the supplementary descriptive measures, for 6,856 institutions in total. Press coverage combines 16 CHCD press institutions with 677 located CHCD publication records and 16 unique print entities in the reviewed McIntosh data, including 7 mission presses across 10 prefectures. These source counts are complementary and are not deduplicated one-for-one.

Table A.2: CMS/Stauffer and CHCD Missionary Coverage, 1920

	Count in 1920	Prefectures with presence	Correlation with CMS/Stauffer count
CMS/Stauffer: Protestant missionaries (1920)	10,715	227	–
CHCD institutions: Protestant	1,860	211	0.737
CHCD institutions: any denomination	1,958	212	0.705
CHCD people: Protestant	3,715	183	0.623
CHCD people: any denomination	4,147	184	0.581

Notes: The comparison is at the 1911 prefecture level and uses the 245 prefectures present in both the CMS/Stauffer source and the main CHCD panel. CMS/Stauffer reports Protestant missionary persons in 1920 only; accordingly, the CHCD “any denomination” rows are broader CHCD measures correlated with the Protestant CMS/Stauffer count, not an all-denomination external benchmark. CHCD institution and person rows count stocks active in 1920. The correlation column reports Pearson correlations between the CMS/Stauffer Protestant missionary count and the corresponding CHCD prefecture-level count. The CMS/Stauffer headcount may be somewhat understated or imprecise as a 1920 snapshot because the survey identifies missionaries who were on the ground without always recording a fully precise reference date. For our purposes, the more important comparison is geographic coverage, which is very similar across the sources: CMS/Stauffer records presence in 227 prefectures, compared with 211–212 for CHCD institutions and 183–184 for CHCD people. The presence column counts prefectures with any recorded measure, using zeros for prefectures without a corresponding CHCD record. Supplementary directory records are not added because their denominational tradition is not coded.

A.2 Summary Statistics and Historical Diffusion

Table A.3: Summary Statistics: Historical DiD Outcomes

Variable	Obs.	Mean	SD	P5	Median	P95
<i>Panel A. Headline outcomes</i>						
Modern banks (stock)	19,760	0.76	6.16	0.00	0.00	3.00
New modern banks	19,760	0.08	0.67	0.00	0.00	0.00
Industrial enterprises (stock)	15,314	4.78	40.10	0.00	0.00	17.00
<i>Panel B. Ownership-specific outcomes</i>						
Private modern banks (stock)	19,760	0.33	4.14	0.00	0.00	0.00
Government modern banks (stock)	19,760	0.29	1.35	0.00	0.00	2.00
Foreign modern banks (stock)	19,760	0.10	1.16	0.00	0.00	0.00
Foreign enterprises (stock)	15,314	0.05	0.48	0.00	0.00	0.00
Government enterprises (stock)	15,314	0.33	1.40	0.00	0.00	2.00
<i>Panel C. Other historical outcomes and infrastructure</i>						
Trade-oriented enterprises (stock)	15,314	0.87	8.33	0.00	0.00	3.00
Domestic-oriented enterprises (stock)	15,314	0.48	5.11	0.00	0.00	1.00
<i>Piaohao</i> branches (stock)	18,320	0.91	3.22	0.00	0.00	7.00
Railway lines (stock)	19,760	0.27	1.34	0.00	0.00	2.00
Government industrial investment (thousands)	18,880	21.65	249.77	0.00	0.00	0.07

Notes: The unit is a 1911-boundary prefecture-year. Statistics use the exact historical panel behind the headline DiD results. The modern-bank sample covers 1857–1936 (247 prefectures and 19,760 prefecture-years). Enterprise variables are observed from 1875–1936, so their maximum sample is 15,314 prefecture-years. Stocks are cumulative within prefecture. Government industrial investment is shown in thousands of nominal historical currency units. Observation counts vary only when the underlying source does not cover the full analysis window.

Table A.4: Summary Statistics: Missionary Treatment, Mechanisms, and Controls

Variable	Obs.	Mean	SD	P5	Median	P95
<i>Panel A. Prefecture-year treatment and mechanism measures</i>						
Missionary institutions (stock)	19,760	3.58	8.34	0.00	0.00	17.00
Mission presence	19,760	0.46	0.50	0.00	0.00	1.00
Missionary people	18,800	5.98	24.12	0.00	0.00	25.00
Mission education institutions (stock)	19,760	0.28	1.47	0.00	0.00	2.00
Mission schools (stock)	19,760	0.23	1.26	0.00	0.00	1.00
Mission higher-education institutions (stock)	19,760	0.03	0.22	0.00	0.00	0.00
Mission publication activity	19,760	0.04	0.20	0.00	0.00	0.00
Protestant mission institutions (stock)	19,760	3.21	7.88	0.00	0.00	15.00
Catholic mission institutions (stock)	19,553	0.37	2.23	0.00	0.00	1.00
Christianity-related bankers / all bankers	19,760	0.030	0.100	0.000	0.000	0.237
Church-school-educated bankers (stock)	19,760	4.30	19.26	0.00	0.00	25.00
<i>Panel B. Prefecture-level treatment and heterogeneity measures</i>						
Ever received a mission by 1936	247	0.94	0.24	0.00	1.00	1.00
Year of first mission (treated prefectures)	232	1,898	17	1,857	1,902	1,926
Ever had mission education	247	0.54	0.50	0.00	1.00	1.00
Ever had mission publication activity	247	0.22	0.42	0.00	0.00	1.00
<i>Panel C. Predetermined controls used in balance tests</i>						
Population, 1880 (thousands)	247	1,408.31	1,273.26	143.90	1,018.48	4,008.94
Provincial capital, 1880	231	0.16	0.36	0.00	0.00	1.00
Main river	247	0.63	0.48	0.00	1.00	1.00

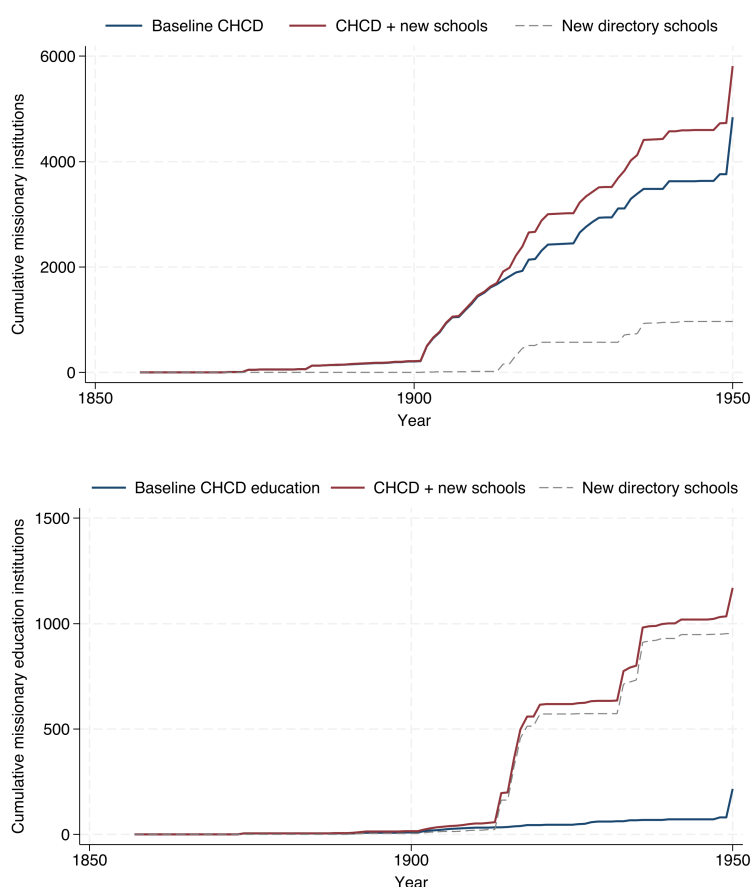
Notes: Panel A uses the exact 1857–1936 prefecture-year panel behind the headline modern-bank DiD. CHCD institution stocks define treatment timing; education adds unmatched directory schools, publication activity defines the school-by-publication split, and the banker share is set to zero when no banker is recorded, matching the corresponding figures. Panels B and C report one observation per 1911-boundary prefecture. First-treatment year is reported only for treated prefectures. Population is in thousands; provincial-capital status is measured in 1880 and available for 231 prefectures; main-river access is time invariant.

Table A.5: Summary Statistics: Main Bank IV Samples

Variable	Obs.	Mean	SD	P5	Median	P95
<i>Panel A. Historical-bank IV cross-section (1936)</i>						
Missionary institutions, 1850–1936	247	10.28	9.06	0.00	8.00	32.00
Predicted missionary exposure	247	0.473	1.138	0.000	0.000	3.351
Railway connection duration (years)	247	24.11	39.59	0.00	0.00	97.00
Cumulative railway density	247	0.0018	0.0051	0.0000	0.0000	0.0117
Modern banks, 1936	247	5.77	13.02	0.00	1.00	26.00
Standardized modern banks, 1936	247	0.000	1.000	-0.443	-0.366	1.553
<i>Panel B. Long-run bank IV cross-section (2020)</i>						
Missionary institutions, 1850–1936	247	10.28	9.06	0.00	8.00	32.00
Predicted missionary exposure	247	0.473	1.138	0.000	0.000	3.351
Railway connection duration (years)	247	24.11	39.59	0.00	0.00	97.00
Cumulative railway density	247	0.0018	0.0051	0.0000	0.0000	0.0117
Total financial-institution offices, 2020	247	713.30	724.25	0.00	473.00	1,895.00
Standardized financial-institution offices, 2020	247	0.000	1.000	-0.985	-0.332	1.632

Notes: The unit is a 1911-boundary prefecture. Both panels report the 247-prefecture complete-case sample used by the headline bank IV regressions, including province fixed effects, missionary exposure, predicted missionary exposure, railway duration, and railway density. Cumulative railway density is cumulative railway stock divided by prefecture area.

Figure A.2: Evolution of Missionary Institutions Over Time



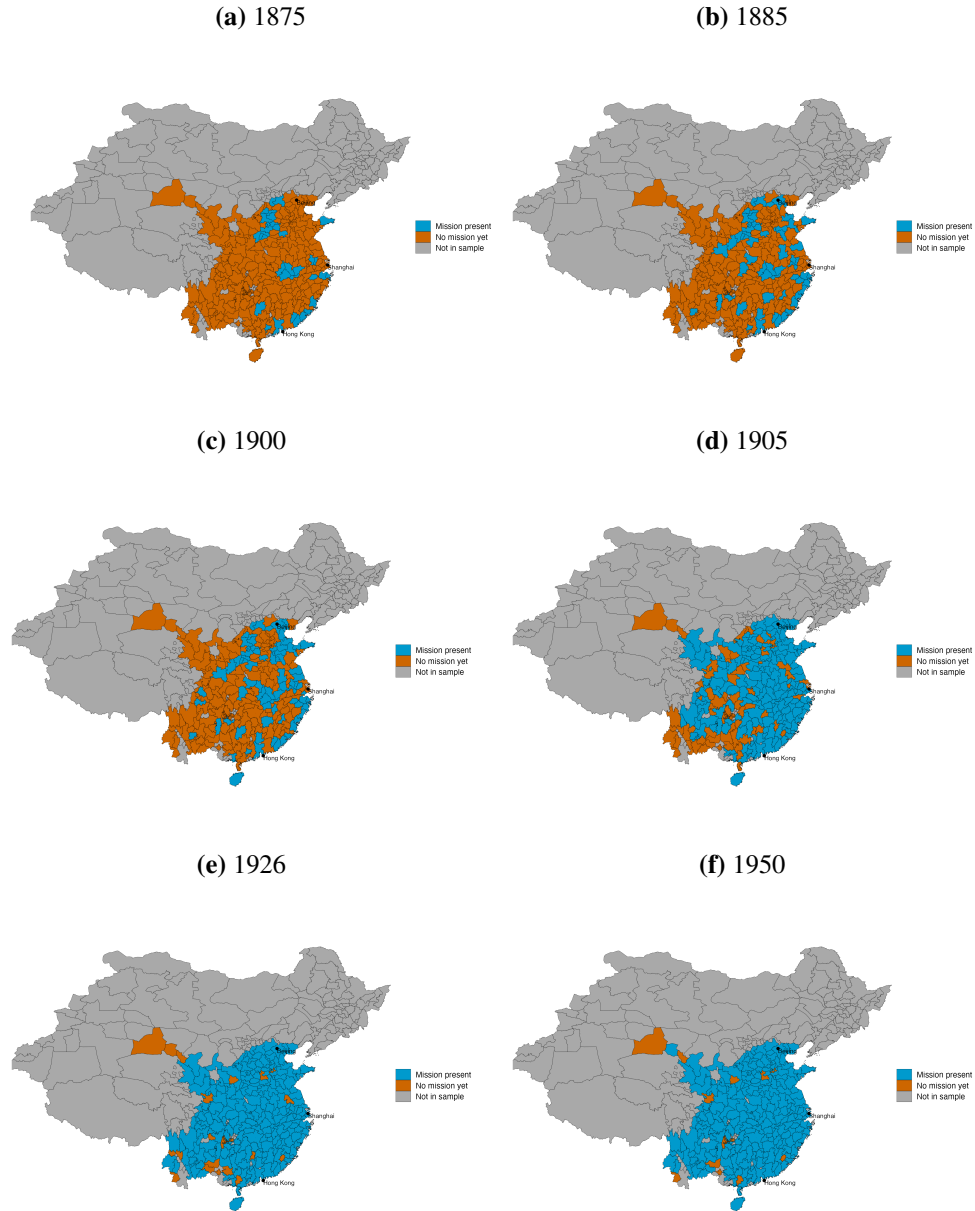
Notes: The first panel plots cumulative missionary institutions. The second panel plots cumulative missionary education institutions. Baseline counts use the CHCD v3.0.0 institution panel; the added series includes new directory-based school records not duplicated in the baseline panel.

Table A.6: Three Types of Financial Institutions

Feature	Piaohao	Qianzhuang	Modern banks
Main Function	Long-distance remittances	Short-term local lending	Deposit and lending
Clients	Governments, large merchants	Small businesses, artisans	All economic agents
Branch Networks	Many branches	No branches, locally operated	Many branches
Lending Focus	Limited lending, high-trust clients	Small, short-term loans	Large mortgage loans

Notes: The table compares the main functions, clients, branch networks, and lending focus of the three principal financial-institution types in China between 1840 and 1936. Modern banks uniquely combined deposit and lending services with limited-liability organization.

Figure A.3: Expansion of Missionaries

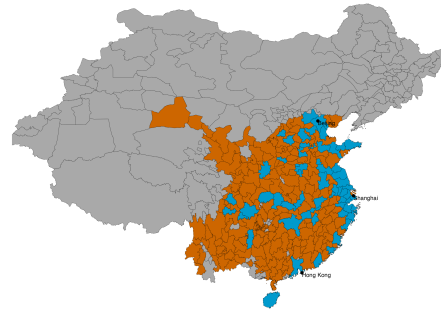
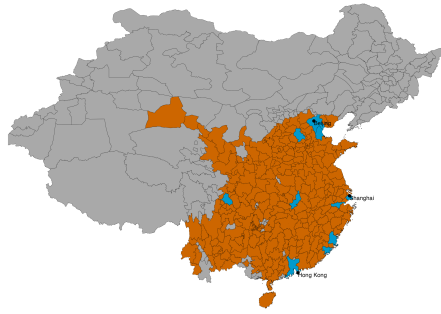


Notes: These maps show the expansion of missionaries over time for selected periods. Prefectures with missionary presence are shown in blue, those without missionaries in orange, and prefectures not included in the main analysis are in gray. Panel A shows prefectures with missionaries by 1875, Panel B by 1885, Panel C by 1900, Panel D by 1905, and subsequent panels continue chronologically. Data on missions are from the CHCD, and prefecture boundaries are from CHGIS 1911.

Figure A.4: Expansion of Modern Banks

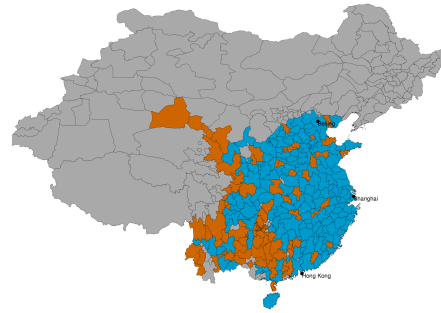
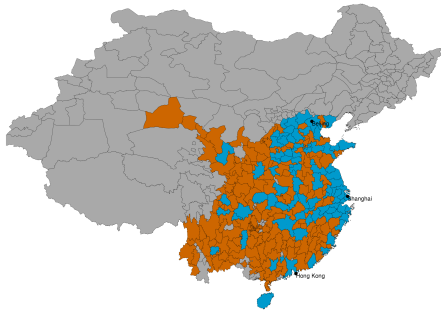
(a) By 1905

(b) By 1915



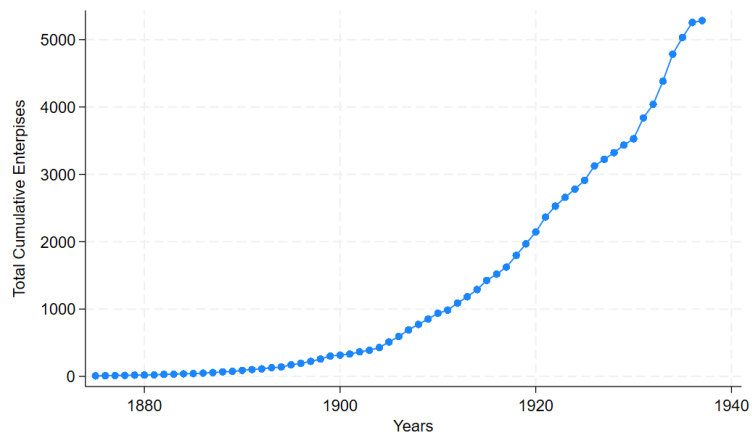
(c) By 1926

(d) By 1936



Notes: These maps show the expansion of modern banks in the current prefecture-year panel. Prefectures with at least one modern bank are shown in blue, sample prefectures without modern banks are shown in orange, and prefectures outside the analysis sample are shown in gray. Data on modern banks are from the Bank of China Economic Research Office, the *National Banking Yearbook*, and [Huang \(1995\)](#). Prefecture boundaries are from CHGIS 1911.

Figure A.5: Evolution of Enterprises Over Time



Notes: The figure plots enterprises recorded in [Du \(1991, 2019\)](#). These sources cover domestic-capital and old Chinese government enterprises between 1840 and 1937.

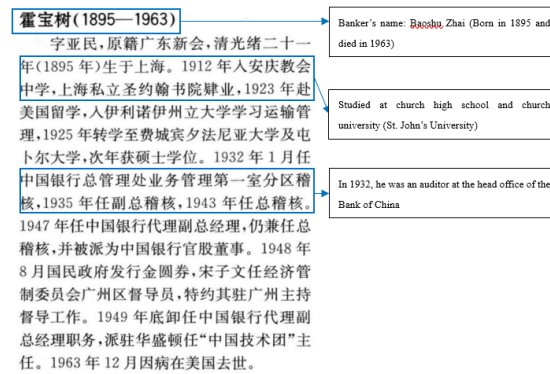
B Data Appendix

B.1 Historical Biographies

Banker biographies. We created a person-level dataset from *Jindai Zhongguo Yinhangye Jigou Renming Dacidian* (Jiang, 2014), a biographical dictionary of people in modern Chinese banking. The file contains 1,041 bankers who held positions such as director, founder, sponsor, or manager at Chinese-funded banks. We transcribed narrative entries and coded names, life dates, native place, education, Christian-school or other Christian connections, comprador experience, banking positions, and work across Chinese and foreign financial institutions. These fields make it possible to examine not only where banks appeared, but also the human-capital and intermediary networks of the people who operated them. Figure B.1 illustrates the underlying narrative-to-field coding.

The dictionary is a selected biographical source rather than a census of all bank employees. It is most informative about people in senior or otherwise documented positions, and missing information in a biography is not interpreted as evidence that an attribute was absent. We therefore use these measures descriptively and for mechanism evidence, rather than treating them as a representative individual-level sample.

Figure B.1: Illustration of Banker-Biography Digitization



Notes: Example of the narrative-to-field workflow used to code banker names, education, and banking positions.

Republican-era elite database. The Republican China Elite Database was assembled by Wang et al. (2026) from the revised edition of Xu Youchun's *Minguo Renwu Dacidian* (Xu, 2007). The source volume covers prominent individuals active between the founding of the Chinese Revolutionary Alliance in August 1905 and September 30, 1949, although individual biographies can extend beyond those dates. Its inclusion criteria span senior political and military figures; prominent businesspeople, bankers, and economists; scientists, engineers, and medical professionals; leading educators, writers, artists, journalists, and lawyers; and prominent religious figures. The resulting individual-level file contains 16,445 biographies and should therefore be interpreted as a selected record of prominent people, not a representative sample of the Chinese population.

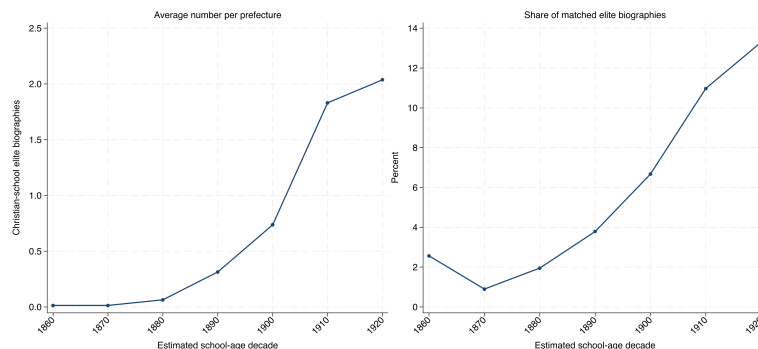
The source PDF was converted to machine-readable text using optical character recognition, manually cleaned and structured, and then parsed with DeepSeek-R1 into person-level fields. The extracted information includes native and childhood place, birth and death years, education and discipline, foreign

study and destination, career history and primary position, regime affiliation, publications, and other biographical markers. We harmonize native-place strings to Qing-era province names and 1911 prefecture boundaries. Of the 16,445 biographies, 14,763 match to one of 236 prefectures in the historical mission panel; 14,113 of those also report a birth year.

For the education proxy, we search the raw education narrative for a dictionary of major Christian institutions and church-related terms. The dictionary includes, among others, St. John's, Ginling, Yenching, Aurora, Fu Jen, West China Union, Lingnan, Cheeloo, Soochow, Hangchow, and the University of Shanghai, as well as generic references to church or Christian schooling. This procedure identifies 1,282 Christian-school-background biographies among the 14,763 geographically matched elites. It measures an education-background connection to Christian institutions, not the individual's religious belief. Because the biographies usually omit the exact date and place of attendance, the event-study proxy assigns each flagged person to the native prefecture at birth year plus 15. This construction assumes that native place is informative about school-age location and uses age 15 as a common marker for possible secondary-school exposure; mobility or attendance at a school elsewhere introduces measurement error.

The foreign-study indicator comes from the person-level biographical coding. The associated year field is free text and sometimes contains ranges or multiple dates, so we extract the first four-digit year between 1800 and 1999 as the departure-year proxy. The event-study outcome then cumulates these departures within the native prefecture.

Figure B.2: Christian-School Background Among Republican-Era Elites by School-Age Cohort



Notes: The sample contains biographies matched to 1911 prefectures with a non-missing birth year. School-age year is defined as birth year plus 15. Panel A reports, for each full decade from 1860 through 1920, the number of biographies coded with a Christian-school background divided by all 236 matched prefectures, including prefectures with zero such biographies. Panel B reports these biographies as a percentage of all matched elite biographies in the same school-age decade. The Christian-school flag is based on the education-field dictionary described in the text. The figure is descriptive: changes can reflect cohort composition and source coverage as well as changes in education, and the selected biographies are not representative of the population.

B.2 Historical Financial Records

Modern-bank balance sheets. We digitized two sources that differ in period, unit of observation, and accounting detail. Republican Ministry of Agriculture and Commerce tables cover the 1910s and list head offices and branches separately ([Ministry of Agriculture and Commerce, 1912–1924](#)). We transcribed the reported accounts, geocoded each office, assigned it to a 1911 prefecture, and aggregated branch-level deposits, capital, paid-in capital, banknote issuance, and surplus reserves. Coverage is not complete in every year, so these records are used primarily as supporting evidence rather than as a balanced panel.

The *National Banking Yearbook* provides bank-level balance sheets for 1932–1936 ([Bank of China Economic Research Office, 1937](#)). We digitized deposits, loans, assets, income, net income, cash, securities, paid-in capital, surplus reserves, and related accounting ratios. Reporting varies across banks and items, and the source is not a complete opening-and-closure panel. Specifications using these accounts therefore report the available outcome-specific sample and do not treat unreported values as zeros.

Piaohao and Qianzhuang. The *Piaohao* data combine the branch histories assembled by [Huang \(2002\)](#) and [Liu \(2018\)](#). The underlying materials draw on museum holdings, archival chamber-of-commerce files, recovered historical records, founders’ descendants, private collections, and original Shanxi merchant documents.⁵⁸ We retain the institution name, founder-origin affiliation, location, head-office or branch status, and opening, suspension, and closure year-month. Because branches have their own dated entry and exit histories, these data permit the stock, openings, and closures of *Piaohao* to be reconstructed with substantially less survivor bias than a single directory snapshot.

We separately transcribed and geocoded *Qianzhuang* entries from the *National Banking Yearbook*. The source records establishment dates, addresses, and the names and positions of core staff. The empirical series covers eight major provinces rather than all of China and lacks the complete branch-level closure histories available for *Piaohao*. We therefore use it to study financial interaction within its observed geography, not to infer a national census of local money houses.

B.3 Other Historical Outcomes and Controls

Enterprises and trade. The domestic-enterprise data record head offices of registered Chinese enterprises from [Du \(1991, 2019\)](#). The source applies capitalization thresholds: more than 10,000 silver dollars for shipping, industrial, and mining enterprises and more than 50,000 silver dollars for financial enterprises. It therefore captures relatively formal and capitalized firms while underrepresenting smaller businesses. We geocode head offices to 1911 prefectures and complement the source with foreign-enterprise directories compiled by [Huang \(1995\)](#). In the combined historical enterprise file, approximately 70% of establishments are private Chinese enterprises, 5% are government enterprises, and 24% are foreign enterprises.

The trade data come from the 1934–1936 postal census compiled in [Ministry of Communications](#)

⁵⁸The sources described by [Huang \(2002\)](#) include materials from the Chinese Piaohao Museum in Pingyao, files of the Fengtian Chamber of Commerce in the Liaoning Provincial Archives, historical materials recovered from Japan, and collections held by founders’ descendants and private collectors. [Liu \(2018\)](#) collates original Shanxi merchant documents from multiple collections.

(1937). The Republican government entrusted local postal offices with collecting information at the county and township levels on product types, production volumes, prices, and sales destinations. We digitize domestic and international export values and the number and composition of commodities traded, then aggregate the underlying local records to 1911 prefectures. These outcomes capture recorded trade flows and product coverage, not informal exchange.

Railways, geography, and clans. We reconstruct railway access from station locations and opening dates in Ma et al. (1983). Stations are geocoded and assigned to 1911 prefectures, producing time-varying indicators, railway duration, counts, and exposure measures. Baseline geographic and economic controls from Bai and Jia (2016) include historical population, coastal and major-river access, rice suitability, provincial-capital status, and treaty-port status. Selected balance and mechanism exercises add genealogy-book counts from Dong and Zhang (2024) as a proxy for clan intensity. These variables are treated as predetermined controls; potentially endogenous historical market outcomes are introduced only in explicitly labeled mechanism or robustness exercises.

B.4 Long-Run National Outcomes and Other Measures

Financial institutions, GDP, and population. For the post-1949 bank series, we standardize the administrative names in the government license registry, assign each modern county or district to the 1911 prefecture containing its centroid, and aggregate institution openings and closures to historical-prefecture years. This preserves a consistent historical geography as modern administrative boundaries change. As described in the main data section, the registry covers a broader modern regulatory category than the Republican-era modern-bank records, so levels across the two sources are not interpreted as directly comparable.

The GDP data of Chen et al. (2022a) provide 1-km gridded estimates for 1992–2019 based on nighttime lights. We intersect and sum the grid cells directly within accepted CHGIS 1911 prefecture polygons, avoiding an intermediate modern-prefecture crosswalk and the measurement error it would introduce. Population controls and denominators combine historical population with county-level totals constructed from the National Bureau of Statistics’ 2000 Population Census and 2015 one-percent population sample survey. Modern counties are assigned to 1911 prefectures by centroid so the population and bank geographies match.

Industrial enterprises and foreign investment. The Annual Survey of Industrial Production is a survey of industrial enterprises above the applicable reporting threshold, not the universe of registered firms. We use the number of surveyed enterprises, rather than their sales, employment, output, or borrowing, and distinguish private firms using the paid-in-capital ownership classification. Geocoded firm locations are assigned directly to accepted 1911 prefecture polygons; records without a usable direct match use a modern county or district centroid, followed by an area-weighted city crosswalk as a final fallback. The private-enterprise series omits 2009 because the ownership field mechanically assigns all observations to the same category in that year; the all-enterprise count remains usable. Foreign direct investment in 2017 comes from the *China City Statistical Yearbook* and is mapped to the same historical prefecture geography.

Biographies and household surveys. The Chinese Engineers Relational Database structures education, employment, and location histories for engineers in Republican China (1912–1949) and early post-war Taiwan (1945–1974) (Pelzer et al., 2025). We use it to study geographic origins, foreign education, and later location while retaining its source-defined coverage of documented engineers. Population Census and intercensal microdata from 1990 to 2015 provide educational attainment. The China Family Panel Studies provide panel measures of religious affiliation, values, and household financial behavior (Xie and Hu, 2014), while the 2007 *Spiritual Life Study of Chinese Residents* provides complementary cross-sectional measures of religiosity and beliefs. Survey locations are linked to 1911 counties or prefectures at the finest disclosed geography; the resulting regressions retain the survey-specific sampling and geographic limitations described with the mechanism tables.

B.5 Shaanxi Provincial Data

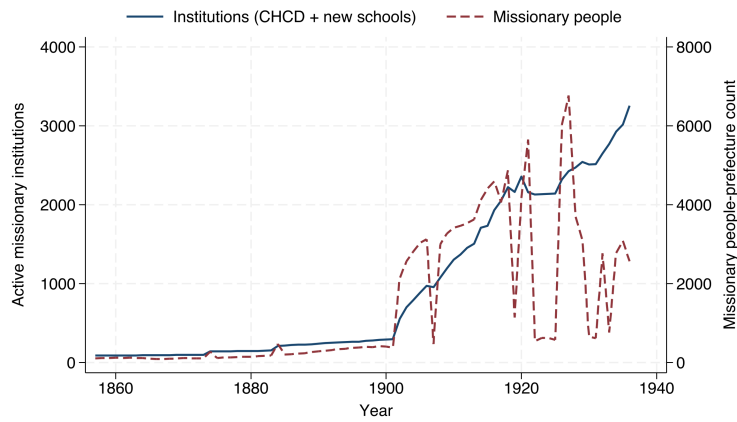
The county panel is transcribed from the Shaanxi Provincial Bureau of Statistics' *Historical Statistical Data Compilation for the Prefectures, Cities, and Counties of Shaanxi Province, 1949–1990* ([Shaanxi Provincial Bureau of Statistics, 1991](#)). It reports benchmark years before reform (1949, 1952, 1957, 1962, 1965, 1970, 1975, and 1978) and annual observations from 1978 through 1990. We use its county-level measures of population, students, fiscal revenue and expenditure, fixed-asset investment, electricity, health infrastructure, industrial and township enterprises, retail sales, savings, and production.

We exclude provincial, prefectural, and broad city aggregates, harmonize the remaining county, district, and county-level-city names to CHGIS 1911 counties using reviewed Chinese-name aliases, and aggregate multiple successor units assigned to the same historical county-year. The resulting panel covers 88 historical counties in the pre-1978 benchmark years and 89 from 1978 onward. Per-capita outcomes use population reported in the same source. Mission exposure is defined by whether a CHCD institution was recorded in the historical county before 1950; the railway and predicted-exposure measures are assigned to the same 1911 county geography. For this exercise, we also conducted a focused recovery of Shaanxi CHCD geography: unresolved province- or prefecture-level place records were reviewed using institution names, historical place-name variants, directory and gazetteer evidence, and the full CHGIS 1911 county reference, allowing additional institutions to be assigned to counties when the evidence was sufficiently specific. A national county-level design is not currently feasible without substantially greater record-by-record recovery, because many CHCD locations elsewhere are identified only at the prefecture or broader level; restricting the national analysis to county-resolved records would discard considerable information and produce a selected sample. Because the source is confined to one province, these estimates are complementary to the national prefecture analysis.

For the religion series, we separately digitize dated references to Christian institutions, activity, closures, persecution, and Cultural Revolution disruption from Shaanxi county and city gazetteers. We aggregate these source references into county-period mention shares through 1998. They measure the content recorded in local histories, not a complete annual census of religious activity.

B.6 Supplementary Missionary Data

Figure B.3: Missionary People and Institutions Over Time



Notes: The figure compares active missionary institutions and missionary people in the CHCD prefecture-year panel through 1936. The institution series includes new directory-based CHCD school records not duplicated in the baseline v3.0.0 panel. The figure is descriptive; the main treatment is based on institutional arrival.

Table B.1: Distribution of Protestant and Catholic Mission Institutions

Institution Tradition	Pre-1857		1857–1899		1900–1904		1905–1925		1926–1936		1937–1950		Undated		Total	
	Obs.	Perc.	Obs.	Perc.	Obs.	Perc.	Obs.	Perc.	Obs.	Perc.	Obs.	Perc.	Obs.	Perc.	Obs.	Perc.
Protestant	4	4.1	204	99.5	548	100.0	1,664	99.9	1,024	100.0	1,292	99.8	265	36.5	5,001	89.9
Catholic	93	95.9	1	0.5	0	0.0	1	0.1	0	0.0	2	0.2	462	63.5	559	10.1

Notes: The table counts unique CHCD v3.0.0 mission institutions with prefecture matches, using the institution tradition field and the cleaned start-year measure used to construct treatment timing. Percentages are within columns. The “Undated” column reports institutions with known tradition and location but no usable cleaned start year. Orthodox, non-religious, unknown, and missing traditions are excluded from the two-way Protestant–Catholic split. Source: CHCD v3.0.0.

C Balance & Identification

Figure C.1: Distribution of Treatment Variable G_i

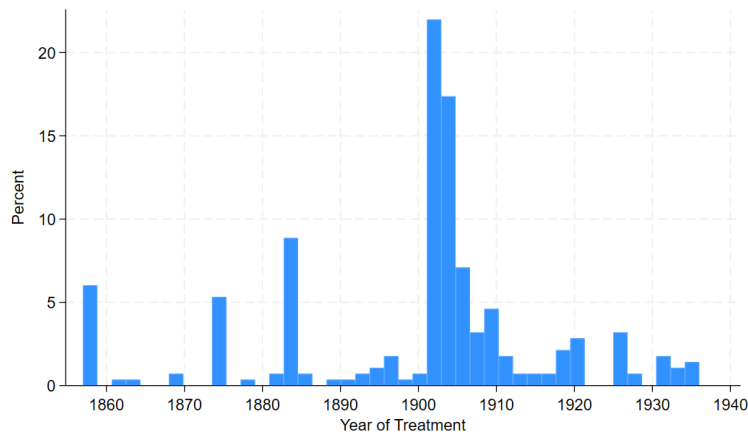
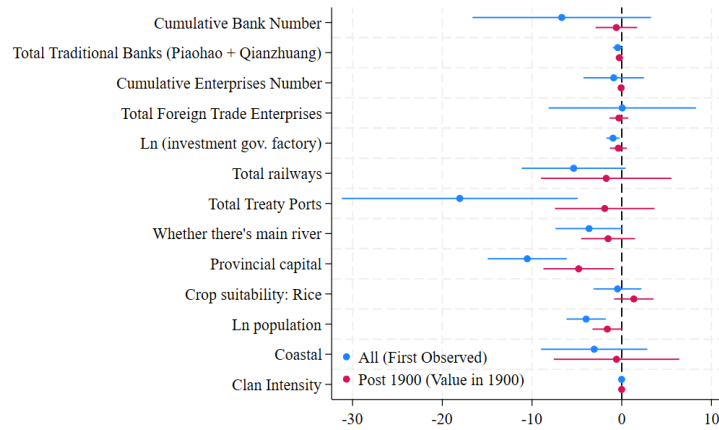


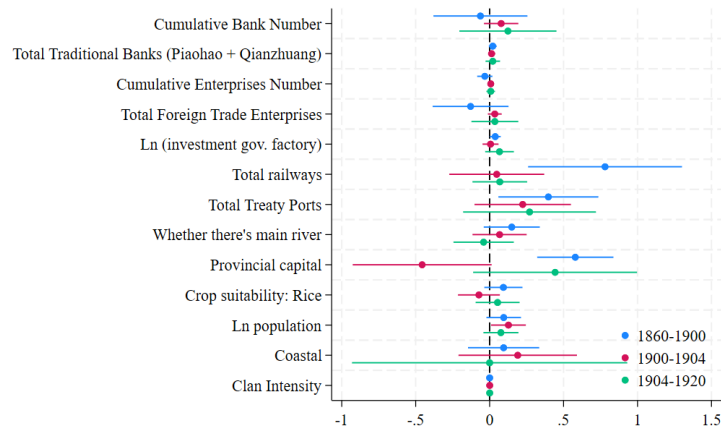
Table B.2: Distribution of Missionary Subdenominations

Religious Family	Pre-1857		1857–1899		1900–1904		1905–1925		1926–1936		1937–1950		Total	
	Obs.	Perc.	Obs.	Perc.	Obs.	Perc.	Obs.	Perc.	Obs.	Perc.	Obs.	Perc.	Obs.	Perc.
Interdenominational	4	33.3	63	12.5	368	12.4	1,710	17.9	1,047	17.0	0	0.0	3,192	16.4
Presbyterian	0	0.0	116	23.1	500	16.8	1,323	13.8	838	13.6	0	0.0	2,777	14.3
Methodist	1	8.3	82	16.3	358	12.0	1,214	12.7	678	11.0	2	1.0	2,335	12.0
Anglican	0	0.0	67	13.3	385	13.0	872	9.1	506	8.2	0	0.0	1,830	9.4
Lutheran	0	0.0	10	2.0	167	5.6	806	8.4	626	10.1	0	0.0	1,609	8.3
Baptist	1	8.3	23	4.6	519	17.5	630	6.6	356	5.8	0	0.0	1,529	7.9
Congregational	1	8.3	91	18.1	267	9.0	751	7.9	353	5.7	1	0.5	1,464	7.5
Other Protestant	3	25.0	24	4.8	396	13.3	2,111	22.1	1,414	22.9	10	5.1	3,958	20.4
Maryknoll	0	0.0	0	0.0	0	0.0	12	0.1	30	0.5	42	21.5	84	0.4
Passionist	0	0.0	0	0.0	0	0.0	0	0.0	56	0.9	21	10.8	77	0.4
Other Catholic	2	16.7	27	5.4	12	0.4	131	1.4	266	4.3	119	61.0	557	2.9

Notes: The table counts unique missionary people in the most recent CHCD person-prefecture file. It uses people data because detailed religious-family information is recorded at the person level. Smaller Protestant families are grouped as other Protestant, and smaller or missing Catholic families are grouped as other Catholic. Source: CHCD.

Figure C.2: Predictors of Mission Establishment Year: One-by-One Regressions

Notes: Predetermined covariates are separately regressed on missionary establishment year. The figure plots the coefficients corresponding to columns 1 and 2 of Table 2. All regressions include province fixed effects. Covariates are in raw units.

Figure C.3: Predictors of Mission Establishment by Broad Treatment Cohort: One-by-One Regressions

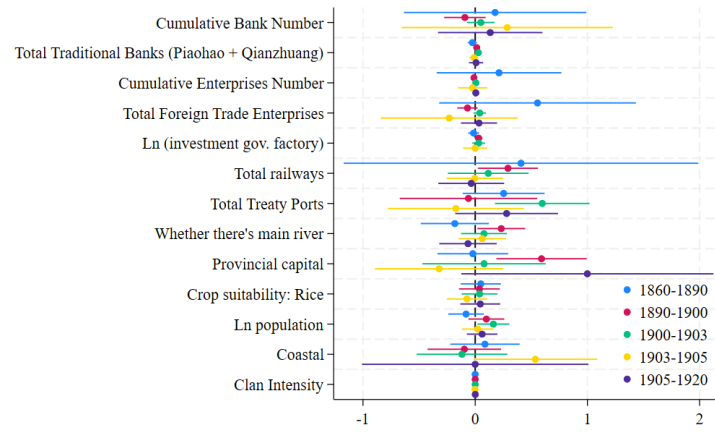
Notes: Univariate regressions of the treated-cohort dummy on each covariate. This figure reports adjacent broad-cohort comparisons as a complementary local-timing diagnostic. All regressions include province fixed effects. Covariates are in raw units.

Table C.1: Predictors of Mission Establishment by Cohort: Prefecture-Level Estimates

VARIABLES	(1) 1860-1880	(2) 1880-1890	(3) 1890-1900	(4) 1900-1903	(5) 1903-1904	(6) 1904-1905	(7) 1905-1910	(8) 1910-1920
Total Modern Banks	-0.830 (0.664)	-0.873 (1.874)	0.732 (1.313)	-0.086 (0.407)	0.107 (0.667)			1.359 (0.824)
Total Enterprises	0.840 (0.540)	-1.115 (0.780)	-0.085 (0.116)	-0.006 (0.047)	0.218 (0.265)	-0.057 (0.102)	-0.018 (0.042)	0.053 (0.285)
Ln (investment gov. factory)	0.005 (0.040)	-0.000 (0.045)	0.046 (0.030)	0.106 (0.105)	-0.096 (0.086)	0.076 (0.087)		
Total railways		0.414 (0.396)	-0.016 (0.353)	0.120 (0.204)	-0.588 (0.628)	-0.117 (0.158)	0.096 (0.201)	-0.034 (0.288)
Total Treaty Ports	0.491 (0.297)	0.252 (0.381)	0.164 (0.374)	0.492 (0.354)		0.093 (0.560)	0.665 (0.488)	-0.330 (1.117)
Whether there's main river	0.035 (0.221)	0.254 (0.329)	0.106 (0.110)	-0.034 (0.118)	-0.117 (0.246)	0.069 (0.183)	0.203 (0.189)	-0.026 (0.205)
Provincial capital	0.257 (0.215)	0.094 (0.304)	-0.018 (0.171)	-0.185 (0.167)	0.156 (0.306)	-0.118 (0.433)	0.609 (0.376)	-0.989* (0.507)
Crop suitability: Rice	0.201 (0.124)	-0.178 (0.239)	-0.009 (0.094)	-0.018 (0.091)	0.182 (0.209)	-0.196 (0.182)	0.140 (0.161)	-0.026 (0.117)
Ln population	0.106 (0.142)	-0.372** (0.151)	0.135* (0.069)	0.031 (0.082)	0.026 (0.177)	-0.031 (0.131)	-0.004 (0.113)	0.028 (0.141)
Coastal	-0.282 (0.200)	0.574 (0.346)	-0.217 (0.221)	-0.002 (0.250)	0.898 (1.518)	0.532 (0.677)	0.122 (0.606)	
Province FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Obs.	42	36	68	79	38	44	36	30
Adj. R sq.	0.272	0.133	0.214	0.185	-0.279	-0.0376	0.218	0.510
Mean Dep. Var.	0.317	0.481	0.157	0.591	0.390	0.339	0.473	0.286
S.D. Dep. Var.	0.466	0.500	0.364	0.492	0.488	0.474	0.500	0.452
P-value F-stat.	0.0543	0.263	0.0414	0.896	0.926	0.913	0.670	0.350

Notes: The historical panel includes 247 prefectures, of which 232 receive missionaries during the sample period. Regressions are conducted at the prefecture level. In each column, the dependent variable is whether a mission was established in the given range of years, with the sample including only prefectures that received missionaries during that time frame and the subsequent period. Controls are from [Bai and Jia \(2016\)](#) and are measured in the pre-period for each column. For example, banks are measured in 1860 in column 1, meaning we compare how the number of banks in 1860 differs between prefectures with missionary presence in 1860-1880 and those with presence in 1880-1890. Obs. refers to the number of observations; Adj. R^2 is the adjusted R^2 . Fixed effects may vary; refer to the table. Standard errors are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively. Appendix Figure C.4 reports the corresponding one-by-one coefficient plot.

**Figure C.4: Predictors of Mission Establishment by Narrow Treatment Cohort:
One-by-One Regressions**



Notes: Univariate regressions of the treated-cohort dummy on each covariate. This figure uses a coarser five-bin cohort partition. All regressions include province fixed effects. Covariates are in raw units.

D Historical DiD Results and Robustness

D.1 Main Estimates, Pre-Period Variation, and Outcome Transformations

Table D.1: Pre- and Post-Treatment Estimates for Main Historical Outcomes

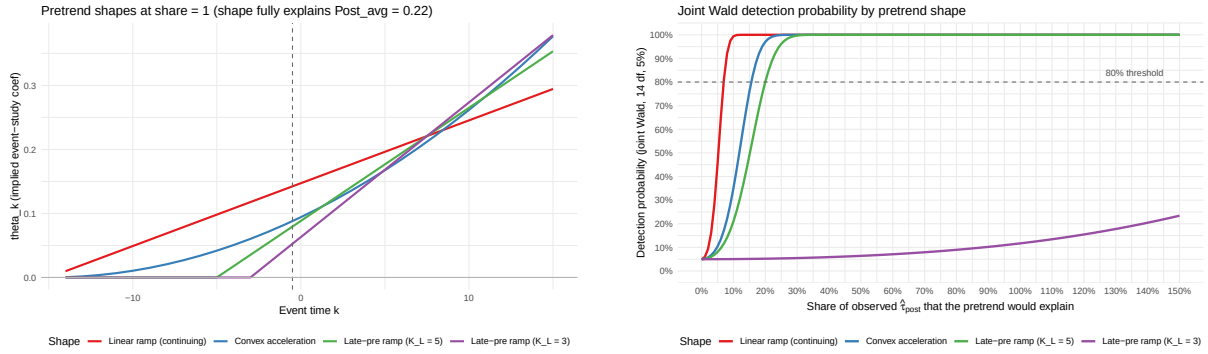
VARIABLES	(1) Modern banks	(2) Enterprises	(3) Log banks	(4) Bank growth	(5) New banks
Pre-treatment ATT	0.024 (0.016)	0.060* (0.036)	0.013 (0.009)	0.007 (0.004)	0.011 (0.007)
Post-treatment ATT	0.221*** (0.071)	1.292** (0.610)	0.075*** (0.021)	0.019*** (0.005)	0.059** (0.024)
Prefecture FE	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes
Period	1857–1936	1875–1936	1857–1936	1857–1936	1857–1936
Observations	18,560	13,268	18,560	18,328	18,560
Prefectures	232	214	232	232	232
Years	80	62	80	79	80
Mean Dep. Var.	0.761	4.072	0.164	0.014	0.082
S.D. Dep. Var.	6.320	41.477	0.540	0.090	0.680

Notes: The first two rows report average pre-treatment and post-treatment ATT estimates from the event-study aggregation over the event-time window $[-15, 15]$. Standard errors are in parentheses. Column 1 corresponds to Figure 6; column 2 corresponds to Figure 7. Columns 3–5 report alternative modern-bank outcomes: $\log(\text{total modern banks} + 1)$, one-year growth in this log outcome, and new modern banks. The post-treatment estimates imply about 31% more modern banks and 27% more enterprises using the outcome means reported in Figures 6 and 7. The log specification implies a 7.8% increase in total modern banks plus one; the growth specification implies a 1.9 percentage-point increase in annual log growth; and new-bank flows rise by about 73% relative to their mean. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Pre-period variation. A natural concern is that modern banks are rare in the early part of the sample, so flat pre-treatment coefficients may partly reflect limited pre-period variation. We therefore complement the event-study pre-trend tests with a design-based minimum detectable effect exercise. The exercise uses the estimated pre-period coefficients and covariance matrix from the baseline modern-bank event study. It then asks how large different hidden pre-trend shapes would need to be for the design to detect them. The design is informative against several smooth pre-trend shapes, although a pre-trend that activates only in the final few pre-treatment years is harder to detect. This exercise is intended as a diagnostic for the baseline design, not as a separate identification assumption.

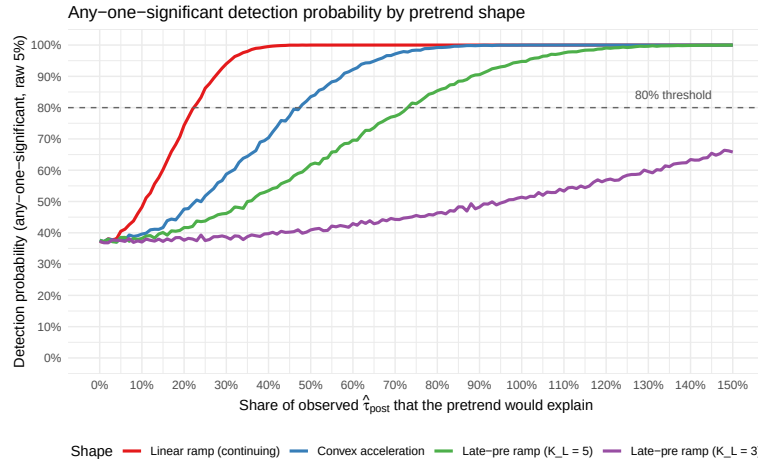
Alternative transformations. We also re-estimate the historical DiD using alternative transformations of the modern-bank outcome. These specifications address the concern that the baseline level outcome may be affected by zeros or limited early variation. Table D.2 reports the post-treatment ATT for each specification, and Figures D.3–D.5 plot the corresponding event studies.

Figure D.1: Design-Based Power Against Hidden Pre-Trends



Notes: The left panel shows candidate hidden pre-trend shapes, scaled relative to the observed post-treatment effect. The right panel reports how often the joint Wald test over pre-treatment coefficients would reject for each candidate shape. The exercise uses the estimated pre-period coefficients and covariance matrix from the baseline modern-bank event study.

Figure D.2: Alternative Pre-Trend Detection Rule



Notes: The figure reports rejection probabilities under an alternative rule based on whether any individual pre-treatment coefficient is statistically significant. This rule is more mechanical than the joint Wald test but gives a useful check that the pre-period is not completely uninformative.

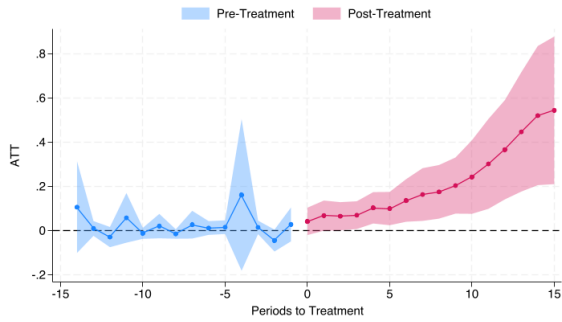
Table D.2: Post-Treatment ATT Under Alternative Modern-Bank Outcomes

Specification	ATT	SE	<i>p</i> -value
Baseline: total modern banks	0.221	0.071	0.002
$\log(y + 1)$	0.075	0.021	< 0.001
Growth rate	0.019	0.005	< 0.001
Extensive margin ($y > 0$)	0.110	0.046	0.017
New modern banks	0.059	0.024	0.014
Province-conditioned levels	0.221	0.071	0.002
Drop pre-1900, levels	0.258	0.092	0.005
Drop pre-1900, $\log(y + 1)$	0.084	0.026	0.001

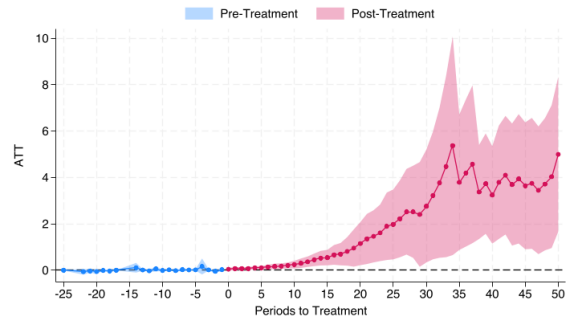
Notes: The table reports post-treatment ATT estimates averaged over event times 0 through 15. The baseline reproduces the main effect reported in Figure 6. The remaining rows use alternative transformations or definitions of the modern-bank outcome.

Figure D.3: Alternative Outcome Transformations: Baseline, Logs, and Growth

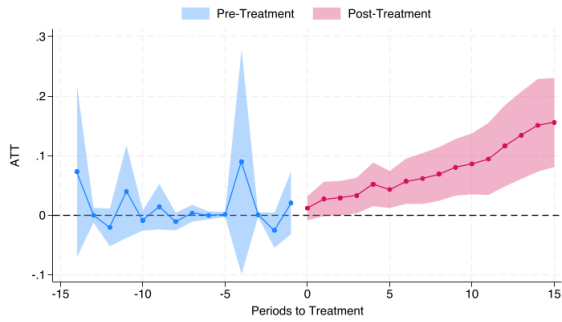
(a) Baseline, 15 periods



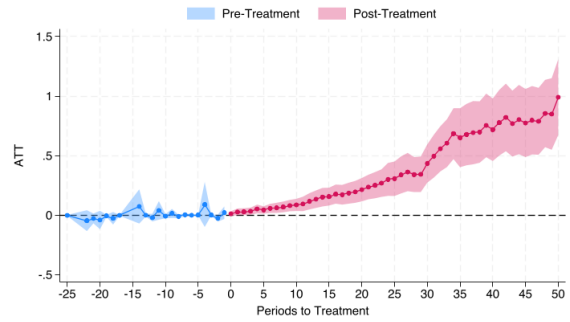
(b) Baseline, 50 periods



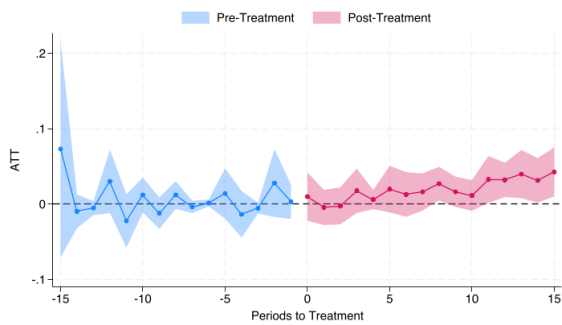
(c) $\log(y + 1)$, 15 periods



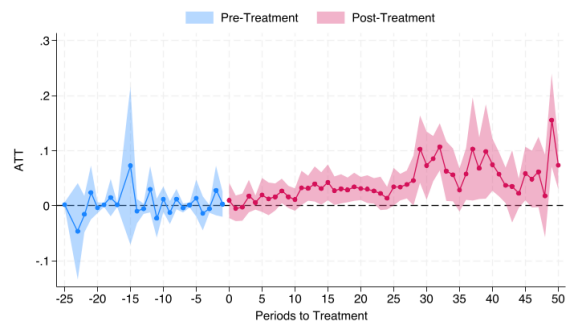
(d) $\log(y + 1)$, 50 periods



(e) Growth rate, 15 periods

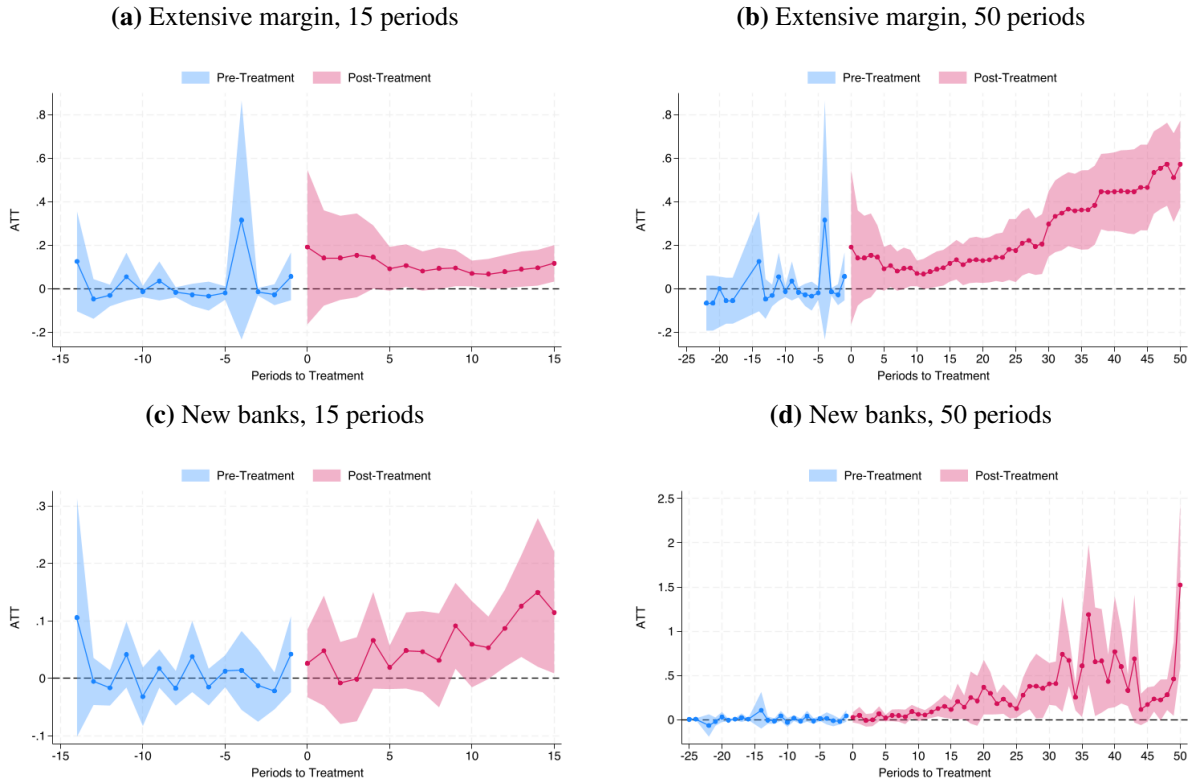


(f) Growth rate, 50 periods



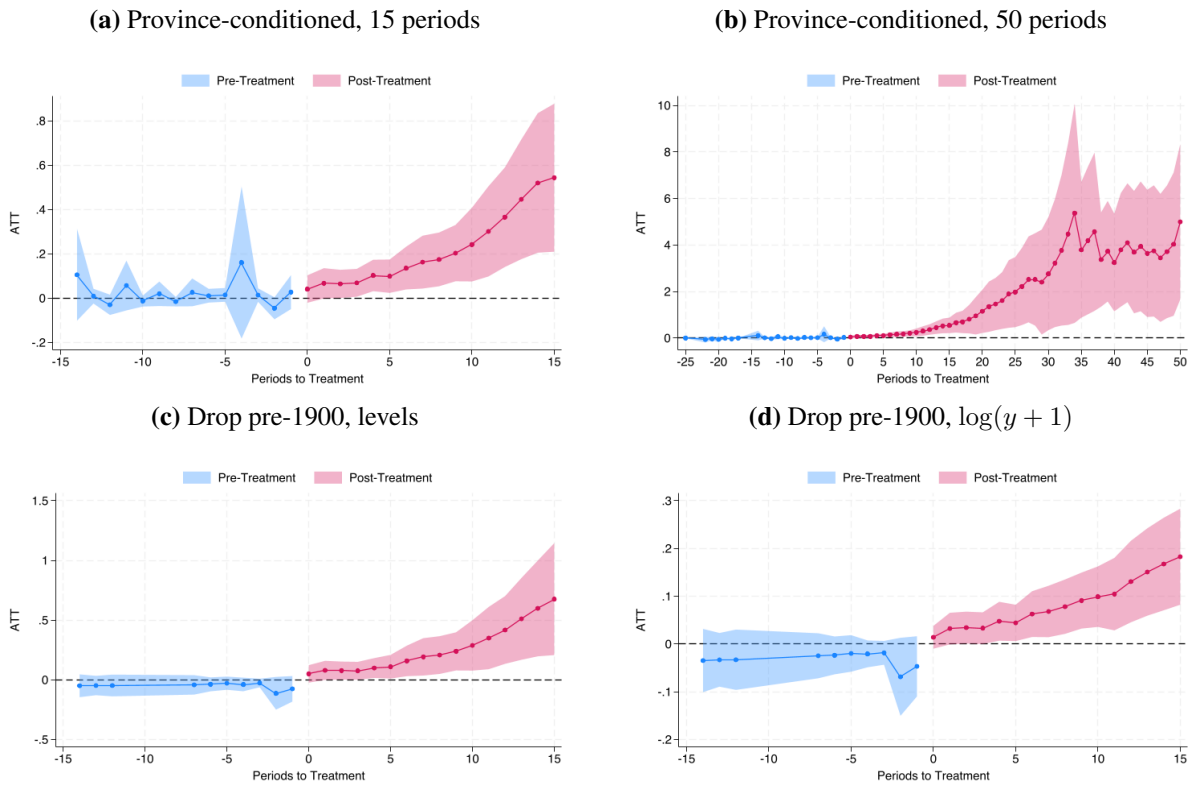
Notes: The figure reports event studies for alternative transformations of the modern-bank outcome. Panels use the same treatment definition as the baseline specification.

Figure D.4: Alternative Outcome Transformations: Extensive Margin and New Banks



Notes: The figure reports additional outcome transformations. The post-treatment averages corresponding to these figures are reported in Table D.2.

Figure D.5: Alternative Outcome Transformations: Province-Conditioned and Restricted Samples



Notes: The figure reports province-conditioned specifications and sample restrictions. The post-treatment averages corresponding to these figures are reported in Table D.2.

D.2 Propagation Across the Banking Sector

The baseline outcome counts all modern banks. We next show that the result is not driven by a single bank type or by a purely extensive-margin count. Figure D.6 reports event studies separately for private, government, and foreign banks. All three outcomes move in the same direction after missionary arrival, suggesting that missionary exposure affected the broader modern-banking sector rather than only one ownership category.

Figure D.6: Missionary Presence and Modern Banks by Ownership Type



Notes: Each panel reports the event-study estimates from the historical DiD design using the indicated bank type as the outcome. The estimation window is 15 periods before and after missionary arrival.

We also use balance-sheet data to study whether missionary exposure is associated with the scale of banking activity. These exercises are supporting evidence rather than the baseline DiD specification, but they point in the same direction: places with greater missionary exposure have larger deposits, loans, assets, and related measures of bank size. The prefecture-level table uses the branch/location balance-sheet records from the 1910s, aggregated to 1911 prefectures, and relates them to the predicted missionary exposure constructed from the railway-flow instrument in Section 5. The bank-level table uses the 1932–1936 bank balance sheets and constructs each bank’s missionary exposure from its branch network across prefectures.

Table D.3: Missionary Exposure and Bank Balance Sheets: Prefecture Level

Variables	(1) Ln Deposits	(2) Ln Banknotes	(3) Ln Surplus Reserves	(4) Ln Capital Paid	(5) Ln Capital	(6) Total Banks (1936)
Predicted Tot Missions	0.448** (0.175)	0.498*** (0.183)	0.396*** (0.151)	0.379** (0.166)	0.368** (0.180)	1.634* (0.959)
Railway Duration	-0.0268 (0.0266)	-0.0396 (0.0282)	-0.0311 (0.0227)	-0.0261 (0.0251)	-0.0245 (0.0270)	-0.179 (0.130)
Railway Density	323.8*** (109.6)	259.9** (124.2)	265.9** (113.8)	259.3** (108.7)	269.2** (113.2)	1,438** (597.7)
Obs.	247	247	247	247	247	247
Mean Dep. Var.	2.271	2.062	0.848	1.458	1.520	3.622

Notes: The table reports prefecture-level balance-sheet regressions using the 1910s branch/location records. Outcomes are measured using historical bank balance-sheet data described in the data section. Predicted total missions is constructed using the railway-flow instrument described in Section 5. Table D.4 reports the corresponding specification with log 1880 population included as an additional control. Standard errors are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table D.4: Missionary Exposure and Bank Balance Sheets: Prefecture Level with 1880 Population Control

Variables	(1) Ln Deposits	(2) Ln Banknotes	(3) Ln Surplus Reserves	(4) Ln Capital Paid	(5) Ln Capital	(6) Total Banks (1936)
Predicted Tot Missions	0.474** (0.227)	0.542** (0.243)	0.461** (0.209)	0.430* (0.221)	0.407* (0.236)	1.888 (1.292)
Railway Duration	-0.0261 (0.0262)	-0.0384 (0.0284)	-0.0293 (0.0238)	-0.0247 (0.0255)	-0.0234 (0.0269)	-0.172 (0.133)
Railway Density	317.3*** (107.5)	249.1** (125.4)	250.0** (121.2)	246.7** (109.6)	259.6** (111.1)	1,376** (620.4)
Ln Population (1880)	-0.577 (1.127)	-0.970 (1.223)	-1.434 (1.035)	-1.126 (1.080)	-0.856 (1.144)	-5.615 (6.151)
Obs.	247	247	247	247	247	247
Mean Dep. Var.	2.271	2.062	0.848	1.458	1.520	3.622

Notes: The table copies the prefecture-level specification in Table D.3 and adds log 1880 population as a control. Missionary institutions are instrumented by the railway-flow prediction described in Section 5. Every column also includes railway duration, railway density, and province fixed effects. The 1880 measure is the project historical population series assigned to 1911 prefectures. Consistent with the original table, unreported monetary balance-sheet items are coded as zero before applying $\ln(1 + x)$. Standard errors are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table D.5: Missionary Exposure and Bank Balance Sheets: Bank Level

Variables	(1) Ln Deposits	(2) Ln Loans	(3) Ln Assets	(4) Ln Income	(5) Ln Security	(6) Ln Surplus Reserves	(7) Tot Branches	(8) Ln Net Income
Mission Exposure (STD)	0.954*** (0.125)	0.858*** (0.107)	0.867*** (0.117)	0.689*** (0.106)	0.844*** (0.143)	1.262*** (0.141)	0.676 (0.799)	0.585*** (0.120)
Exposure to Population (STD)	0.488*** (0.0867)	0.472*** (0.0784)	0.479*** (0.0824)	0.396*** (0.0782)	0.457*** (0.120)	0.268*** (0.0849)	17.47*** (0.776)	0.329*** (0.0669)
Obs.	155	156	157	156	136	144	160	153

Notes: The table reports bank-level regressions using 1932–1936 historical balance-sheet data. Mission exposure is standardized and is based on the historical missionary exposure of the prefectures in which each bank operated branches. The baseline population-exposure variable sums 1820 log population density over active bank–prefecture–year records. Table D.6 replaces that legacy proxy with log 1880 population in each bank’s pre-outcome geographic branch footprint. Standard errors are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table D.6: Missionary Exposure and Bank Balance Sheets: Bank Level with 1880 Population Control

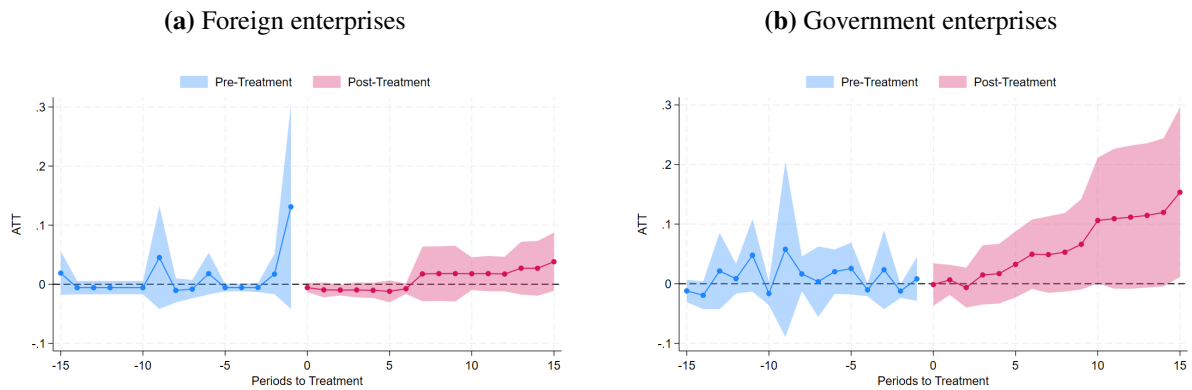
Variables	(1) Ln Deposits	(2) Ln Loans	(3) Ln Assets	(4) Ln Income	(5) Ln Security	(6) Ln Surplus Reserves	(7) Tot Branches	(8) Ln Net Income
Mission Exposure (STD)	0.651*** (0.124)	0.585*** (0.111)	0.584*** (0.124)	0.392*** (0.106)	0.848*** (0.162)	0.791*** (0.130)	1.910 (1.151)	0.325*** (0.110)
Ln 1880 Network Population	0.859*** (0.140)	0.770*** (0.147)	0.819*** (0.153)	0.761*** (0.142)	0.921*** (0.173)	0.712*** (0.128)	7.131*** (1.762)	0.650*** (0.172)
Obs.	97	97	97	97	91	94	98	97

Notes: The table copies the bank-level specification in Table D.5, replacing its legacy population-exposure proxy with log 1880 network population. For each bank, the population measure sums the project 1880 population across the distinct 1911 prefectures in which the bank had established a branch by the end of 1931. A prefecture is counted once even when a bank had multiple branches there. Banks are retained only when every prefecture in their branch network has a valid positive 1880 population, so the denominator never represents a partial network. Of the 160 banks in the collapsed 1932–1936 balance-sheet cross-section, 108 link to the historical branch network and 98 have complete 1880 population coverage; outcome reporting produces the column-specific samples shown. Mission exposure is standardized over the original 160-bank cross-section. Estimates are robust OLS with an intercept and are not instrumented. Monetary outcomes are $\ln(1 + x)$ and total branches is in levels. Standard errors are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

D.3 Other Market Outcomes

The main text reports the effect of missionary presence on total enterprises. This appendix provides additional evidence on other dimensions of market activity. First, we separate enterprises by ownership type. The positive enterprise result is mainly driven by private enterprises, which account for most of the enterprise panel. Government and foreign enterprises move in the same direction but are less precisely estimated.

Figure D.7: Missionary Presence and Enterprises by Ownership Type



Notes: The figure reports event studies for foreign and government enterprises. The total enterprise result is reported in the main text. Private enterprises account for most of the enterprise panel.

Second, we use trade data from 1934–1936 to examine whether missionary exposure is associated with broader market exchange. We treat this as evidence on another market outcome rather than as a mechanism test. A downstream response through modern banks is one possible interpretation; direct effects from practical education and the diffusion of commercially useful knowledge are also plausible. The trade data come from a government census of trade product types, production volumes, prices, and sales areas. In 1934, local postal offices were entrusted with collecting the census at the county and township levels, and the data collection was completed in 1936 ([Ministry of Communications, 1937](#)). The data report domestic and international export activity, including values and commodity types.

We use these data to classify enterprise industries as trade-oriented or domestic-oriented. The idea is that treaty ports were the main departure points for overseas-bound goods, so products disproportionately exported from treaty ports are more likely to be linked to international trade. We compare export composition across treaty and non-treaty ports, compute the relative importance of each product category, and identify products with the largest positive and negative differences in treaty-port export shares. Products with high treaty-port export intensity are classified as “trade-likely,” while products with low treaty-port export intensity are classified as “domestic-likely.” We then map these products to enterprise industries in the historical enterprise panel. This gives us two measures of market activity: trade-likely enterprises and domestic-likely enterprises.

Table D.7 reports the resulting classification, and Figure D.8 estimates the historical DiD separately for trade-likely and domestic-likely enterprises. The trade-likely response appears only after a delay, mirroring the timing of the total-enterprise result in the main text. Table D.8 also reports the relationship between missionary exposure and export transactions in 1936. These exercises show that missionary exposure is associated with stronger trade-related activity. This is consistent with modern banks sup-

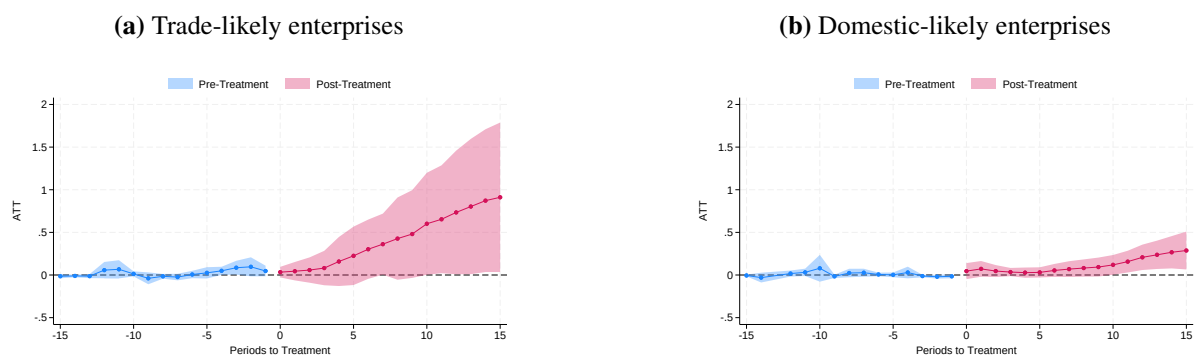
porting broader market integration, although this may also reflect practical education and knowledge diffusion.

Table D.7: Trade-Likely and Domestic-Likely Enterprise Categories

Trade-Likely Enterprises	Domestic-Likely Enterprises
Silk	Brick
Tea	Cement
Tobacco	Construction
Seafood	Building
Fish	Real Estate
Herbal Products	Services
Shipping	Textile
Foreign Trade	Animal Products
Transshipment	Agriculture
Beverages	Farming
Porcelain	Clothing
Cigarette	Entertainment
	Rice

Notes: Product groups are classified using differences in export composition between treaty ports and non-treaty ports in the 1930s trade data. We then map these products to enterprise industries.

Figure D.8: Missionary Presence and Trade-Likely Enterprises



Notes: Panel A uses trade-likely enterprises as the outcome; Panel B uses domestic-likely enterprises. The categories are defined in Table D.7.

Table D.8: Missionary Exposure and Export Transactions in 1936

Variables	IV	OLS
	Export Transactions	Export Transactions
Tot Miss Inst	0.865* (0.496)	0.289*** (0.0893)
RR Duration	0.00212 (0.146)	0.185** (0.0719)
RR Density	-595.0 (637.4)	-617.4 (491.6)
Province FE	Yes	Yes
Observations	231	231
Adj. R-squared	-0.214	0.385
Mean Dep. Var.	16.07	16.07
S.D. Dep. Var.	33.66	33.66
K-P F-Stat	11.68	NA

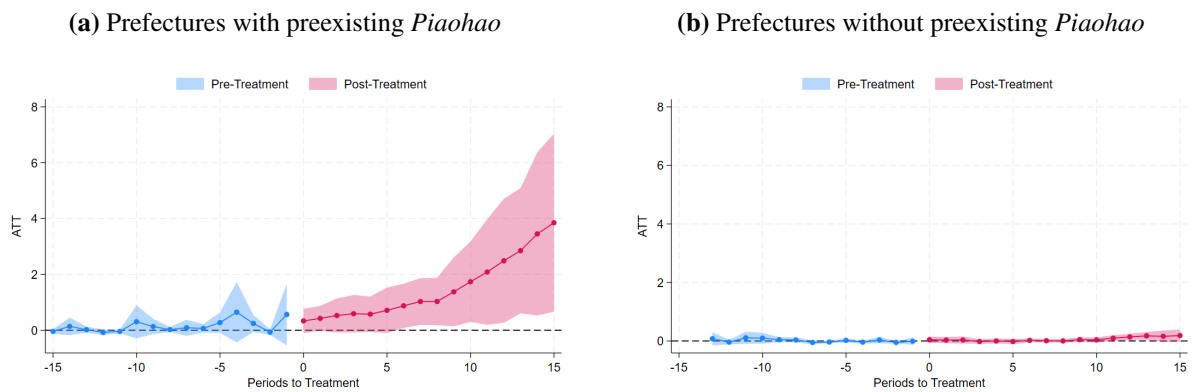
Notes: The dependent variable is export transactions in 1936. The IV instruments missionary institutions with the railway-flow instrument from Section 5. Standard errors are in parentheses; ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

D.4 Traditional Finance and Financial Interaction

This section studies how missionary exposure interacted with traditional financial institutions. The main text emphasizes the transition from traditional finance to modern banking. We therefore begin with *Piaohao*, where the substitution pattern is clearest.

Modern banks in places with preexisting *Piaohao*. We first split the sample by whether a prefecture had *Piaohao* before missionary arrival. Prefectures with preexisting *Piaohao* experienced a sharper rise in modern banks following missionary entry, suggesting that modern banks expanded where financial demand was already present.

Figure D.9: Modern Banks in Places With and Without Preexisting *Piaohao*

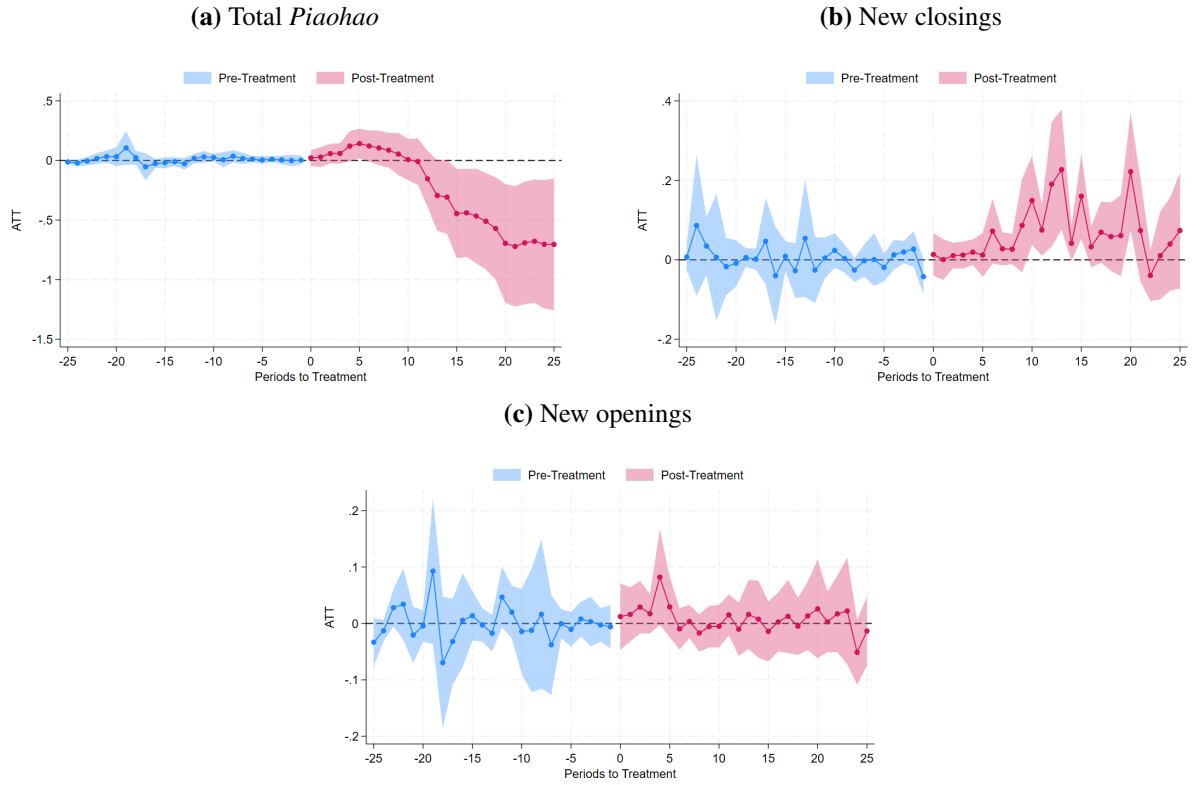


Notes: The outcome is the number of modern banks. Panel A includes prefectures with *Piaohao* before missionary arrival and never-treated prefectures. Panel B includes prefectures without *Piaohao* before missionary arrival and never-treated prefectures. Split-group sizes: 37 treated prefectures with preexisting *Piaohao* and 195 treated prefectures without preexisting *Piaohao*.

***Piaohao*.** We then use *Piaohao* as the outcome. The event study shows that *Piaohao* initially rise but later decline after missionary arrival, consistent with modern banks eventually replacing older remittance institutions. The lack of a clear *Piaohao* pre-trend is also useful because these institutions were already active before the modern-bank wave. Unlike the modern-bank outcome, the pre-period has meaningful variation, so the flat pre-period is less likely to be only a mechanical consequence of early zeros.

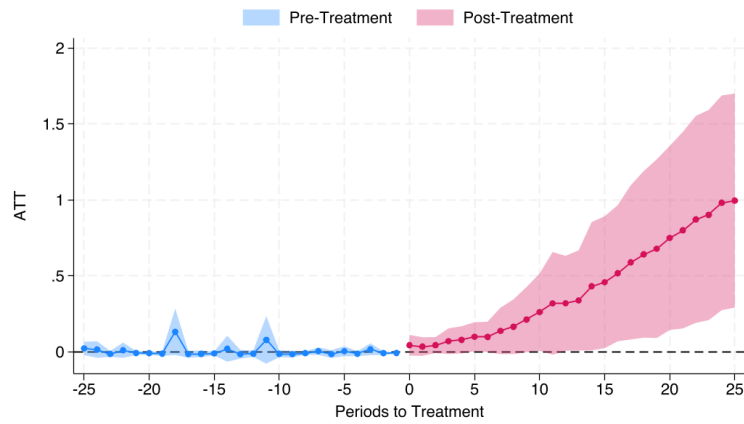
***Qianzhuang*.** The pattern for *Qianzhuang* is different. Because *Qianzhuang* were local money shops with dense merchant networks, they could complement modern banks that initially lacked local trust and information. We therefore treat the *Qianzhuang* results as evidence on financial interaction rather than simple substitution. Taken together, the traditional-finance results suggest that missionary exposure did not erase all older institutions; the substitution pattern is clearest for *Piaohao*, while *Qianzhuang* appear more complementary.

Figure D.10: Missionary Presence and *Piaohao*



Notes: The figure reports event studies using total *Piaohao*, new *Piaohao* closings, and new *Piaohao* openings as outcomes. The absence of a clear pre-trend is informative because *Piaohao* have substantial pre-period variation and branch-level opening and closing dates.

Figure D.11: Missionary Presence and *Qianzhuang*

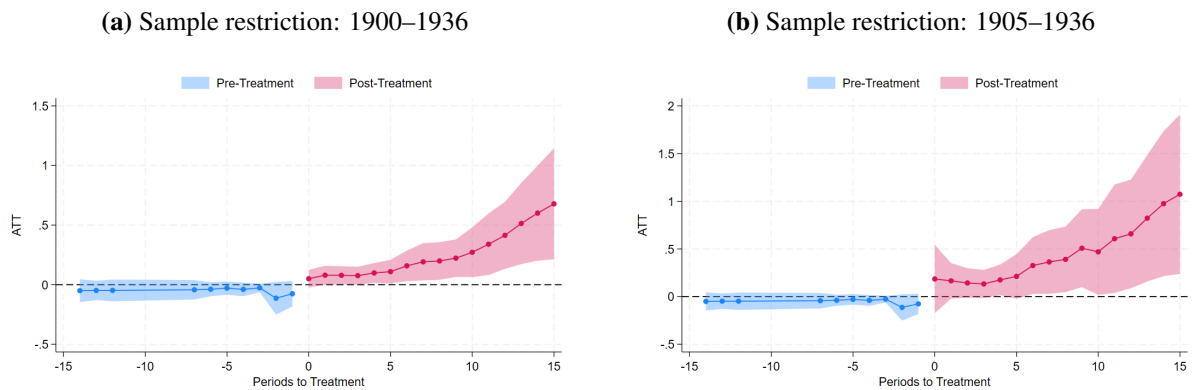


Notes: The figure reports the event study using total *Qianzhuang* as the outcome.

D.5 Identification Robustness

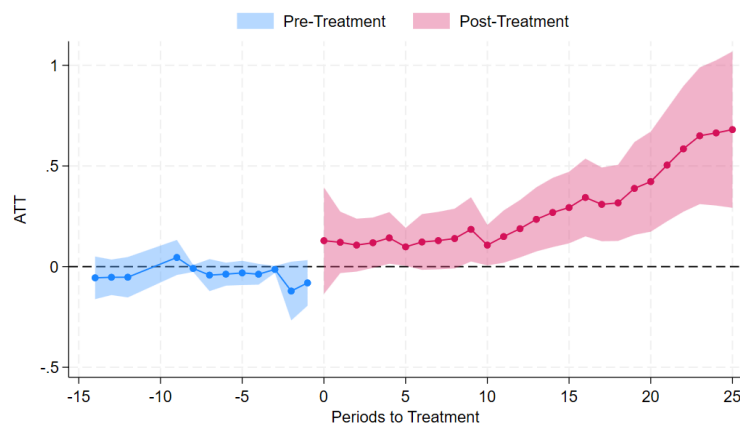
This section reports additional checks for the historical DiD design. First, we restrict the sample to the post-1900 and post-1905 periods, when selection into missionary entry is weaker and the modern-bank outcome has more pre-period variation. Second, we exclude treaty ports to address the concern that the result is driven by locations with early foreign presence. Third, we use longer event windows, balanced pre/post windows, and group-specific ATTs. Fourth, we add railway controls and examine pre-trends in other predetermined outcomes. Across these exercises, the qualitative pattern remains similar to the baseline result: the effect is not driven by the earliest part of the sample, treaty-port prefectures, one particular cohort, or one event-window choice.

Figure D.12: Missionary Presence and Modern Banks: Post-1900 and Post-1905 Samples



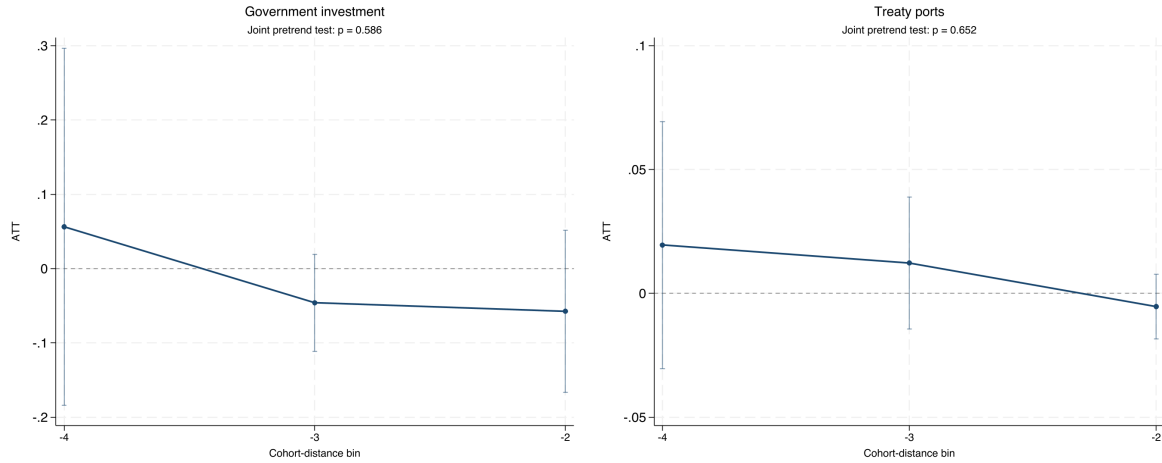
Notes: The figure reports the baseline event study after restricting the estimation sample to later periods. The dependent variable is the total number of modern banks. The positive post-treatment pattern remains when the early years with sparse modern-bank activity are excluded.

Figure D.13: Missionary Presence and Modern Banks: Excluding Treaty Ports



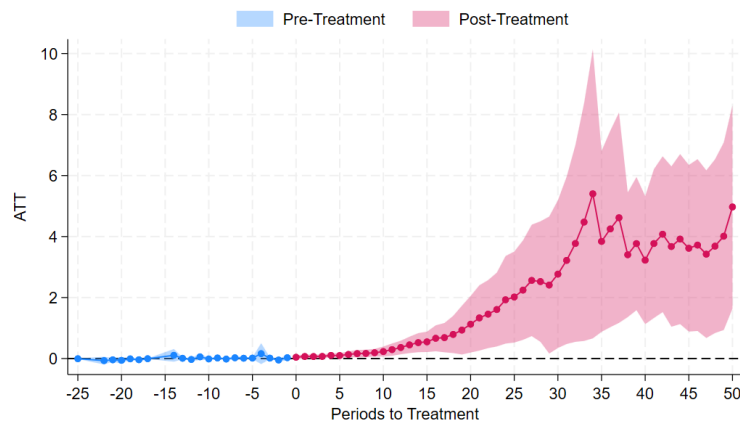
Notes: The figure reports the baseline event study after excluding treaty-port prefectures. The dependent variable is the total number of modern banks. The persistence of the effect outside treaty ports suggests that the result is not only a treaty-port foreign-presence effect.

Figure D.14: Pre-Treatment Institutional Conditions in Aggregated Cohorts



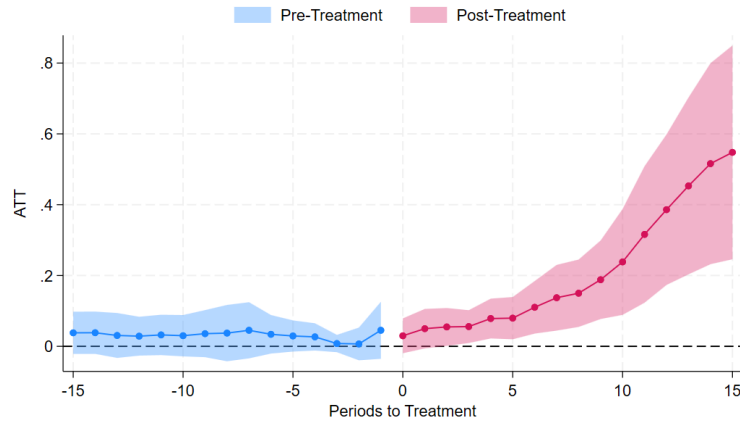
Notes: Annual data are not consistently available for government investment and treaty ports over the historical period, so the estimates pool years into cohort-distance bins. The underlying treatment cohorts are the same as those used in the balance exercise in Table C.1. Points report pooled pre-treatment coefficients with 95% confidence intervals; the reported p -values are covariance-based joint Wald tests that all displayed coefficients equal zero.

Figure D.15: Missionary Presence and Modern Banks: Longer Event Window



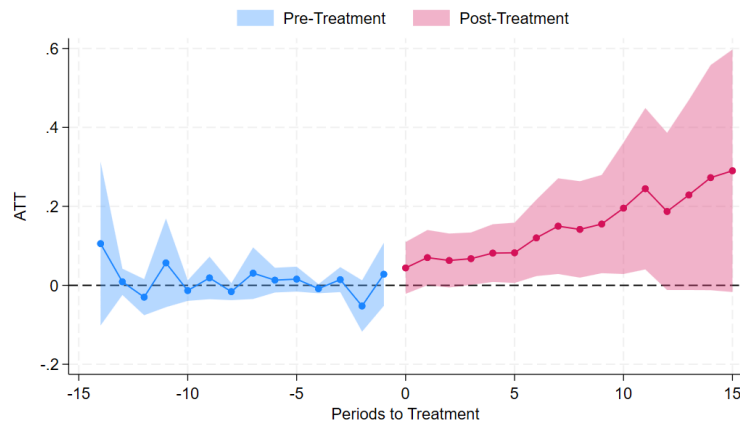
Notes: The figure reports the event study using the longer event window. The dependent variable is the total number of modern banks. The longer window shows a similar post-treatment increase.

Figure D.16: Missionary Presence and Modern Banks: Long Gaps in the Pre-Period



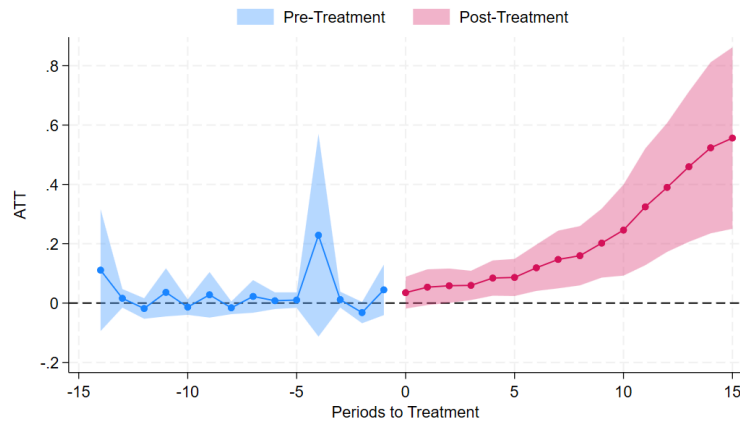
Notes: The figure reports the event study allowing for longer gaps in the pre-period. The exercise checks that the result is not an artifact of how long pre-treatment gaps are handled.

Figure D.17: Missionary Presence and Modern Banks: Number of Railways as Control



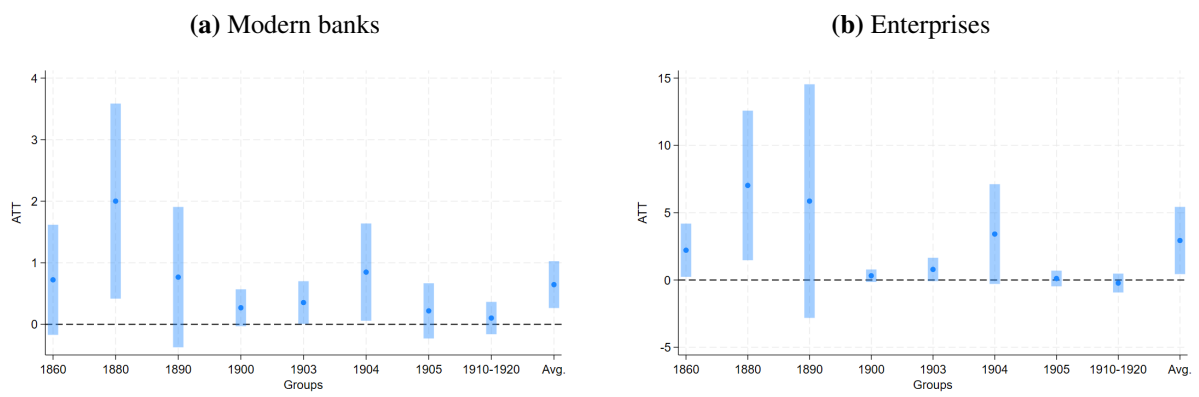
Notes: The figure reports the baseline event study after adding the number of railways as a control. The result is similar, suggesting that railway expansion is not driving the historical DiD pattern.

Figure D.18: Missionary Presence and Modern Banks: Balanced Pre/Post Window



Notes: The figure reports the event study using a balanced number of pre- and post-treatment periods. The pattern remains similar after imposing the balanced event window.

Figure D.19: ATT by Cohort Group



Notes: The figure reports average treatment effects by treatment-cohort group for modern banks and enterprises. The estimates do not point to the result being driven by a single treatment cohort.

E Mechanisms

E.1 Historical Narratives of the Mechanism

Local embeddedness and knowledge translation. This appendix provides brief historical background for the claim in Section 2.1 that mission organizations were locally embedded and could help translate foreign knowledge to the Chinese context. The point is not that missionaries alone possessed local knowledge, nor that foreign knowledge automatically generated institutional change. The argument is narrower. Mission organizations often combined access to foreign ideas and networks with the local linguistic and social knowledge needed to translate those ideas into forms that Chinese actors could adapt and use.

This adaptive logic was consistent with the late-Qing *ti-yong* idea, associated with Zhang Zhidong: Chinese learning would provide the moral and philosophical foundation, while Western learning could be used for practical development. Missionaries did not create this formulation, but mission-educated Yung Wing and collaborations between Chinese scholars and missionary translators provide examples of this selective adaptation (Spence, 1990).

Language and contextual learning. Language acquisition was often systematic rather than incidental. Mission societies commonly required newly arrived personnel to devote substantial time to Chinese language study before taking on independent responsibilities. In some cases, missionaries were examined in the vernacular before being allowed to lead local work. Training could also include geography, government, religion, etiquette, and other aspects of local society. In Hakka-speaking areas, for example, mission work involved extended language study with Chinese teachers, regular examinations, and the production of dictionaries, translations, and other vernacular materials. These practices made linguistic and contextual knowledge more reproducible across successive cohorts of workers.

Chinese personnel. Mission organizations also relied heavily on Chinese personnel. Chinese pastors, teachers, translators, evangelists, printers, doctors, nurses, and other workers helped link mission institutions to local communities. A Protestant count for 1921 reported 6,636 foreign missionaries and 24,732 paid Chinese workers. The Chinese workforce therefore substantially exceeded the foreign missionary presence.

Women and social reach. Mission organizations were also unusually female. In our linked person data, 54 percent of named missionary personnel are women. Historical counts show a similar pattern among foreign Protestant missionaries, and they also record a large number of Chinese Bible women. This composition allowed mission organizations to reach a broader audience that would otherwise have been less accessible.

E.2 Missionary Interview Evidence

As additional qualitative evidence, we reviewed 44 interviews with Christian missionary workers in China held in the Claremont Colleges Special Collections. To summarize the material systematically, we also coded the short archival summary associated with each interview into broad topics. The counts

below are based on those summaries, while the selected passages and photographs draw on the underlying interview materials. This exercise is descriptive and does not enter the causal analysis, but it helps discipline the historical interpretation of the education mechanism.

Table E.1: Topics in Missionary Interviews

Topic	% of archival summaries mentioning
Economic contribution to local economy	18%
Banking link: banks, credit, or bookkeeping	9%
Enterprise link: firms or entrepreneurship	7%
Mentions mission schools	62%
Curriculum: modern subjects, science, math, or English	27%
Girls' education	16%

Notes: The table reports preliminary AI-assisted topic coding of short archival summaries associated with 44 interviews. The underlying interviews are from the Claremont Colleges Special Collections.

Selected interviews. Three interviews illustrate how mission education could build skills valued in modern organizations:

- **Glenn Fuller** arrived in China in 1921 under Methodist sponsorship to establish business-administration training. In 1922, he opened a high-school-level program at Hwei Wen Middle School in Tientsin. He reported that most graduates entered Western firms and several joined the Bank of China or Central Bank of China, a direct link between mission training and modern banking.
- **Francis Price Jones** served in China from 1915 to 1950 as a Methodist educational missionary. A middle-school principal, Nanking professor, and editor of more than forty books in Chinese, he described an early gap in public education and argued that mission schools' "superior instruction in English" helped them compete with government schools.
- **Katherine Ward** arrived in 1925 and taught English at a mission-supported girls' high school in Nanking from 1928 to 1937. She recalled that government and military leaders, including anti-Christian officials, sent daughters to mission high schools because they saw them as academically stronger. Her account shows demand for girls' schooling beyond Christian families.

Source: China Missionaries Oral History Project, Claremont Colleges Special Collections. Quotations are shortened from the interview transcripts.

Figure E.1: Mission Education: Classroom and College Study

(a) Classroom lecture scene



(b) Lingnan college library, 1920s



Notes: Panel A is a classroom photograph from the China Missionaries Oral History Project materials at The Claremont Colleges Library. Panel B shows the Lingnan University college library in the 1920s. Source for Panel B: Lingnan University Digital Commons, “Library Reading Room in Grand Hall.”

Figure E.2: Mission Education: Adult Workers and Practical Training

(a) West China pastors and women workers, 1937



(b) Outdoor practical-demonstration scene



Notes: Panel A shows West China pastors and women workers in 1937. Panel B is an outdoor demonstration photograph illustrating the practical training described in the mission manuscripts. Sources: F. Olin Stockwell and Louise Hathaway Stanley manuscripts, China Missionaries Oral History Project, The Claremont Colleges Library.

Figure E.3: American Presbyterian Mission Press in Shanghai



Notes: The photograph shows the front building and staff of the American Presbyterian Mission Press, reproduced from McIntosh (1895). The press began in Macao in 1844, moved to Ningbo, and relocated to Shanghai in 1860. It combined Chinese typefounding, typesetting, editing, mechanical printing, binding, sales, and distribution under one organization and trained Chinese printers in these activities. Its personnel and skill network later helped seed the Chinese-owned Commercial Press. The Commercial Press's four founders (Xia Ruifang, Bao Xian'en, Bao Xianchang, and Gao Fengchi) came out of the Presbyterian Qingxin school and Huamei Press printing ecosystem, and the firm's official history describes its early operation as deliberately modeled on Huamei Press (Lu, 2017). This is evidence of skill and organizational diffusion rather than missionary ownership of the Commercial Press.

E.3 Additional Historical Examples

- Zifeng Lin, the founder of Kiawah Savings Bank (It's now known as China CITIC Bank International Limited). He studied at the Christian-founded Baptist Jieyang Truth High School, Shantou Queshi High School and the Peking Union Medical College. He became a Christian at the age of 14. He was recruited because of his knowledge of English and the West and was sent to Hong Kong to help develop the local market for the Rice Company. He then recognized the opportunity to expand his banking business.
- The secondary school curriculum of the Tengchow College in Shandong recorded that in the sixth year of students' enrolment, the school would offer a course on the "policy of enriching the country", which taught Western economics. Calvin Wilson Mateer, founder of Tengchow College, argues that the Chinese are eager to learn what makes the West so powerful in their quest to make China rich and strong, and that studying economics is necessary.
- Yuan Cudeng (1874 - 1954), formerly known as Xianan, a native of Yin County, Zhejiang Province, affiliated with the Anglican Church and the YMCA, was a famous entrepreneur in Shanghai's commerce and industry. When he was young, his mother worked as a maid in the house of the British missionary Hai He De in the Ningbo Christian Association (the predecessor of the Methodist Church). Mrs. Hai sent Xian An to the Kai Ming Street Fidelity School, which was run by the Methodist Church, to study free of charge. Two years later, she also sent him to Fidei Secondary School on the north bank of the river, where he was baptized as a believer in Christ.
- Kung Hsiang-hsi, a famous banker and politician in the Republic of China, suffered from mumps in 1889, and was cured by Ding Jiali's Renjiu Hospital in Taigu. 1890, Kung was enrolled in the Chinese-American Public School run by the church, and after graduation, he went to the Luhe Academy run by the American Congregationalist Church of North China in Tongzhou, Zhili, and was baptized as a Christian in 1894.
- Feng Yuxiang, a Christian general and one of the leaders of the Kuomintang (Nationalist Party) in the Republic of China era, began to believe in Christianity in the second half of March 1918 when he was stationed at Changde with his troops, and under the influence of the doctor who treated him, he often went to church to listen to sermons by the pastor. In April 1925, he founded the Northwest Bank in Zhangjiakou, Hebei Province.
- Kesheng Xu, born in Shanghai to a Supervisory Board Christian family. In his early years of business in Shanghai, he recognized the market potential of the emerging insurance industry and participated in the founding of Zhaotai Water and Fire Insurance Company, one of the earliest private insurance companies in China.
- Charles Sung, the father-in-law of Dr. Sun Yat-sen, President of the Republic of China, is engaged in church-related businesses, having established the Meihua Bookstore, specializing in the printing of printed Bibles, and a flour mill, among others.

E.4 Empirical Evidence on the Mechanism

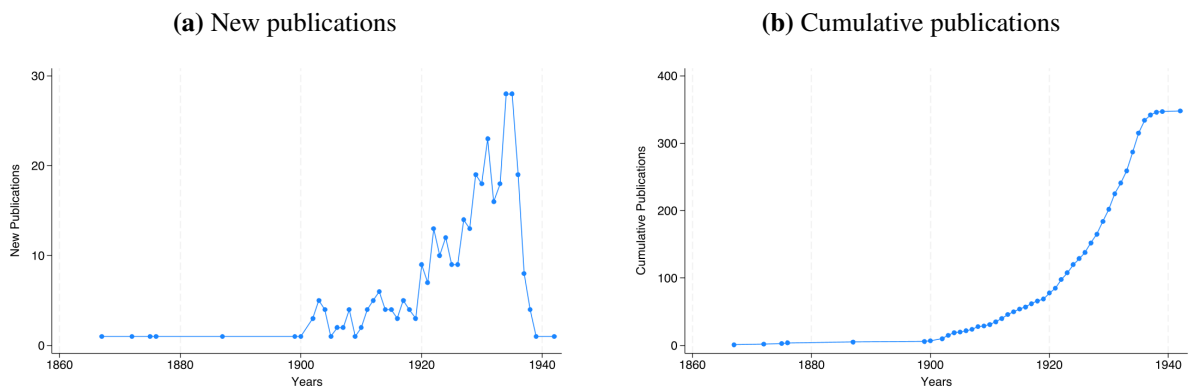
Publication descriptives, coverage, and school complementarity.

Table E.2: Features of Missionary Publications

Publication feature	Freq.	Percent
Religious / missionary focus	276	67.0%
Vernacular / romanized Chinese	182	42.3%
High circulation (>900)	147	34.2%
Western language present	122	28.4%
English present	65	15.1%
Educational / student-oriented	68	16.5%
High frequency (24+ issues per year)	60	14.0%
Bilingual (Chinese + Western)	33	7.7%
Scientific / technical	24	5.8%

Notes: The table reports descriptive features of missionary publications. Percentages use publications for which the relevant feature is classified; categories are not mutually exclusive.

Figure E.4: Missionary Publications Over Time



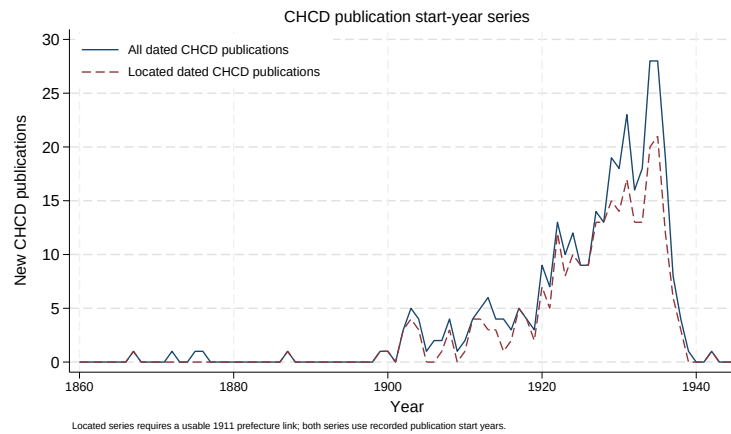
Notes: Panel A plots the number of new missionary publications by start year. Panel B plots the cumulative number of missionary publications over time.

The located CHCD file contains 677 publication records, with usable links spanning 74 historical prefectures. These data are not intended to be a universal bibliography of missionary publishing in China. Most of the underlying publications come from Lowenthal’s Catholic and Protestant religious-periodical tables. The records therefore cover newspapers, magazines, journals, bulletins, newsletters, and related religious or educational periodicals, but may miss smaller titles, some early publications, and other print output. McIntosh provides a complementary source on mission-press infrastructure rather than another title list. This limitation matters more for counting the complete flow of titles than for the mechanism exercise, which uses the geography of any recorded publication or press activity.

Among the 348 CHCD publications with a recorded start year, 268 also have a usable 1911-prefecture link. Figure E.5 shows that the located series is lower, especially in the 1920s and 1930s, but follows the same broad time pattern. Because the heterogeneity split does not require publication timing, our

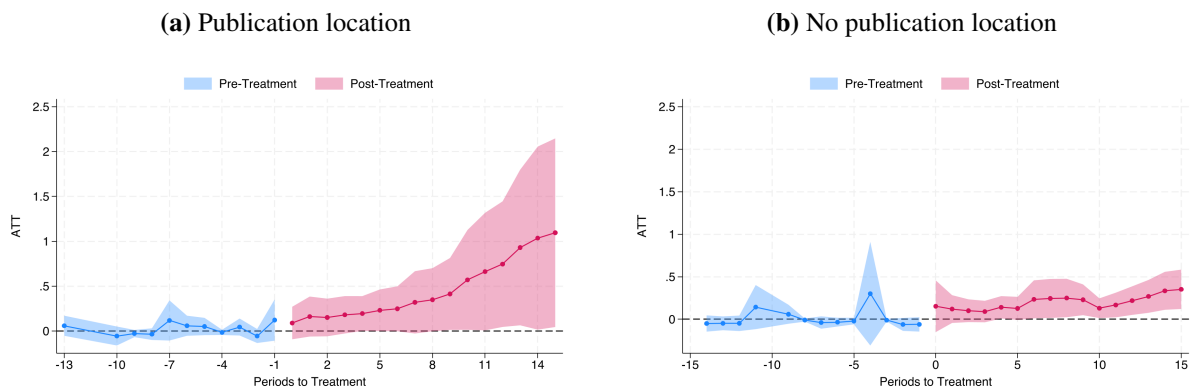
preferred geographic measure retains all located publication records even when start year is missing. This classifies 61 treated prefectures as publication locations; adding McIntosh mission-press locations raises the total to 62. Figure E.6 shows that the publication-only result is nearly identical to the combined measure in Figure 12.

Figure E.5: Coverage of Dated CHCD Publications



Notes: The solid series uses all 348 CHCD publications with a recorded start year. The dashed series uses the 268 publications that also have a usable 1911-prefecture link. The comparison describes coverage within the CHCD source; it does not establish that CHCD is a complete bibliography of missionary publications.

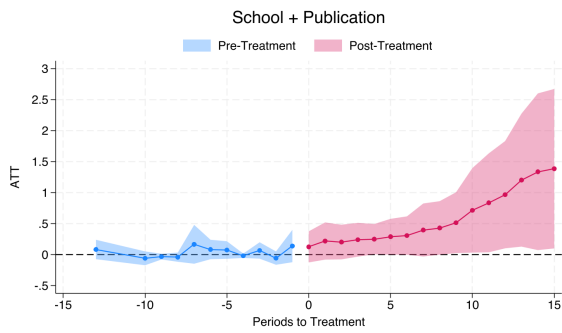
Figure E.6: Publication-Geography Robustness Without a Start-Year Requirement



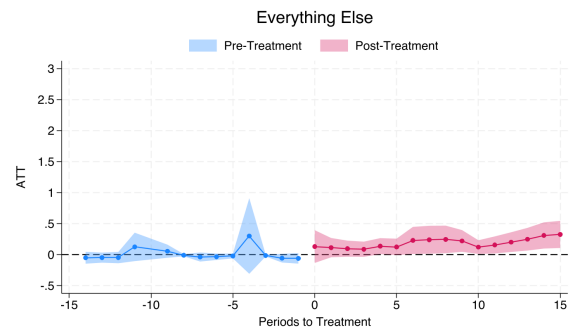
Notes: The outcome is the total number of modern banks. The left panel reports treated prefectures with any located CHCD publication record, regardless of whether its start year is known; the right panel reports the other treated prefectures. Split-group sizes are 61 and 171 treated prefectures, respectively. Figure 12 adds McIntosh mission-press locations, which reclassifies only one treated prefecture.

Figure E.7: School and Publication Complementarity

(a) School and publication

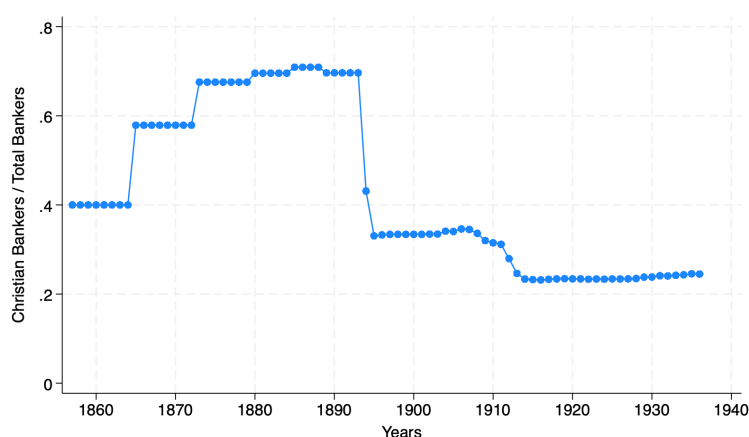


(b) Other treated prefectures



Notes: Split-group sizes: 49 treated prefectures with both a missionary school and missionary publication activity and 183 other treated prefectures.

Figure E.8: Evolution of the Ratio of Total Christian Bankers to Total Bankers



Notes: The figure plots the ratio of Christianity-related bankers to all bankers over time. Changes in the ratio partly reflect increases in the number of non-Christian bankers around the establishment of the first government banks and after the founding of the Republic in 1911.

Heterogeneity magnitudes and difference tests. This appendix summarizes the magnitude comparisons behind the heterogeneity discussion in the main text. The table reports average post-treatment differences between high-exposure and low-exposure groups. The figures then plot coefficient-by-coefficient differences at each event time. These exercises are best read as suggestive heterogeneity checks that compare magnitudes across groups.

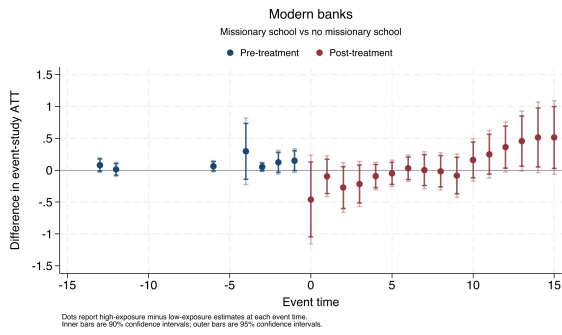
Table E.3: Average Heterogeneity Differences

Outcome	High-exposure group	Low-exposure group	Post diff.
Modern banks	Missionary school	No missionary school	0.063
Modern banks	School and publication	Neither	0.441
Enterprises	Missionary school	No missionary school	1.998**
Enterprises	School and publication	Neither	5.141
Trade-likely enterprises	Missionary school	No missionary school	0.679*
Domestic-likely enterprises	Missionary school	No missionary school	0.006

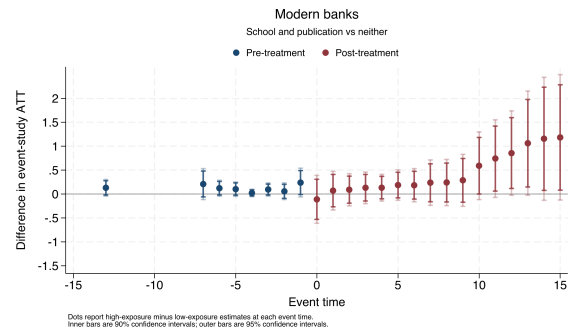
Notes: The table reports high-exposure minus low-exposure differences in average post-treatment event-study estimates. Standard errors for the differences are computed as the square root of the sum of the two group-specific variances; this is conservative because the groups share the never-treated comparison units. Stars denote two-sided significance using this conservative standard error. The table is intended to summarize magnitudes, so it should be read together with the event-time plots in Figure E.9. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Figure E.9: Coefficient-by-Coefficient Heterogeneity Differences

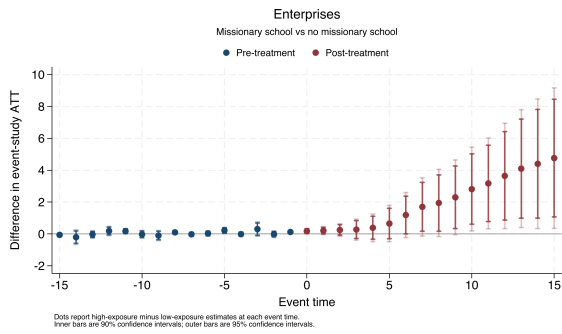
(a) Modern banks: school vs no school



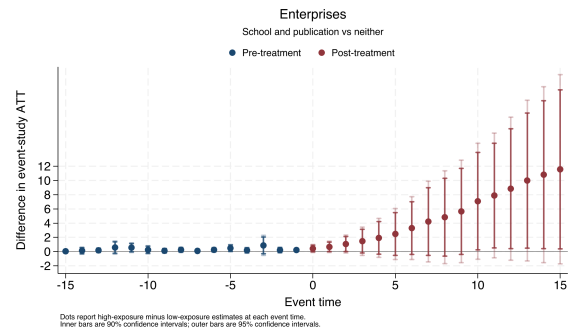
(b) Modern banks: school and publication vs neither



(c) Enterprises: school vs no school



(d) Enterprises: school and publication vs neither



Notes: Split-group sizes: 135 treated prefectures with a missionary school and 97 without a missionary school; 49 treated prefectures with both a missionary school and publication activity and 90 with neither.

Compradors and foreign-bank intermediaries. This section reports descriptive evidence on a possible intermediary-network channel. Compradors were Chinese intermediaries who worked with foreign firms and banks. They could therefore serve as bridges between foreign financial practices and Chinese financial institutions. The evidence in this section is descriptive.

Our data contain comprador information at two levels. First, the foreign-enterprise data include a firm-level flag for whether a foreign firm involved a recorded comprador. Second, the banker biography data include an individual-level flag for whether a Chinese banker had a comprador background. The individual-level data are especially useful because they can be linked to the same banker biographies used in the Christian-banker analysis.

Table E.4: Comprador Data and Descriptive Patterns

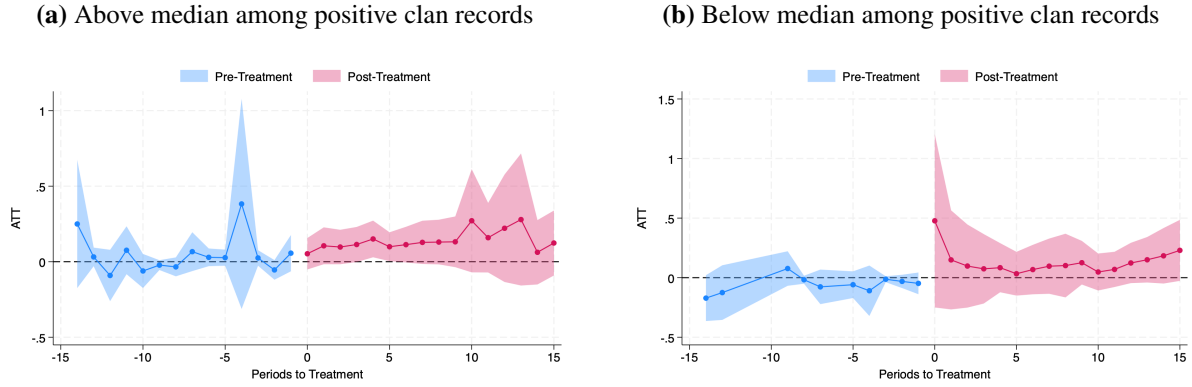
Measure	Value
Share of foreign firms with a recorded comprador	10.9%
Bankers flagged as comprador-related	4.1%
Share Christianity-related among comprador-flagged bankers	28%
Comprador-flagged bankers who founded a Chinese bank, <i>Qianzhuang</i> , <i>Yinhao</i> , or <i>Piaohao</i>	63%
Comprador-flagged bankers who worked at both foreign and Chinese banks	72%

Notes: The table summarizes descriptive evidence from the foreign-enterprise data and the banker biography data. The foreign-firm row reports the share of firms with a recorded comprador. The banker data contains 1,041 bankers; 43 are flagged as comprador-related.

Several patterns are consistent with the idea that compradors could act as intermediaries between foreign and Chinese finance. First, 10.9% of foreign firms in the data had a recorded comprador. Second, Christianity-related bankers constitute a sizable share of comprador-background bankers (28%). Third, most comprador-background bankers worked at both foreign and Chinese banks. Fourth, a large share later founded or invested in Chinese financial institutions. This suggests that compradors may have transmitted institution-specific knowledge from foreign firms and banks into Chinese banking, complementing the main-text education and publication channels.

Clan institutions as an alternative explanation. One alternative explanation is that missionary activity weakened kin-based institutions, which then created space for modern banks. This is relevant because Confucian clan institutions and financial markets may operate as substitutes (Chen et al., 2022b). Missionary activity may also weaken kin-based institutions more generally (Schulz et al., 2019; Henrich, 2020). The exercises below suggest that the bank effect is not simply due to clan substitution.

Figure E.10: Missionary Presence and Modern Banks by Clan Intensity



Notes: The figure reports historical DiD estimates for the effect of missionary presence on the total number of modern banks, separately for prefectures above and below the median of clan intensity among prefectures with positive clan records. Never-treated prefectures are included as controls in both panels. Split-group sizes: 105 prefectures above the median and 101 prefectures below the median.

Table E.5: Missionary Exposure and Clan Intensity: Horse Races*Panel A. Clan records per 100 genealogy books*

VARIABLES	(1) Modern banks in 1936	(2) Modern banks in 1936	(3) Modern banks in 1936	(4) Modern banks in 1936
Missionary Count	0.252*** (0.063)	0.245*** (0.065)	1.634* (0.917)	1.678* (0.952)
Clan Intensity (per 100 books)		0.315 (0.261)		-0.695 (1.080)
Estimator	OLS	OLS	2SLS	2SLS
Clan Control	No	Yes	No	Yes
Railway Controls	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes
Obs.	247	247	247	247
R sq.	0.452	0.453	-0.423	-0.472
Mean Dep. Var.	5.850	5.850	5.850	5.850
S.D. Dep. Var.	13.59	13.59	13.59	13.59

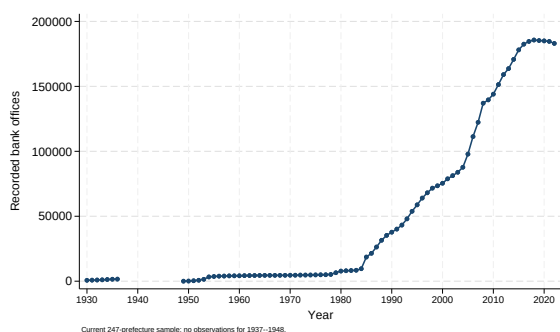
Panel B. Clan records per 10,000 people

VARIABLES	(1) Modern banks in 1936	(2) Modern banks in 1936	(3) Modern banks in 1936	(4) Modern banks in 1936
Missionary Count	0.252*** (0.063)	0.255*** (0.065)	1.634* (0.917)	1.675* (0.932)
Clan per 10,000 People		-1.029 (1.742)		-5.634 (5.134)
Estimator	OLS	OLS	2SLS	2SLS
Clan Control	No	Yes	No	Yes
Railway Controls	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes
Obs.	247	247	247	247
R sq.	0.452	0.452	-0.423	-0.463
Mean Dep. Var.	5.850	5.850	5.850	5.850
S.D. Dep. Var.	13.59	13.59	13.59	13.59

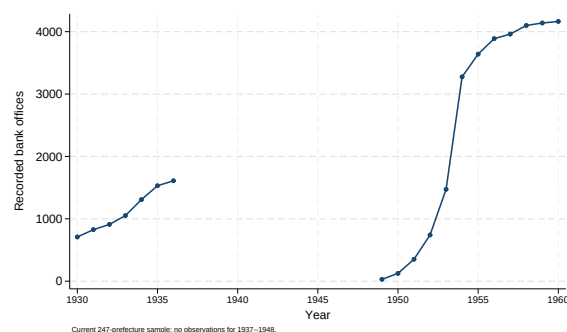
Notes: Cross-sectional regressions for modern banks in 1936. Missionary exposure is the total number of missionary institutions. Panel A scales clan records by genealogy books; Panel B scales them by population. The IV columns use the railway-based missionary-exposure instrument from Section 5. Standard errors are in parentheses; ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

Figure F.1: Raw Bank-Office Dynamics, 1930–1936 and 1949–2022

(a) All Years

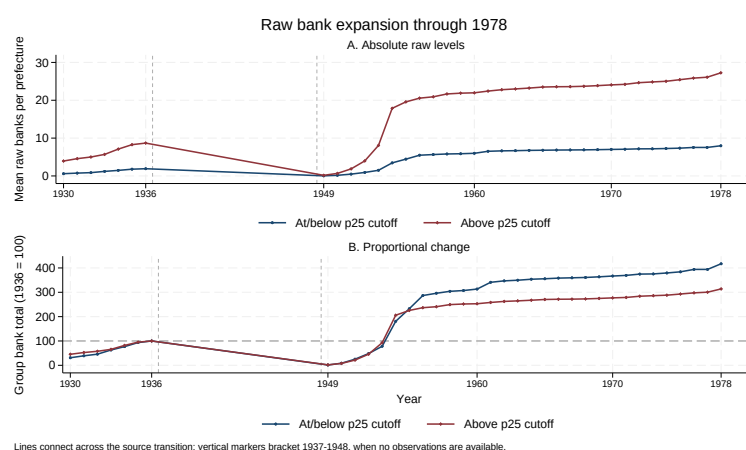


(b) 1930–1960



Notes: The figure sums raw recorded bank-office counts over the same 247 prefectures used by the baseline IV. Recorded bank offices rise from 1,610 in 1936 to 4,164 in 1960, an increase of 159%. The series contains no observations from 1937 to 1948.

Figure F.2: Raw Bank Expansion Through 1978 in Absolute and Proportional Terms



Notes: Panel A reports mean raw banks per prefecture. Panel B indexes each group's raw bank total to 100 in 1936. The lower-exposure group contains prefectures at or below the empirical 25th percentile of missionary exposure; because ties at four missions are retained, it includes 79 of 247 prefectures. The comparison group contains the remaining 168 prefectures. Lines connect the 1936 and 1949 observations, while vertical markers bracket 1937–1948, when no observations are available.

F Long-Term Analysis

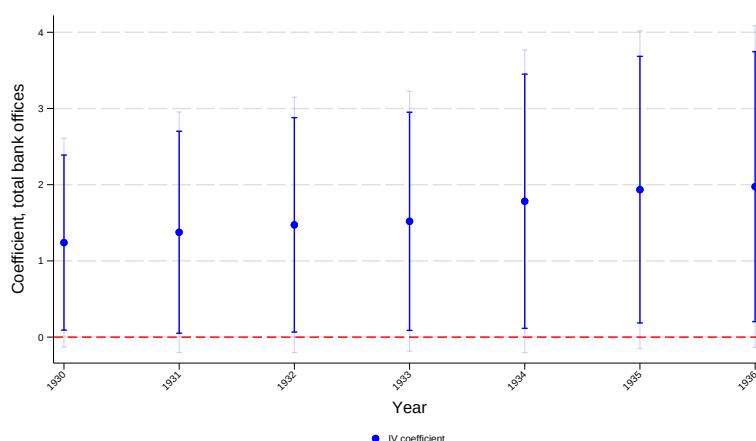
F.1 Bank Dynamics and Headline Estimates

Table F.1: IV Estimates of Mao-Period Bank Changes

Outcome	Start point	Endpoint	IV coefficient	Robust SE	p	Implied change
Raw bank change	1936	1960	1.836	2.957	0.535	
Log bank change	1936	1960	−0.077*	0.045	0.086	−7.4%
Raw bank change	1936	1978	2.216	3.190	0.488	
Log bank change	1936	1978	−0.066*	0.039	0.089	−6.4%

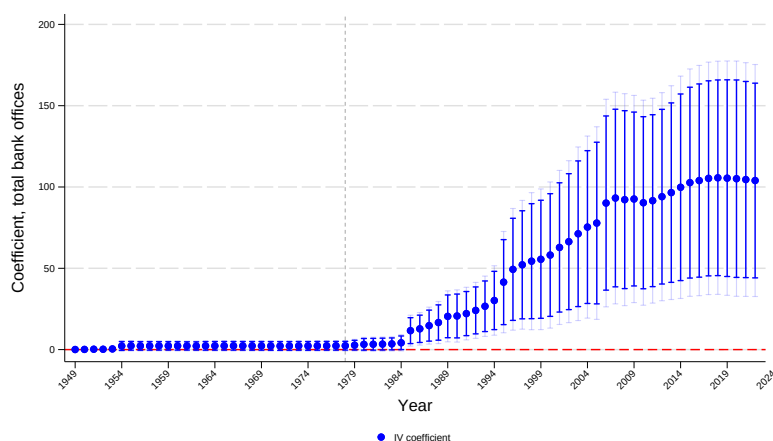
Notes: The table reports 2SLS estimates in the 247-prefecture baseline-IV sample. Missionary exposure is instrumented with predicted missionary exposure from the railway-flow instrument. The raw outcome is the endpoint count minus the 1936 count. The log outcome is the difference in $\ln(1 + \text{banks})$. All regressions include the baseline railway controls and province fixed effects and use robust standard errors. The Kleibergen–Paap first-stage F-statistic is 14.87 in every row. The implied percentage change is $100[\exp(\hat{\beta}) - 1]$. * indicates significance at the 10% level.

Figure F.3: 2SLS Estimates: Total Modern Banks, 1930–1936



Notes: The figure reports coefficients from separate year-specific regressions based on Equation 5, using the historical modern-bank data from the difference-in-differences design. The outcome and all IV-side variables use the 1st–99th percentile convention. The average post-treatment ATT in the historical design is 0.22. The dependent-variable means are 2.4 in 1930 and 5.8 in 1936. The IV estimates correspond to increases of approximately 51% and 34%, respectively. Dark and light intervals show 90% and 95% confidence intervals.

Figure F.4: 2SLS Estimates: Total Banks, 1949–2022



Notes: The figure reports coefficients from separate year-specific regressions based on Equation 5. Modern county- and district-level bank counts are assigned to 1911 prefectures by each modern unit's centroid, and the annual outcome and all IV-side variables use the 1st–99th percentile convention. Mean total-bank counts are 145 in 1990 and 707 in 2022. The 1990 and 2022 estimates correspond to increases of approximately 14.2% and 14.7%, respectively. Dark and light intervals show 90% and 95% confidence intervals. Source: China Financial Supervision and Administration Bureau.

Table F.2: Headline IV Estimates

	(1) Std. modern banks 1936	(2) Std. total banks 2020	(3) GDP 2019
Missionary institutions	0.152* (0.083)	0.145*** (0.051)	7,581*** (2,745)
RR Duration	-0.0125 (0.0095)	-0.0046 (0.0065)	-304.1 (330.5)
RR Density	105.5** (48.0)	47.50 (32.14)	1,972,945 (1,574,214)
Province FE	Yes	Yes	Yes
Observations	247	247	242
K–P F-stat	14.87	14.87	15.01
Mean dep. var.	0.000	0.000	46,051
SD dep. var.	1.000	1.000	52,206

Notes: The table reports 2SLS estimates from Equation 5. Treatment is total missionary institutions, instrumented with predicted missionary exposure from the railway-flow instrument. Column (1) uses standardized modern banks in 1936, the last pre-war year in the historical panel. Column (2) uses standardized 2020 bank counts constructed by assigning modern counties and districts to 1911 prefectures by their centroids. Column (3) uses 2019 GDP in millions of 2017 US dollars, constructed by summing 1-km grid cells directly within accepted 1911 prefecture polygons. The mission stock, instrument, railway controls, and outcomes use the 1st–99th percentile convention. RR Duration is years connected to the railway network as of 2000. RR Density is cumulative railway stock divided by prefecture area. Robust standard errors are in parentheses. K–P F-stat is the Kleibergen–Paap first-stage F-statistic. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

F.2 Population Controls and Per-Capita Outcomes

Population adjustment produces a similar long-run bank result. Table F.3 adds two alternative population measures to both stages of the IV specification or uses population to construct banks-per-capita outcomes. The estimated effect remains positive across all four specifications. Our preferred population-control and per-capita checks use the same county-centroid Census 2015 scale, and Figure F.5 shows the corresponding year-by-year per-capita estimates. The post-reform coefficients remain positive and are statistically distinguishable from zero in recent years.

These checks are not the baseline specifications because population need not be predetermined with respect to missionary exposure. Missions and the institutions that followed them may have affected subsequent population growth, migration, or urban concentration. Controlling for later population can therefore condition on a potential outcome of treatment, while dividing by it changes the estimand. We report both approaches as robustness checks rather than treating population adjustment as conceptually neutral.

Table F.3: Long-Run Bank Results with Population Controls and Per-Capita Outcomes

	Population controls		Per-capita outcomes	
	1880 population	Census 2015 county-centroid scale	1880 denominator	Census 2015 denominator
Missionary institutions	0.146** (0.069)	0.143* (0.085)	0.042 (0.027)	0.052** (0.024)
Outcome	Standardized 2020 bank presence		Standardized banks per million	
Province fixed effects	Yes	Yes	Yes	Yes
Railway controls	Yes	Yes	Yes	Yes
Observations	247	231	247	231
First-stage F-statistic	8.59	4.18	14.87	15.67

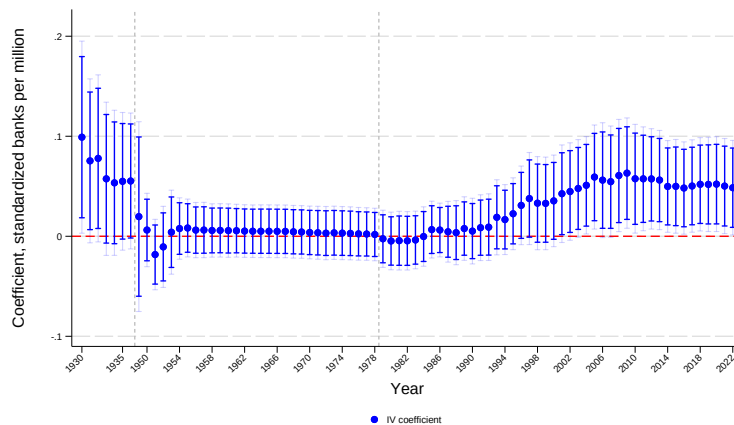
Notes: The table reports IV estimates from Equation 5. The 2020 bank outcome assigns modern county- and district-level bank counts to historical prefectures according to the modern unit's centroid. The mission stock, instrument, railway controls, and bank outcomes use the 1st–99th percentile convention. The first two specifications add the indicated log population measure or scale to both stages. The last two use bank counts per million denominator units and standardize the resulting outcome. The 1880 measure is the historical baseline population. The Census 2015 measure is constructed by summing normalized person weights within modern counties and districts and assigning each modern unit to its historical prefecture by its centroid; it is therefore a weighted relative population scale rather than a literal full-population count. All specifications include province fixed effects and the railway controls used in the baseline IV design. Robust standard errors are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table F.4 applies the same four-specification structure to the direct-grid GDP outcome in 2019. The estimated effect is positive under every population adjustment. It is significant at the 5% level when 1880 population is added as a control and when GDP per capita uses either denominator, and at the 1% level when GDP per capita uses the Census 2015 denominator.

Table F.4: Long-Run GDP Results with Population Controls and Per-Capita Outcomes

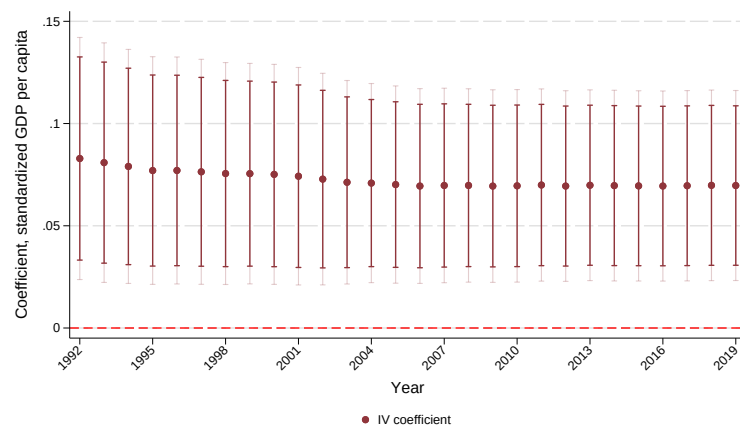
	Population controls		Per-capita outcomes	
	1880 population	Census 2015 county-centroid scale	1880 denominator	Census 2015 denominator
Missionary institutions	8,358** (3,922)	8,190 (5,049)	0.055** (0.025)	0.070*** (0.024)
Outcome	GDP, 2019 (millions of 2017 USD)		Standardized GDP per million	
Province fixed effects	Yes	Yes	Yes	Yes
Railway controls	Yes	Yes	Yes	Yes
Observations	242	231	242	231
First-stage F-statistic	8.64	4.18	15.01	15.67

Notes: The table reports IV estimates from Equation 5. The 2019 GDP outcome sums 1-km grid-cell GDP directly within accepted 1911 prefecture polygons. The mission stock, instrument, railway controls, and GDP outcome use the 1st–99th percentile convention. The first two specifications add the indicated log population measure or scale to both stages. The last two express GDP per million denominator units and standardize the resulting outcome. The 1880 measure is the historical baseline population. The Census 2015 measure is constructed by summing normalized person weights within modern counties and districts and assigning each modern unit to its historical prefecture by its centroid; it is therefore a weighted relative population scale rather than a literal full-population count. All specifications include province fixed effects and the railway controls used in the baseline IV design. Robust standard errors are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Figure F.5: 2SLS Estimates: Banks Per Capita, 1930–1936 and 1949–2022

Notes: The figure reports year-specific estimates from Equation 5. Modern county- and district-level bank counts are assigned to 1911 prefectures by each modern unit's centroid. The outcome is bank counts per million county-centroid Census 2015 scale units, winsorized at the 1st and 99th percentiles and standardized within year; the same Census 2015 denominator is used in every year. Blue points are IV estimates, and dark and light intervals show 90% and 95% confidence intervals. The red dashed horizontal line marks zero, and gray vertical dashed lines mark 1949 and 1978. The coefficient is 0.052 in 2020 ($p = 0.032$) and 0.049 in 2022 ($p = 0.045$); both estimates use 231 historical prefectures.

Figure F.6: 2SLS Estimates: GDP Per Capita, 1992–2019



F.3 Industrial Enterprises and External Finance Dependence

We complement the bank and GDP results with industrial-enterprise outcomes from the Annual Survey of Industrial Production (ASIP). The outcome is the number of industrial enterprises observed in the survey, not sales, employment, output, borrowing, or the universe of registered firms. We report all ASIP enterprises and private ASIP enterprises, where private enterprises are identified using the survey’s paid-in-capital ownership classification. The yearly outcomes are expressed per million residents and standardized within year before estimating Equation 5.

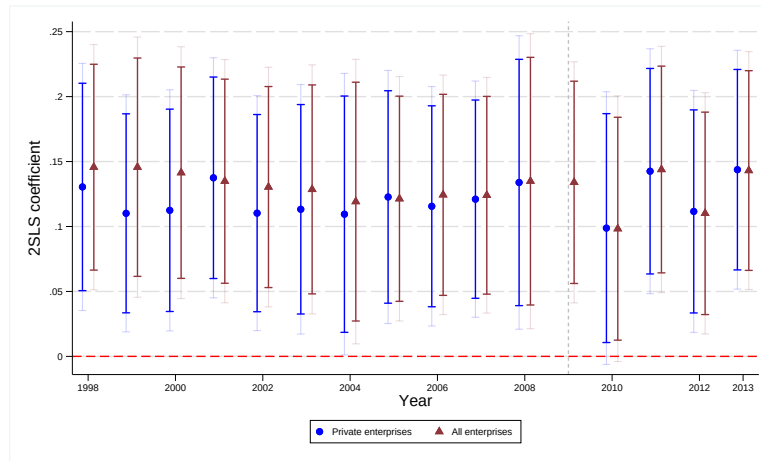
To study whether the enterprise response is related to financial intermediation, we construct an industry-level measure of external finance dependence following the logic of [Rajan and Zingales \(1998\)](#). For firm i in three-digit industry j , we first calculate

$$\text{EFD}_{it} = \frac{\text{Investment}_{it} - \text{Operating Cash Flow}_{it}}{\text{Investment}_{it}}$$

over 2004–2006, requiring positive investment. Investment is the ASIP real-investment field. Operating cash flow is the direct operating-cash-flow field when available and operating cash inflows minus outflows otherwise. We then take the median of this ratio across firm-year observations within each three-digit industry. “High finance dependence” denotes industries at or above the median of these industry values; “low finance dependence” denotes industries below the median. This classification is therefore a predetermined industry characteristic measuring the need for outside finance, not a firm-level measure of realized borrowing.

We assign geocoded ASIP firm locations directly to accepted 1911 prefecture polygons. For records without a usable direct match, we use the modern county or district centroid and then an area-weighted modern-city crosswalk as a final fallback. Firm counts are aggregated to historical prefecture-year cells before the per-capita transformation, 1st–99th percentile winsorization, and within-year standardization.

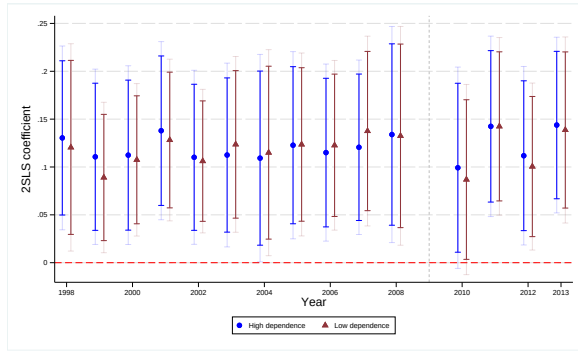
Figure F.7: Historical Missionary Exposure and Industrial Enterprises



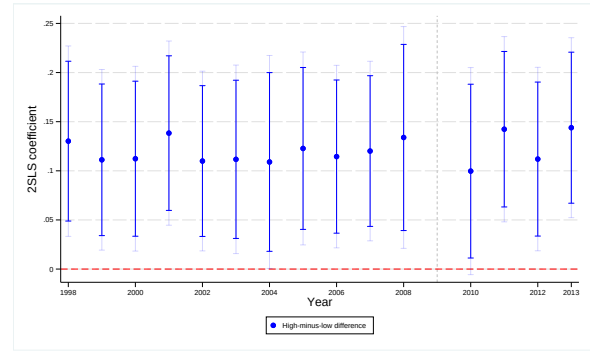
Notes: The figure reports year-specific 2SLS estimates from Equation 5. Geocoded ASIP firms are assigned directly to accepted 1911 prefectures, with county-centroid and area-weighted city fallbacks when a direct match is unavailable. Outcomes are private and all ASIP industrial enterprises per million residents, winsorized at the 1st and 99th percentiles and standardized within year. Dark and light intervals show 90% and 95% confidence intervals. In the 2007–2013 period-average specification, the coefficient is 0.128 standard deviations for private enterprises and 0.129 for all enterprises. The corresponding raw private-enterprise estimate is approximately 40 firms per million residents, or 15.8% of the sample mean. The private-enterprise outcome is omitted in 2009 because the ASIP ownership field mechanically assigns every observation to the same category in that year; the all-enterprise outcome remains defined.

Figure F.8: Historical Missionary Exposure and Private Enterprises by External Finance Dependence

(a) High- and low-dependence industries



(b) High-minus-low difference



Notes: The figure reports year-specific 2SLS estimates from Equation 5. Geocoded ASIP firms are assigned directly to accepted 1911 prefectures, with county-centroid and area-weighted city fallbacks when a direct match is unavailable. The outcomes are private ASIP industrial enterprises per million residents in high- and low-external-finance-dependence industries, winsorized at the 1st and 99th percentiles and standardized within year. Panel B directly estimates the high-minus-low difference. Dark and light intervals show 90% and 95% confidence intervals. In the 2007–2013 period-average specification, the high-minus-low estimate is 0.128 standard deviations; the corresponding raw estimate is approximately 37 private enterprises per million residents, or 15.8% of the sample mean of the difference outcome. The private-enterprise outcome is omitted in 2009 because of the ASIP ownership-coding anomaly described in Figure F.7.

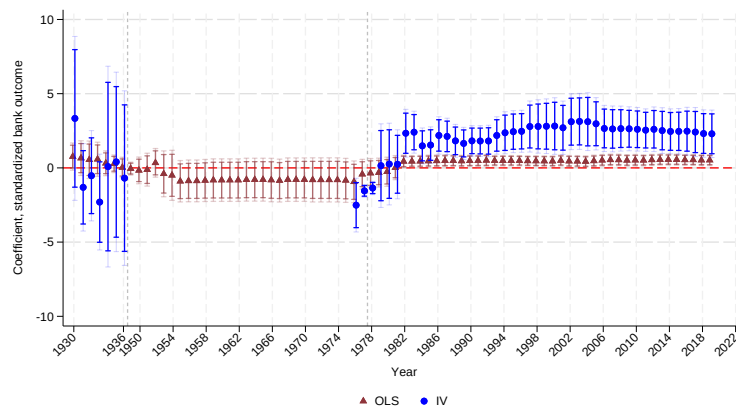
Together, the enterprise results indicate that the long-run effect is not limited to the number of banks. Missionary-exposed prefectures have more later industrial activity, including private industrial enterprises. The high-minus-low finance-dependence outcome is also positive, although the independently standardized high- and low-dependence coefficients are similar in magnitude. We interpret this as complementary evidence consistent with a finance channel, while recognizing that ASIP covers surveyed industrial enterprises above the applicable size threshold.

F.4 Long-Run Mechanism Evidence

This section reports the detailed evidence summarized in Section 6. The exercises use several units of observation and therefore cannot be reduced to one identical regression. Wherever the unit is a prefecture, we retain the baseline long-run specification: railway controls and province fixed effects in OLS, with historical missionary exposure instrumented by predicted missionary exposure in IV. Individual-level exercises add the fixed effects and controls appropriate to the underlying biographies or surveys. Shaanxi exercises use counties nested in 1911 prefectures, absorb historical-prefecture fixed effects, and instrument pre-1950 mission presence with predicted missionary exposure. We report these results as a structured set of mechanism checks rather than as independent causal estimates.

F.5 Shaanxi County-Level Long-Run Replication

Figure F.9: Historical Missionary Exposure and Long-Run Financial Institutions in Shaanxi



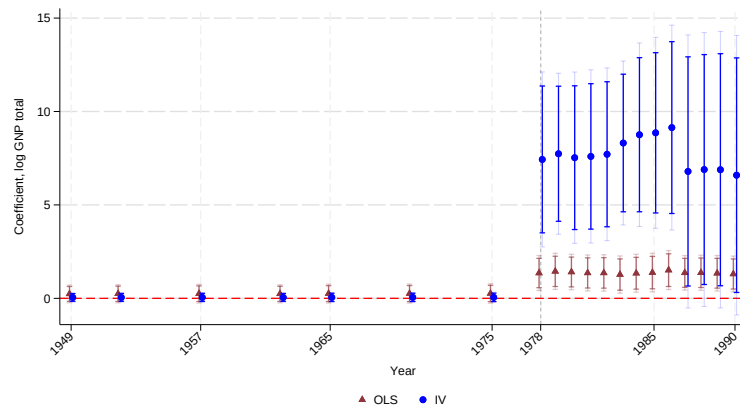
Notes: The figure reports year-specific OLS and IV estimates for standardized modern banks or financial institutions in Shaanxi counties. Blue circles show IV estimates and maroon triangles show OLS estimates; light and dark intervals are 95% and 90% confidence intervals. OLS includes 1911 historical-prefecture fixed effects. IV instruments pre-1950 mission presence with predicted missionary exposure and adds railway duration and cumulative railway-area exposure. Early post-1949 financial-institution coverage is sparse, so Mao-period IV estimates are generally not estimable.

Table F.5: Shaanxi County-Level Bank and GNP Estimates by Period

Outcome	Period	OLS	OLS SE	IV	IV SE	First-stage F	Obs.
Std. financial institutions	1930–1936	0.452	0.407	-0.144	1.966	15.88	30
Std. financial institutions	1949–1976	-0.763	0.665	—	—	—	—
Std. financial institutions	1978–2022	0.430**	0.165	2.047***	0.671	14.93	87
Log GNP	1978–1990	1.370***	0.460	7.711***	2.708	15.00	89

Notes: Outcomes are averaged within county over the indicated period. The financial-institution outcome is standardized within year; log GNP is $\log(1 + \text{GNP})$ and uses 1978–1990, the years with broad county coverage. Specifications follow Figures F.9 and F.10. The Mao-period financial-institution IV is not estimable because early post-1949 coverage is too sparse after fixed effects and railway controls. The historical financial-institution series records cumulative establishments rather than a complete active-stock panel. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

Figure F.10: Historical Missionary Exposure and Log GNP in Shaanxi, 1949–1990



Notes: The figure reports year-specific OLS and IV estimates for log total GNP in Shaanxi counties. Specifications, markers, and confidence intervals follow Figure F.9; the dashed vertical line marks 1978. Broad county-level GNP coverage begins in 1978, so the earlier coefficients are retained for continuity but should be interpreted descriptively.

F.6 Returning Talent and Elite Networks

Table F.6: Mission-Exposed Origins and the Technical Diaspora in Taiwan

Origin-place exposure	Any Taiwan record		Observed in Taiwan, 1974 or later	
	(1)	(2)	(3)	(4)
Any mission institution by 1880	2.73* (1.65)		2.45** (1.18)	
Any mission school by 1900		6.23* (3.33)		5.36** (2.37)
Native-province FE	Yes	Yes	Yes	Yes
Historical controls	Yes	Yes	Yes	Yes
Observations	11,657	11,657	11,657	11,657
Origin-prefecture clusters	205	205	205	205
Mean dependent variable	5.1%	5.1%	3.0%	3.0%

Notes: Each observation is an engineer/scientist origin record. Coefficients are percentage-point changes from origin-prefecture OLS regressions. The 1974-or-later outcome is the stricter near-reform Taiwan measure. Historical controls include coastal location, main-river access, treaty-port and provincial-capital status, city size, log population, river length, 1820 tax per capita, *jinshi*, and official counts. Standard errors are clustered by origin prefecture. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

Table F.7: Mission Origins, Foreign Exposure, and Mainland Institutional Roles

	Outcome: mainland or PRC institutional role	
	Mao period	Reform period
Mission-exposed origin	-7.05 (5.08)	-24.10* (14.09)
Foreign exposure	-8.38 (8.12)	-49.63*** (17.16)
Mission-exposed origin $\times$ foreign exposure	14.77* (8.09)	54.64*** (17.33)
Province FE	Yes	Yes
Birth-decade FE	Yes	Yes
Historical controls	Yes	Yes
Observations	6,357	2,510
Origin-prefecture clusters	222	202
Mean dependent variable	65.2%	71.9%

Notes: Coefficients are percentage points. Mission-exposed origin equals one if the elite's birth or origin prefecture contained a mission institution during ages 0–25. Foreign exposure is biographical evidence of study, residence, or work outside mainland China. In the reform-period regression, adjusted mainland-role probabilities are 93.5% for neither exposure, 69.4% for mission exposure only, 43.8% for foreign exposure only, and 74.4% for both. The implied mission association among foreign-exposed elites is 7.72 (SE 6.67) during the Mao period and 30.53 (SE 18.54) during the reform period. In a pooled person-by-period specification, the reform-minus-Mao difference in this full association is 24.68 percentage points ($p = 0.117$), while the corresponding triple interaction is 42.53 percentage points ($p = 0.005$). Historical controls include log population, examination quota, treaty-port status, coast, main river, rebellion, Boxer activity, and 1820 tax per capita. Standard errors are clustered by origin prefecture. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

F.7 Commercial Persistence and Foreign Investment

Table F.8: Historical Missionary Exposure and Foreign Direct Investment

	Log FDI, 2017		Log FDI per capita, 2017	
	OLS	IV	OLS	IV
Mission count	0.066*** (0.011)	0.129** (0.051)	0.018* (0.009)	0.060* (0.034)
Railway controls	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes
Observations	240	240	240	240
R-squared	0.569		0.483	
First-stage F-statistic		14.22		14.22
Mean dependent variable	9.799	9.799	4.306	4.306
SD dependent variable	2.557	2.557	1.727	1.727

Notes: The unit is a 1911 prefecture. Modern-city FDI is assigned to historical prefectures using area-overlap weights. Raw total and per-capita FDI are winsorized at the 1st and 99th percentiles before applying the reported $\log(1 + x)$ transformations. Mission count is the historical number of missionary institutions. IV instruments mission count with predicted missionary exposure; the mission stock, instrument, and railway controls use the same 1st–99th percentile convention. Robust standard errors are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

F.8 Religious and Cultural Change

Table F.9: Historical Missionary Exposure and Contemporary Christian Affiliation

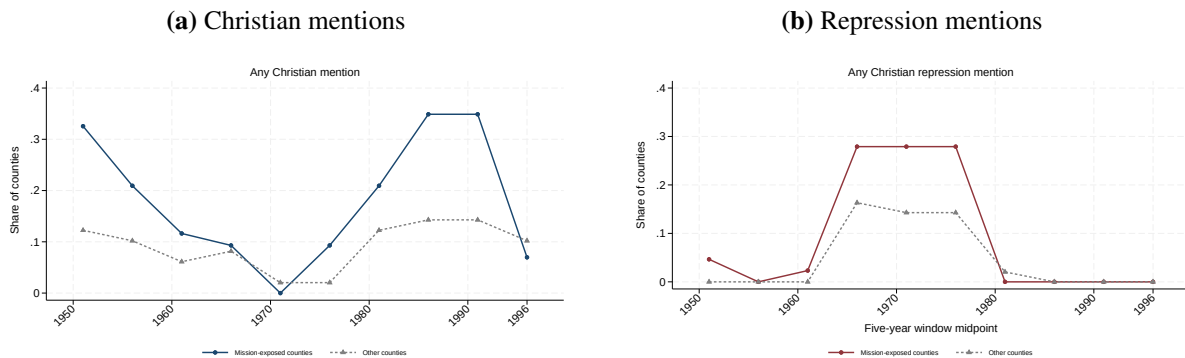
<i>Panel A. China Family Panel Studies</i>			
	Christian belief	Protestant belief	Catholic belief
Mission count	0.0002 (0.0002)	0.0005** (0.0002)	-0.0001 (0.0002)
1911 prefecture FE	Yes	Yes	Yes
Wave FE	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes
Observations	97,509	97,509	97,509
Clusters	122	122	122
Mean dependent variable	0.035	0.028	0.013
<i>Panel B. Chinese Spiritual Life Survey</i>			
	Christian belief	Protestant belief	Catholic belief
Mission count	0.0029* (0.0017)	0.0020 (0.0017)	0.0010 (0.0009)
1911 province FE	Yes	Yes	Yes
Railway controls	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes
Observations	5,660	5,660	5,660
Clusters	40	40	40
Mean dependent variable	0.028	0.026	0.003

Notes: The CFPS unit is a respondent-wave located to a 1911 county; specifications include 1911 prefecture fixed effects, wave fixed effects, and individual controls, with standard errors clustered by 1911 county. The Spiritual Life Survey uses the high-quality geocoded sample; specifications include 1911 province fixed effects, railway controls, and individual controls, with standard errors clustered by assigned 1911 prefecture. Mission count is the historical missionary-institution count divided by 10. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

Table F.10: Historical Missionary Exposure and Contemporary Religiosity

	China Family Panel Studies			Chinese Spiritual Life Survey		
	Religious group member	Religion important	Religious activity frequency	Believes in God	Ever prays	Donated to religion
Mission count	0.0000 (0.0003)	0.0002 (0.0011)	-0.0011 (0.0038)	0.0026 (0.0048)	-0.0038 (0.0065)	0.0020 (0.0076)
Geographic FE	Prefecture	Prefecture	Prefecture	Province	Province	Province
Wave FE	Yes	Yes	Yes	No	No	No
Railway controls	No	No	No	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	67,883	74,561	74,885	5,362	5,660	5,706
Mean dependent variable	0.025	0.218	0.598	0.083	0.114	0.226

Notes: Sample construction, mission-count scaling, and clustering follow Table F.9. The table shows little systematic association between historical missionary exposure and the intensity of contemporary religious practice. Robust standard errors are in parentheses.

Figure F.11: Christianity and Repression in Shaanxi Local Histories

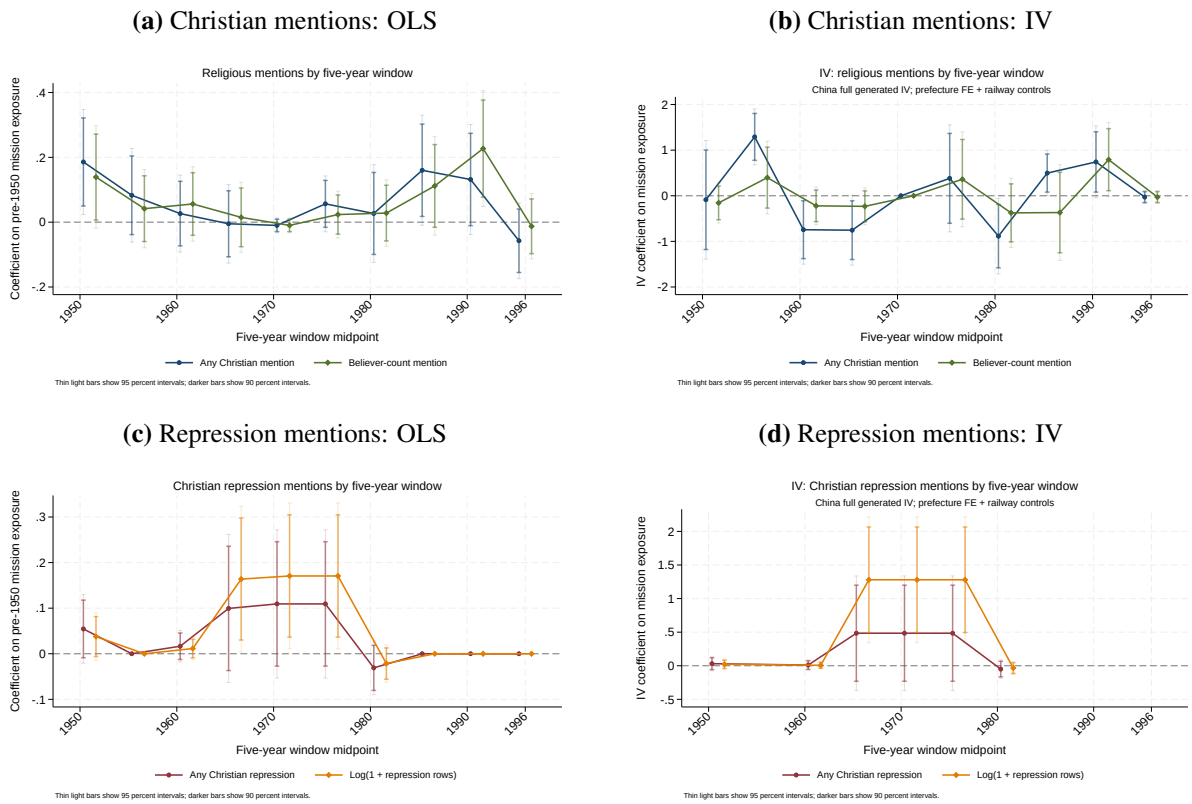
Notes: Split-group sizes: 43 Shaanxi counties with a recorded pre-1950 missionary institution and 49 counties without one.

Table F.11: Historical Missionary Exposure and Trust

	Trust people generally	Trust family	Trust religious groups
Mission count	0.0377** (0.0162)	0.0028 (0.0146)	0.0487 (0.0382)
1911 province FE	Yes	Yes	Yes
Railway controls	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes
Observations	4,389	5,753	2,472
Mean dependent variable	1.781	3.821	1.877

Notes: The data are from the high-quality geocoded Chinese Spiritual Life Survey sample. Mission count is the historical missionary-institution count divided by 10. Standard errors are clustered by assigned 1911 prefecture. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

Figure F.12: Dynamic Estimates for Christian and Repression Mentions in Shaanxi



Notes: Each point is estimated separately by time window. OLS specifications include 1911 historical-prefecture fixed effects. IV specifications instrument the pre-1950 mission-presence indicator with predicted missionary exposure and add railway controls. Bars show 90% and 95% confidence intervals.

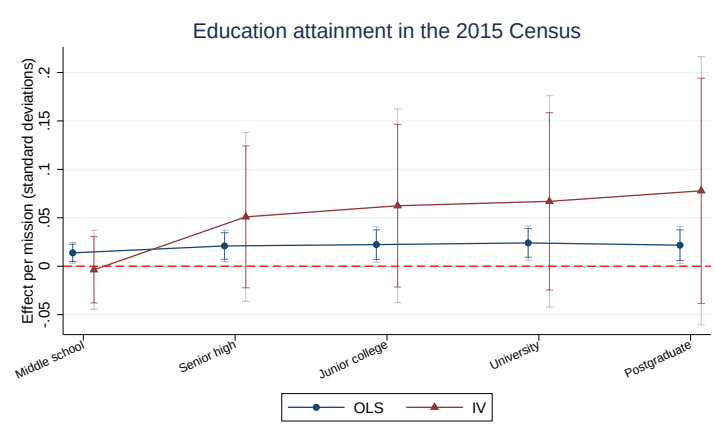
Table F.12: Historical Missionary Exposure and Financial Behavior

	Asset ownership			Risk attitudes and knowledge		
	Has savings	Has financial asset	Has non-deposit asset	Risk tolerance (1–4)	Willing to take investment risk	Knows investment risk
Mission count	0.0020** (0.0010)	0.0019** (0.0009)	0.0051*** (0.0014)	0.0047 (0.0031)	0.0018 (0.0015)	0.0028 (0.0024)
1911 prefecture FE	Yes	Yes	Yes	Yes	Yes	Yes
Wave FE	Yes	Yes	Yes	Yes	Yes	Yes
Individual controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	131,549	130,797	130,797	6,476	6,476	6,525
Mean dependent variable	0.732	0.770	0.215	1.993	0.576	0.856

Notes: The unit is a CFPS respondent-wave located to a 1911 county. Mission count is the historical missionary-institution count divided by 10. Standard errors are clustered by 1911 county. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

F.9 Long-Run Education

Figure F.13: Historical Missionary Exposure and Education in the 2015 Census

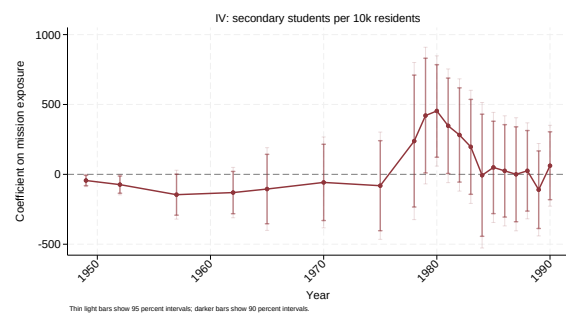
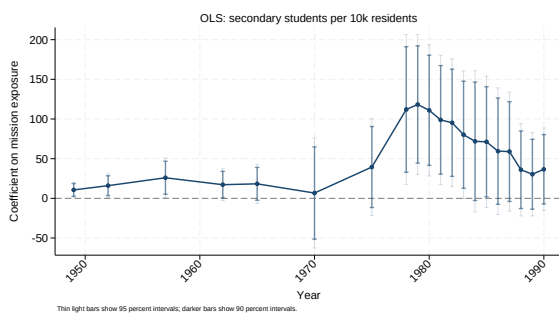


Notes: The unit is a 1911 prefecture and the sample includes adults age 18 and older. Each point comes from a separate regression of a standardized attainment share. Modern counties are assigned using the Census 2015 county-centroid bridge. OLS and IV include province fixed effects and railway controls; IV instruments mission count with predicted missionary exposure. Robust standard errors are used. Dark and light bars show 90% and 95% confidence intervals.

Figure F.14: Historical Mission Exposure and Secondary Students in Shaanxi

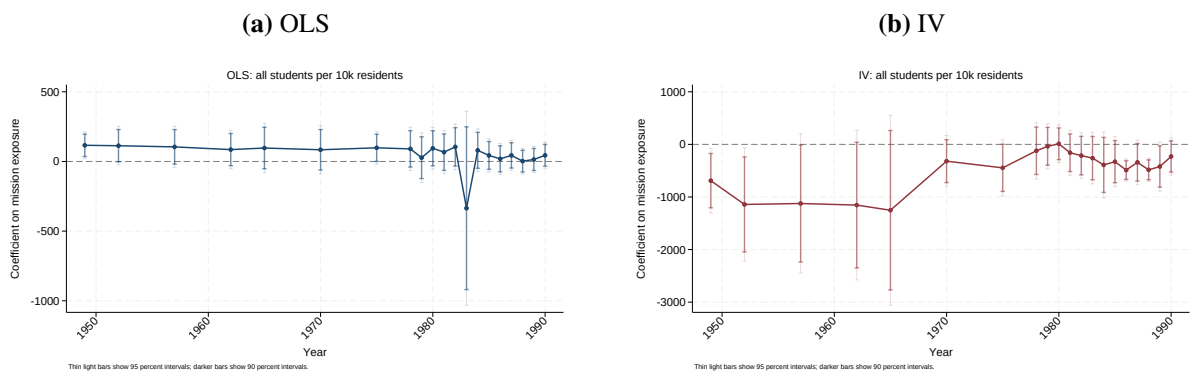
(a) OLS

(b) IV



Notes: The outcome is secondary students per 10,000 residents in the Shaanxi yearbook county panel. OLS absorbs 1911 historical-prefecture fixed effects. IV instruments the pre-1950 mission-presence indicator with predicted missionary exposure and adds railway controls. Bars show 90% and 95% confidence intervals.

Figure F.15: Historical Missionary Exposure and Students in Shaanxi



Notes: The outcome is all students per 10,000 residents in the Shaanxi yearbook county panel. OLS absorbs 1911 historical-prefecture fixed effects. IV instruments the pre-1950 mission-presence indicator with predicted missionary exposure and adds railway controls. Bars show 90% and 95% confidence intervals.

F.10 State Capacity and Infrastructure

Table F.13: Historical Missionary Exposure, State Capacity, and Market Activity in Shaanxi

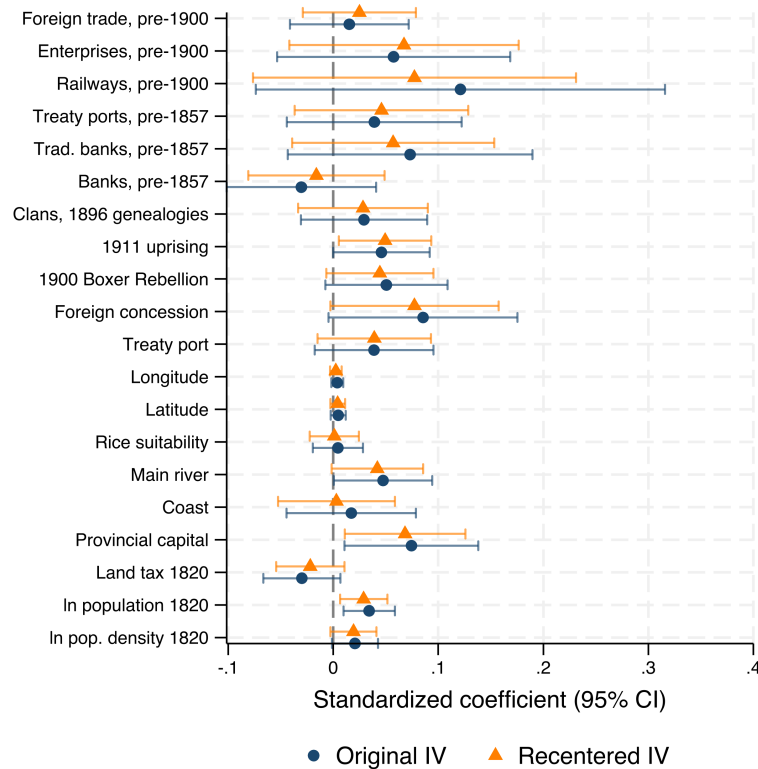
<i>Panel A. Fiscal capacity</i>		
Outcome	1949–1990 IV	1978–1990 IV
Log fiscal expenditure per capita	-1.130*** (0.370)	-0.616*** (0.144)
Log fiscal revenue per capita	-0.845 (0.534)	0.082 (0.286)
Health workers per 10,000 residents	-4.813 (15.494)	-0.294 (17.961)
<i>Panel B. Investment and infrastructure</i>		
Outcome	1949–1975 IV	1978–1990 IV
Log fixed-asset investment per capita	-1.182 (1.231)	-1.509 (1.673)
Log basic-construction investment per capita	-1.167 (1.226)	-1.562 (1.623)
Log machinery power per 10,000 residents	-1.320* (0.735)	-2.327** (0.885)
Log rural electricity per capita	-1.513*** (0.365)	-0.405 (0.431)
Log health institutions per 10,000 residents	-0.385 (0.286)	-0.565* (0.334)
Log health-institution beds per 10,000 residents	-0.726 (0.941)	-0.864 (0.817)
<i>Panel C. Market-facing outcomes</i>		
Outcome	1949–1975 IV	1978–1990 IV
Log industrial enterprise units	1.516*** (0.442)	1.560*** (0.466)
Log state-owned enterprise units	0.275 (0.491)	-1.436 (1.705)
Log retail sales per capita	0.970** (0.421)	0.486*** (0.157)
Log township enterprise employment	–	1.415*** (0.485)
Log township enterprise units	–	1.022* (0.539)

Notes: The unit is a 1911 Shaanxi county. Each cell reports the IV coefficient on pre-1950 mission presence from a separate regression. Mission presence is instrumented with predicted missionary exposure. All specifications absorb 1911 historical-prefecture fixed effects and include railway duration and cumulative railway-area exposure. Panel A uses 89 observations and a first-stage F-statistic of 15.00. Panel B uses 85–88 observations in 1949–1975 and 89 observations in 1978–1990; first-stage F-statistics range from 13.00 to 15.00. Panel C uses 88 observations for the displayed Mao-period outcomes and 89 after reform. Township-enterprise outcomes are available only in the reform period. Robust standard errors are in parentheses. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

F.11 Placebo Outcomes in the IV Second Stage

As an additional diagnostic, we replace the headline bank and GDP outcomes with fixed prefecture characteristics and early historical outcomes. Each point comes from a separate 2SLS regression in which missionary institutions are instrumented with either the original instrument or the full-permutation recentered instrument. The outcome is standardized, and every regression includes railway duration, railway density, and province fixed effects.

Figure F.16: Placebo Outcomes: IV Second-Stage Estimates



Notes: Points report separate 2SLS coefficients on instrumented missionary institutions, with standardized placebo outcomes and robust 95% confidence intervals. Blue circles use the original instrument and orange triangles use the full-permutation recentered instrument. All specifications include railway duration, railway density, and province fixed effects. The figure reproduces the SIOE presentation exhibit exactly and therefore uses its earlier 0.5th–99.5th-percentile instrument construction. The pre-1900 and 1911 outcomes can postdate early missionary exposure and should be read as historical diagnostics rather than uniformly predetermined characteristics.

The instruments are balanced on most of these outcomes when considered individually. We interpret the figure as an outcome-by-outcome placebo diagnostic rather than proof that every historical characteristic is balanced.

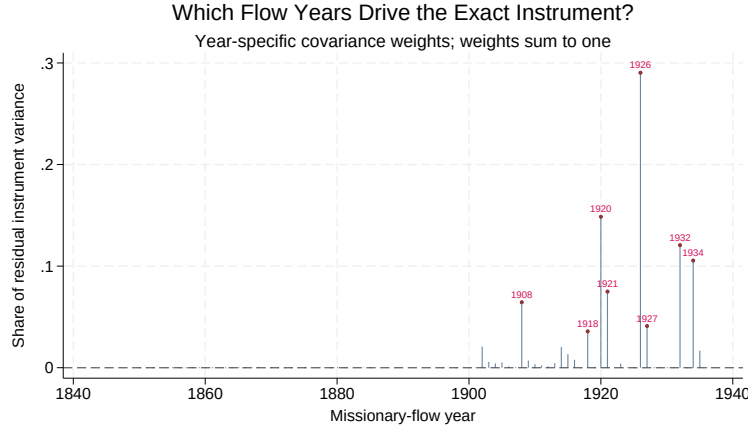
F.12 Year-Specific Contributions and Leave-One-Year-Out Sensitivity

We first examine which annual missionary flows supply the cross-prefecture variation in the exact instrument. Let $c_{it} = \hat{\beta} F_t R_{i,t-1}$ denote year t 's contribution for prefecture i , so that $Z_i = \sum_t c_{it}$. We residualize both c_{it} and Z_i on railway duration, railway density, and province fixed effects. The contribution weight for year t is

$$\omega_t = \frac{\sum_i \tilde{c}_{it} \tilde{Z}_i}{\sum_i \tilde{Z}_i^2},$$

where tildes denote these residualized values. Because the total instrument is the sum of its annual contributions, the covariance weights sum to one. They measure each year's contribution to the identifying variation in the residualized instrument, rather than its share of the raw national missionary flow.

Figure F.17: Year-Specific Contribution Weights in the Exact Instrument



Notes: Bars report the covariance weights ω_t after residualizing the annual contributions and total instrument on railway duration, railway density, and province fixed effects in the 247-prefecture baseline-IV sample. Labels identify the eight years with the largest absolute weights. The weights sum to one; the 1926 flow supplies 29.0% of total absolute weight, the five largest years supply 73.9%, and the ten largest supply 92.2%.

The concentration of these weights does not imply that one flow year determines the IV results. For each of the 59 positive-flow years, we construct $Z_i^{(-s)} = Z_i - c_{is}$ and re-estimate the baseline 2SLS. Table F.14 summarizes all estimates. Omitting 1926, the most influential year, changes each headline coefficient by less than 0.20 baseline standard error; all estimates remain positive and the minimum first-stage F-statistic exceeds 13. Thus no single year dominates, although the IV may still rely on the cluster of high-contribution flows in the 1920s and early 1930s.

Table F.14: Leave-One-Missionary-Flow-Year-Out Robustness

Outcome	Baseline coefficient (SE)	Leave-one-year-out range	Largest change	Omitted year	Minimum K-P F
Modern banks, 1936 (SD)	0.152 (0.083)	0.140–0.168	0.196 SE (10.7%)	1926	13.30
Modern banks, 2020 (SD)	0.145 (0.051)	0.138–0.153	0.145 SE (5.1%)	1926	13.30
GDP, 2019 (million 2017 USD)	7,581 (2,745)	7,181–8,092	0.186 SE (6.7%)	1926	13.45

Notes: The baseline column reports the headline 2SLS coefficient with its robust standard error in parentheses. The range is the minimum and maximum coefficient across 59 specifications, each of which removes one positive missionary-flow year’s contribution from the exact instrument and otherwise retains the baseline sample, railway controls, and province fixed effects. Largest change is the maximum absolute coefficient change in baseline-standard-error units, with the change as a percentage of the baseline coefficient in parentheses. Omitted year identifies the specification producing that largest change. Minimum K–P F is the smallest Kleibergen–Paap first-stage F-statistic across all 59 omissions.

F.13 Recentered IV with Simulated Missionary Flows

This section reports a robustness check for the exposure structure of the long-run IV. The concern is that prefectures with earlier or denser railway histories may be more exposed to any national flow process through transport networks. In this setting, average missionary flows interacted with railway access could therefore proxy for generic railway-mediated exposure rather than the realized timing of missionary inflows. Following [Borusyak and Hull \(2023\)](#), we compute the expected instrument under simulated counterfactual missionary-flow histories and subtract that expected component from the original instrument.

Let F_t be the national missionary flow in year t , and let $R_{i,t-1}$ be prefecture i ’s lagged cumulative railway stock. The original instrument is $Z_i = \sum_t \hat{\beta} F_t R_{i,t-1}$. For each simulation, we permute the national missionary-flow series across years, keep each prefecture’s railway path fixed, and compute the instrument implied by this counterfactual timing. Averaging over 1,000 simulated flow histories gives μ_i , the component of the instrument mechanically predicted by the prefecture’s railway exposure profile under average missionary-flow timing. The recentered instrument is $\tilde{Z}_i = Z_i - \mu_i$. We use two versions: a full-year permutation and a 10-year block permutation that preserves within-decade serial structure in missionary flows.

Table F.15: Recentered IV with Simulated Missionary Flows

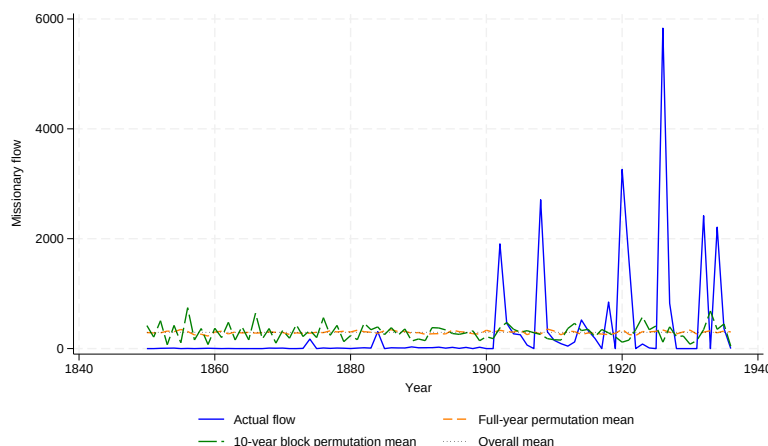
Outcome	Instrument	Coef.	SE	K–P F	Obs.
Std. total banks, 2020	Original IV	0.145***	0.051	14.87	247
Std. total banks, 2020	Recentered, full permutation	0.127***	0.043	16.72	247
Std. total banks, 2020	Recentered, 10-year blocks	0.128***	0.044	16.70	247
Direct-grid GDP, 2019	Original IV	7,581***	2,745	15.01	242
Direct-grid GDP, 2019	Recentered, full permutation	6,548***	2,273	16.87	242
Direct-grid GDP, 2019	Recentered, 10-year blocks	6,616***	2,318	16.84	242

Notes: The table reports headline 2SLS estimates for the original long-run IV and two recentered instruments under the 1st–99th percentile convention. The full permutation treats the realized annual missionary-flow values as exchangeable across years. The 10-year block permutation randomly reorders decade-sized blocks while preserving the within-block sequence of flows. All specifications use the same controls as the baseline long-run IV: railway duration, railway density, and province fixed effects. Modern bank counts are assigned from counties and districts to 1911 prefectures by their centroids. GDP is measured in millions of 2017 US dollars and constructed by summing 1-km grid cells directly within accepted 1911 prefecture polygons. Table [F.16](#) reports the corresponding first and second stages. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Table E.16: First and Second Stages for Recentered Missionary-Flow IV

	Original IV	Recentered, full-year	Recentered, 10-year blocks
<i>Panel A. First stage: missionary institutions</i>			
Instrument	3.400*** (0.882)	5.923*** (1.448)	5.884*** (1.440)
RR Duration	0.0690** (0.0279)	0.0667** (0.0282)	0.0667** (0.0281)
RR Density	-685.3*** (128.9)	-665.5*** (116.0)	-667.3*** (117.3)
Province FE	Yes	Yes	Yes
Observations	247	247	247
K-P F	14.87	16.72	16.70
<i>Panel B. Second stage: standardized total banks, 2020</i>			
Missionary institutions	0.145*** (0.051)	0.127*** (0.043)	0.128*** (0.044)
RR Duration	-0.0046 (0.0065)	-0.0025 (0.0055)	-0.0026 (0.0056)
RR Density	47.5 (32.1)	41.2 (29.6)	41.7 (29.8)
Province FE	Yes	Yes	Yes
Observations	247	247	247
K-P F	14.87	16.72	16.70
Mean dep. var.	0.000	0.000	0.000
SD dep. var.	1.000	1.000	1.000

Notes: Robust standard errors are in parentheses. The first-stage dependent variable is missionary institutions in the outcome year. Modern county- and district-level bank counts are assigned to 1911 prefectures by their centroids. The mission stock, instrument, railway controls, and outcome use the 1st–99th percentile convention. All specifications include railway duration, railway density, and province fixed effects. ***, **, and * indicate significance at the 1%, 5%, and 10% levels.

Figure E.18: Actual and Simulated Missionary-Flow Histories

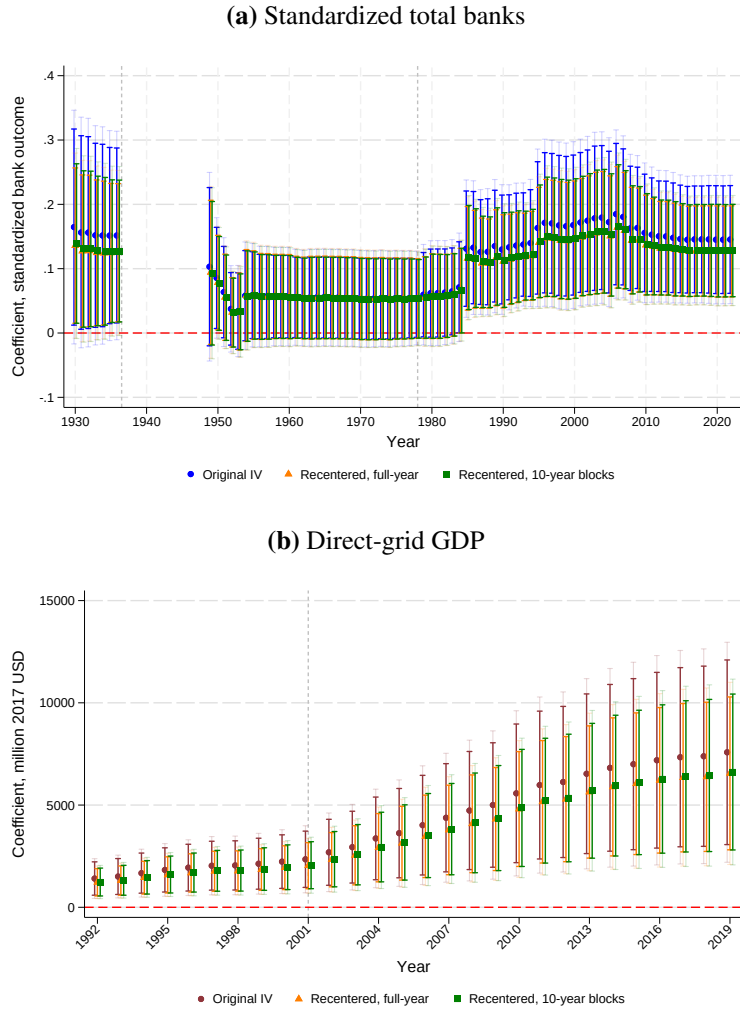
Notes: The figure compares the realized 1st–99th percentile national missionary-flow series with the average counterfactual flow series generated by full-year and 10-year block permutations. These simulated flow histories are used to compute the expected instrument that is subtracted from the original IV.

Table F.16: First and Second Stages for Recentered Missionary-Flow IV (continued)

	Original IV	Recentered, full-year	Recentered, 10-year blocks
<i>Panel C. First stage: missionary institutions, direct-grid GDP sample</i>			
Instrument	3.441*** (0.888)	5.985*** (1.457)	5.946*** (1.449)
RR Duration	0.0675** (0.0280)	0.0653** (0.0283)	0.0653** (0.0282)
RR Density	-694.7*** (131.6)	-674.0*** (119.0)	-675.7*** (120.4)
Province FE	Yes	Yes	Yes
Observations	242	242	242
K-P F	15.01	16.87	16.84
<i>Panel D. Second stage: direct-grid GDP, 2019</i>			
Missionary institutions	7,581*** (2,745)	6,548*** (2,273)	6,616*** (2,318)
RR Duration	-304.1 (330.5)	-187.2 (271.5)	-194.9 (276.9)
RR Density	1,972,945 (1,574,214)	1,618,686 (1,397,870)	1,641,973 (1,411,742)
Province FE	Yes	Yes	Yes
Observations	242	242	242
K-P F	15.01	16.87	16.84
Mean dep. var.	46,051	46,051	46,051
SD dep. var.	52,206	52,206	52,206

Notes: Robust standard errors are in parentheses. The first-stage dependent variable is missionary institutions in the outcome year. A separate GDP first stage is reported because the direct-grid GDP and bank outcomes contain different prefectures. The mission stock, instrument, railway controls, and GDP outcome use the 1st–99th percentile convention. All specifications include railway duration, railway density, and province fixed effects. GDP is measured in millions of 2017 US dollars and constructed by summing 1-km grid cells directly within accepted 1911 prefecture polygons. ***, **, and * indicate significance at the 1%, 5%, and 10% levels, respectively.

Figure F.19: Long-Run IV Estimates with Recentered Instruments

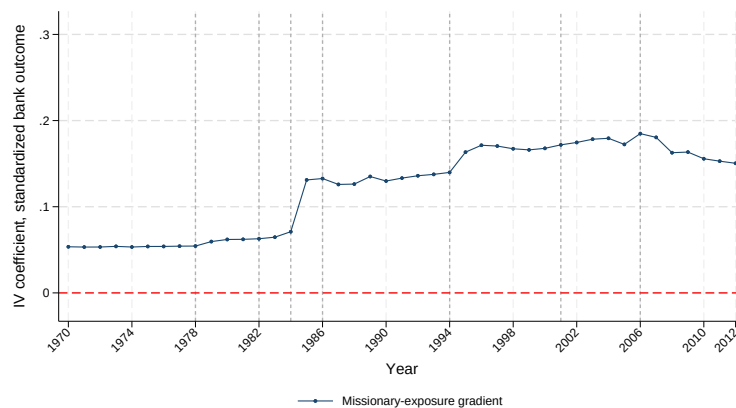


Notes: The figure compares the baseline long-run IV estimates with the recentered-IV estimates based on simulated missionary-flow histories. All inputs and outcomes use the 1st–99th percentile convention. The bank panel uses county- and district-centroid assignments to 1911 prefectures. The GDP panel uses the direct-grid outcome constructed by summing 1-km grid cells directly within accepted 1911 prefecture polygons. The point estimates are slightly smaller after recentering but follow the same time pattern as the baseline estimates.

F.14 Interactions with Opening Policies

The conceptual framework implies that historical missionary exposure should matter more when the national institutional environment raises the return to historically formed market-relevant inputs. We first plot the year-specific coefficients from Equation 5 in a single series, rather than resetting the coefficient to zero before each reform. Figure F.20 shows that the missionary-exposure gradient is largely flat before the opening reforms. It begins to rise after 1978, with the most visible early jump between 1984 and 1985, and rises again during the market-opening sequence of the 1990s and early 2000s.

Figure F.20: Missionary-Exposure Gradient in Standardized Bank Offices, 1970–2012



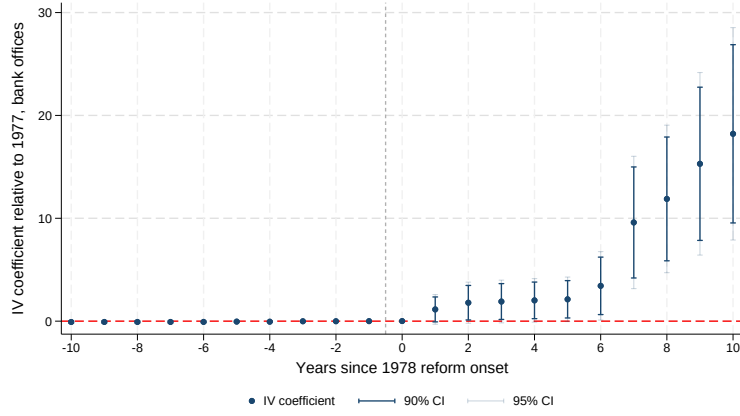
Notes: The figure reports year-specific coefficients from Equation 5. Bank offices are winsorized at the 1st and 99th percentiles and standardized within year. Dashed lines denote 1978 rural reform pilots; 1982 national HRS endorsement; 1984 decollectivization and two-tier banking; 1986 banking-administration rules; 1994 commercialization; 2001 WTO accession; and 2006 completion of the WTO banking phase-in.

Unlike the later policy dates, the years before the 1978 opening-reform onset are not already part of an earlier opening-policy sequence. We therefore also report one event-study normalization around 1978:

$$Y_{i,t} = \alpha_i + \lambda_t + \sum_{\substack{k=-10 \\ k \neq -1}}^{10} \beta_k \text{TotMiss Inst}_i \mathbf{1}\{t = 1978 + k\} + \varepsilon_{i,t}, \quad (6)$$

where each missionary-exposure interaction is instrumented by predicted missionary exposure interacted with the same relative-year indicator, and 1977 is the omitted year. Figure F.21 shows no systematic movement in the ten years before the reform onset; the pre-period coefficients are jointly indistinguishable from zero ($p = 0.465$). The gradient then rises gradually after 1978 and becomes clearly positive from 1984 onward. The later post-period coefficients capture the broader opening sequence, including the decollectivization and banking reforms that followed the initial opening, rather than a single 1978 policy.

Figure F.21: Missionary Exposure and the 1978 Opening-Reform Onset



Notes: The figure reports Equation 6 for the ten years before and after 1978, using 1st–99th percentile-winsorized bank offices. Coefficients are relative to 1977; dark and light intervals are 90% and 95%. Standard errors are clustered by prefecture.

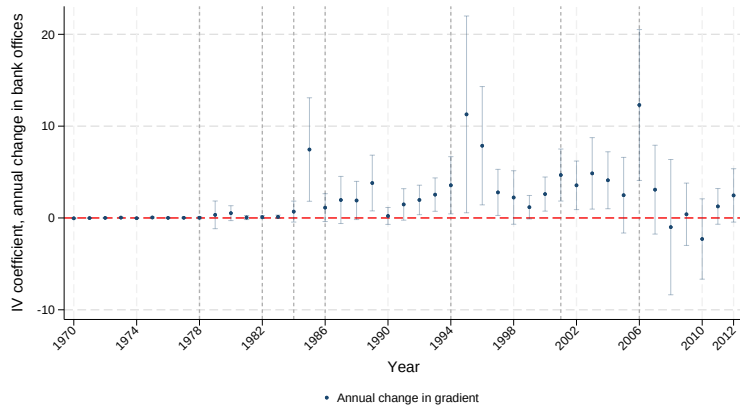
We do not repeat this event-study normalization around the later policy dates. Their pre-periods already contain earlier reforms and a rising missionary-exposure gradient, so resetting the year before each policy to zero would mechanically turn that inherited path into apparent negative pre-trends. For those dates, the absolute gradient and growth-shift comparisons below are more informative.

To examine when this gradient changes, we next replace the bank-office outcome with its annual first difference:

$$\Delta Y_{i,p,t} = \xi_{p,t} + \psi_t \text{TotMiss Inst}_{i,p} + \pi_t \text{RR Duration}_{i,p} + \rho_t \text{RR Density}_{i,p} + \nu_{i,p,t}, \quad (7)$$

where $\Delta Y_{i,p,t} = Y_{i,p,t} - Y_{i,p,t-1}$ and missionary institutions are instrumented by predicted missionary exposure as in Equation 4. Figure F.22 reports $\hat{\psi}_t$ for each year. The largest early increase occurs in 1985, after the 1984 completion of decollectivization and reconstruction of the two-tier banking system. The 2001 coefficient also shows a clear increase around WTO accession. The annual estimates are noisier in the 1990s and 2000s, motivating a more compact comparison around each policy date.

Figure F.22: Annual Change in the Missionary-Exposure Gradient for Bank Offices



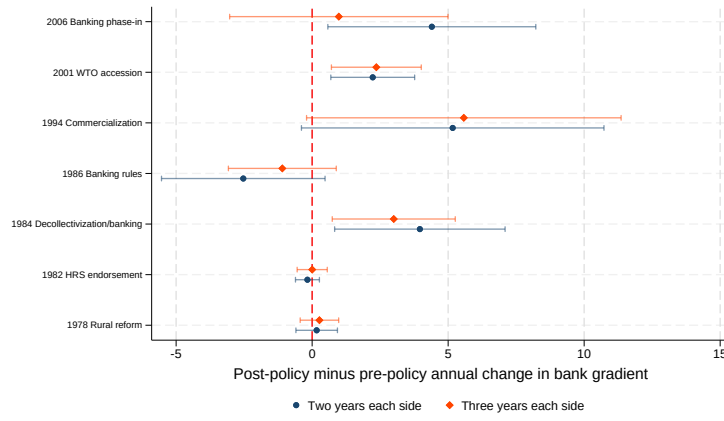
Notes: The dependent variable is the annual change in bank offices. Points report the year-specific IV coefficient from Equation 7; intervals are 95%. Dashed lines denote the same policy dates as Figure F.20.

Finally, for each candidate date T_p , we compare the missionary-exposure gradient in annual outcome growth immediately before and after the policy:

$$\Delta Y_{i,t} = \alpha_i + \lambda_{p(i),t} + \theta_p \text{TotMiss Inst}_i \times \text{Post}_{p,t} + \Gamma'(\text{RR}_i \times \text{Post}_{p,t}) + \eta_{i,t}, \quad (8)$$

where the missionary interaction is instrumented by predicted missionary exposure interacted with the same post indicator. We estimate Equation 8 using either two or three years on each side of the policy date. Figure F.23 shows that the 1984 and 2001 breaks are positive and statistically distinguishable from zero under both window choices. The 2006 estimate is positive using two years on each side but becomes small and imprecise using three, while the other dates provide less consistent evidence of a break.

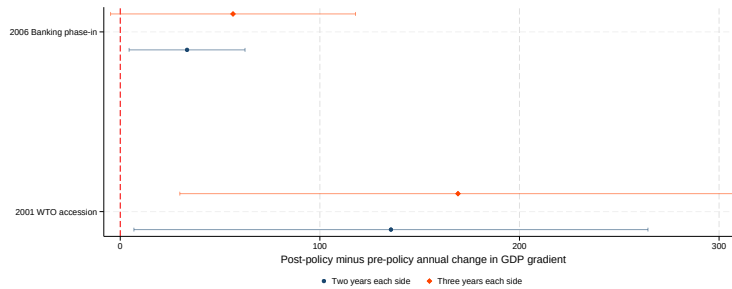
Figure F.23: Change in Annual Bank-Office Growth Around Candidate Policy Dates



Notes: Each point is the post-period minus pre-period missionary-exposure gradient in annual bank-office changes from Equation 8, using two or three years on each side. Models include prefecture and province-by-year fixed effects and post interactions with the railway controls. Intervals are 95%; standard errors are clustered by prefecture.

Figure F.24 applies the same comparison to direct-grid GDP. The GDP data begin in 1992, so annual changes begin in 1993. The 1994 commercialization marker therefore has only one available pre-policy annual change and cannot support the balanced two- or three-year windows used here. For the subsequent policy dates, the increase around 2001 is positive and statistically distinguishable from zero under both windows. The 2006 estimate is positive but again more sensitive to the window.

Figure F.24: Change in Annual GDP Growth Around Candidate Policy Dates



Notes: Each point is the post-period minus pre-period missionary-exposure gradient in annual GDP changes, using two or three years on each side. GDP is measured in millions of 2017 US dollars and is not standardized. Models and inference otherwise follow Figure F.23. The balanced GDP windows are available for 2001 WTO accession and the 2006 banking phase-in.

Taken together, these exercises provide the clearest evidence of changes in the missionary-exposure gradient around 1984 and 2001. The 1984 break coincides with the completion of rural decollectivization, which dismantled the commune system and shifted production decisions to households, and with the reconstruction of a two-tier banking system that separated central-bank functions from lending by specialized state banks. The 2001 break coincides with China's WTO accession, which deepened its integration into global trade and committed China to a staged opening of domestic banking services to foreign institutions.

Figure F.25: Long-Run 2SLS Robustness: Railway Density and Two Railway Bins

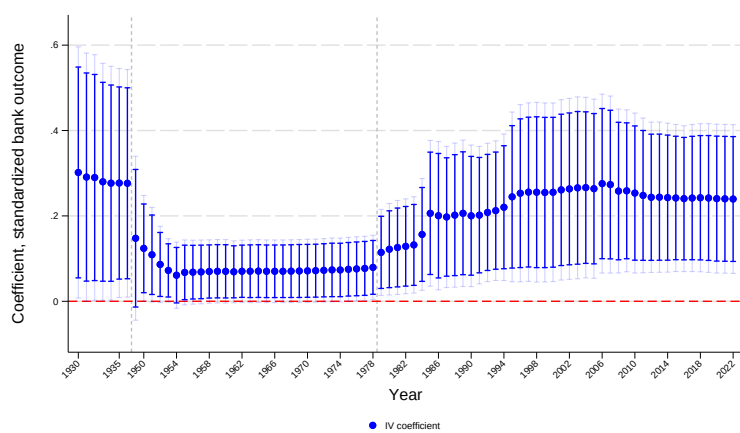
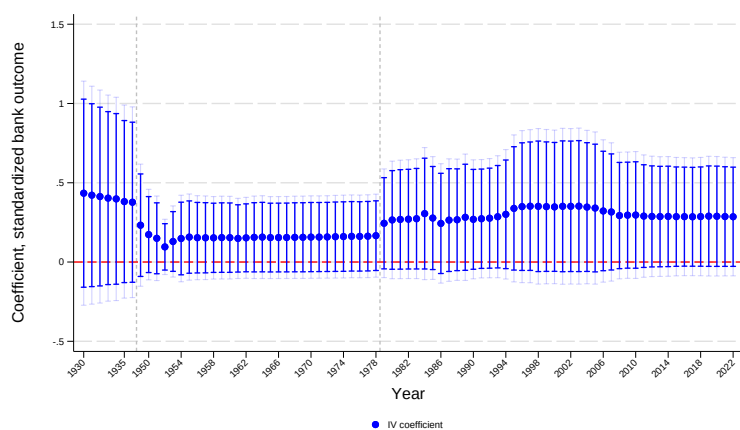


Figure F.26: Long-Run 2SLS Robustness: Railway Density and Six Railway Bins



Notes: Each figure reports year-specific 2SLS coefficients for standardized bank offices using the specification named in its caption. Dark and light intervals show 90% and 95% confidence intervals; dashed lines mark zero, 1949, and 1978. Inputs and outcomes use the 1st–99th percentile convention unless the caption states otherwise. Source: China Financial Supervision and Administration Bureau.

Figure F.27: Long-Run 2SLS Robustness: Railway Mileage per Square Kilometer

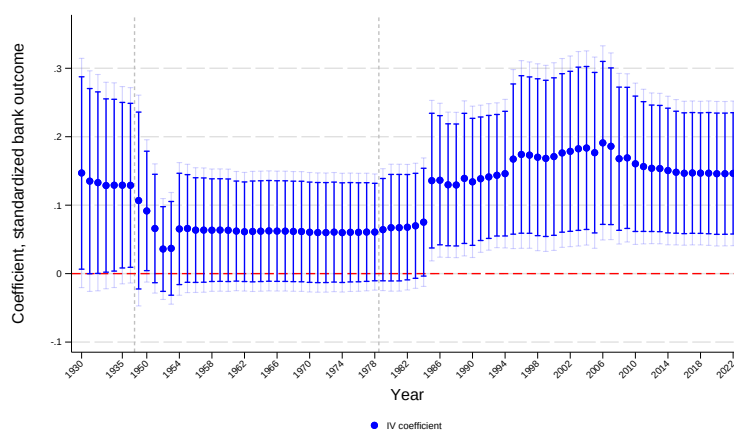
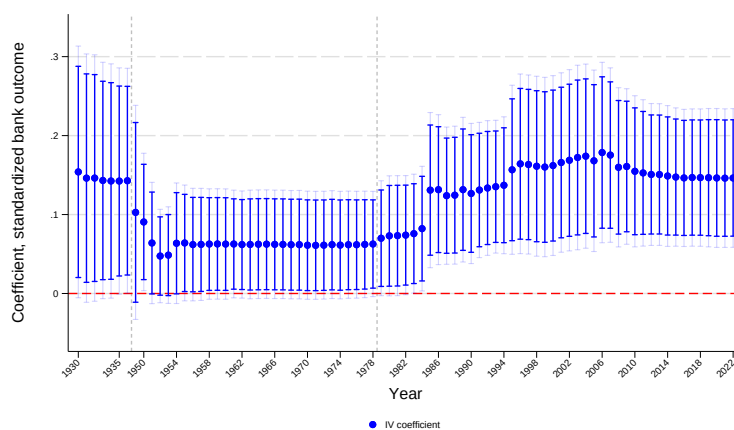


Figure F.28: Long-Run 2SLS Robustness: Latitude and Longitude Controls



Notes: Each figure reports year-specific 2SLS coefficients for standardized bank offices using the specification named in its caption. Dark and light intervals show 90% and 95% confidence intervals; dashed lines mark zero, 1949, and 1978. Inputs and outcomes use the 1st–99th percentile convention unless the caption states otherwise. Source: China Financial Supervision and Administration Bureau.

Figure F.29: Long-Run 2SLS Robustness: Railway Count in 1936

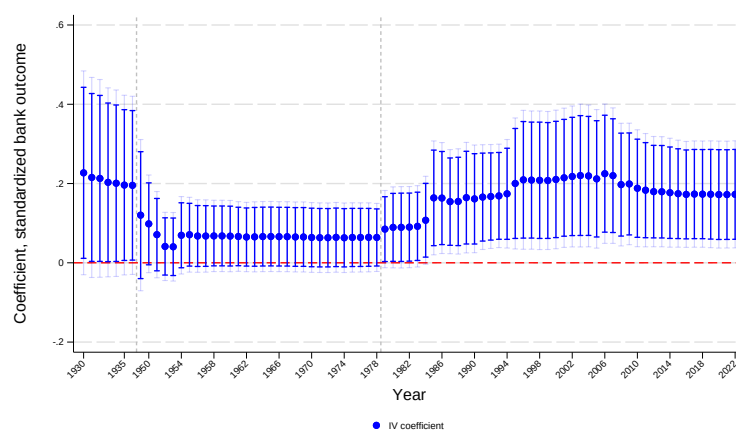
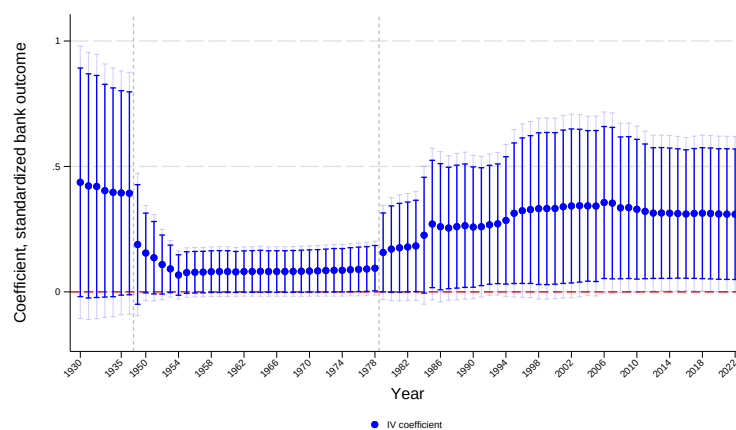


Figure F.30: Long-Run 2SLS Robustness: Railway Duration and Two Railway Bins



Notes: Each figure reports year-specific 2SLS coefficients for standardized bank offices using the specification named in its caption. Dark and light intervals show 90% and 95% confidence intervals; dashed lines mark zero, 1949, and 1978. Inputs and outcomes use the 1st–99th percentile convention unless the caption states otherwise. Source: China Financial Supervision and Administration Bureau.

Figure F.31: Long-Run 2SLS Robustness: Excluding Treaty Ports

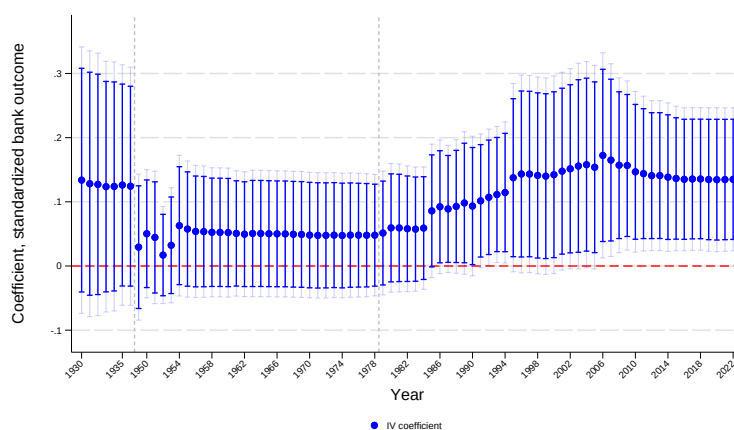
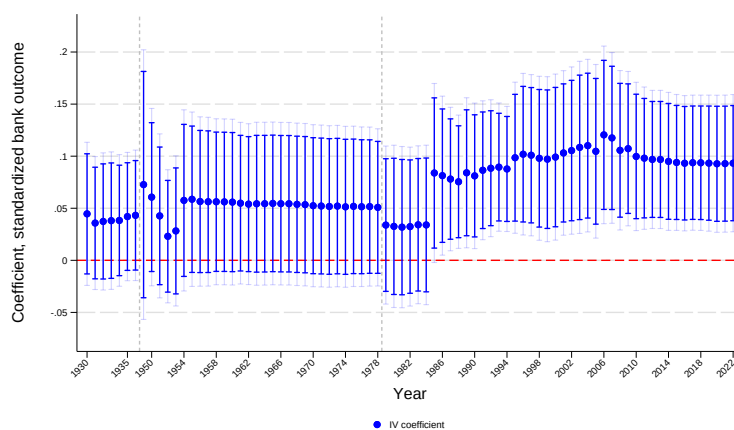


Figure F.32: Long-Run 2SLS Robustness: Excluding the Three Largest Railway Outliers



Notes: Each figure reports year-specific 2SLS coefficients for standardized bank offices using the specification named in its caption. Dark and light intervals show 90% and 95% confidence intervals; dashed lines mark zero, 1949, and 1978. Inputs and outcomes use the 1st–99th percentile convention unless the caption states otherwise. Source: China Financial Supervision and Administration Bureau.

Figure F.33: Long-Run 2SLS Robustness: Alternative Zero-Stage Window, 1850–1950

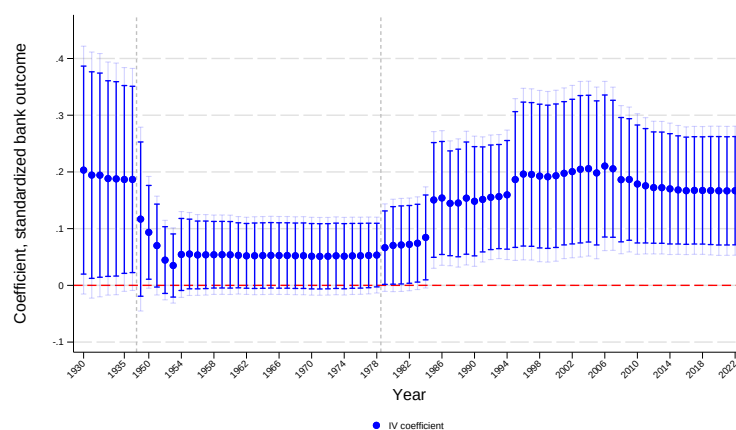
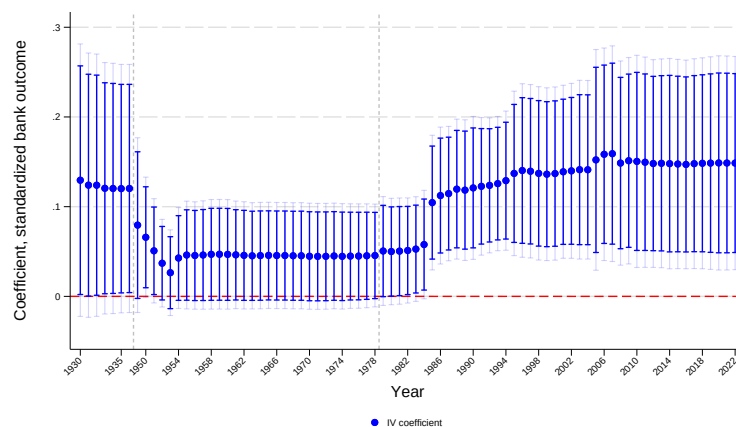
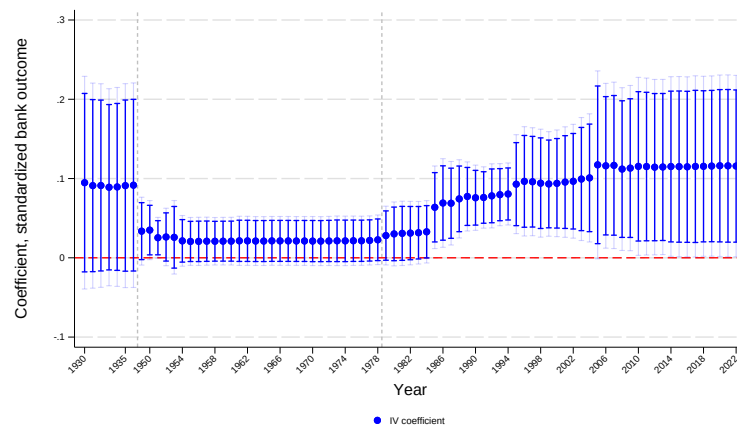


Figure F.34: Long-Run 2SLS Robustness: Alternative Winsorization at the 0.5th and 99.5th Percentiles



Notes: Each figure reports year-specific 2SLS coefficients for standardized bank offices using the specification named in its caption. Dark and light intervals show 90% and 95% confidence intervals; dashed lines mark zero, 1949, and 1978. Inputs and outcomes use the 1st–99th percentile convention unless the caption states otherwise. Source: China Financial Supervision and Administration Bureau.

Figure F.35: Long-Run 2SLS Robustness: No Winsorization



Notes: Each figure reports year-specific 2SLS coefficients for standardized bank offices using the specification named in its caption. Dark and light intervals show 90% and 95% confidence intervals; dashed lines mark zero, 1949, and 1978. Inputs and outcomes use the 1st–99th percentile convention unless the caption states otherwise. Source: China Financial Supervision and Administration Bureau.